

SRINIVAS UNIVERSITY

**INSTITUTE OF COMPUTER SCIENCE & INFORMATION
SCIENCES**

**CITY CAMPUS, PANDESHWAR,
MANGALORE - 575 001**

BACKGROUND STUDY MATERIAL

CLOUD COMPUTING

B. C. A - V SEMESTER



Compiled by
Faculty

CLOUD COMPUTING

CONTENTS

UNIT-1

1. Introduction

Web 2.0 and the cloud
Cloud types
Deployment models
Service models
Exploring the uses of cloud,
Scalability
Virtualization
Collecting processing power through Grid Computing

2. Software as a Service

Getting starting with SaaS
Understanding the Multitenant nature of SaaS solution
Understanding open SaaS Solutions
Understanding service-oriented architecture

Assignment 1

UNIT-2

3. Platform as a Service

IT evolution leading to the cloud
Benefits of Paas Solution
Disadvantages of Paas Solutions

4. Infrastructure as a Service

Understanding IaaS
Improving performance through load balancing, system and storage redundancy
Utilizing cloud based NAS devices
Advantages of IaaS Solutions
Identity as a service
Understanding single sign on

Understanding open ID
Mobile ID management

Assignment 2

UNIT - 3

5. Cloud data storage and virtualization

Examining the evaluation of network storage
Understanding Cloud Based data storage
Advantages and Disadvantages of Cloud Based Storage
Cloud Based backup system
Industry specific Cloud Based data storage
Cloud Based database Solutions
Cloud based Block Storage
Understanding virtualization
History of virtualization
Server virtualization
Desktop virtualisation
Desktop solution on demand
Virtual networks
Data storage virtualization
Assignment 3

UNIT - 4

6. Cloud security and management.

General security advantages of Cloud Based Solutions
Introducing business continuity and Disaster recovery
Know your service level Agreement
Ensure and audit system backup

Know your system's backup
Beware of Vendor lock in
Determine technical support
Determine training Procedures
Monitor capacity planning and scaling capabilities
Monitor Audit Log
Assignment 4

UNIT - 5

7. Migration to the Cloud

- Define the system goals and requirements
- Protect your existing data
- Use an experienced cloud consultant
- Know your application's current characteristics
- Remember Vender Lock-in
- Define your Training requirements
- Establish a Realistic Deployment Schedule
- Identify IT governance issues
- Understanding Cloud Bursting

8.Mobile Cloud Computing

- Understanding 3G and 4G
- The mobile cloud ecosystem
- Introducing mobile players- pages, apps and widgets
- Mobile development considerations

Assignment 5

CLOUD COMPUTING SYLLABUS

Paper Code: 20BCASD54/ 20BCANT55 Theory/Week:4 Hours Credits:4	CLOUD COMPUTING	Hours: 40 IA: 50 Exam: 50
UNIT I: Introduction		8 Hours
Web 2.0 and the cloud, cloud types, deployment models, service models, Exploring the uses of cloud, scalability, virtualization, collecting processing power through Grid Computing. Software as a Service: Getting starting with SaaS, Understanding the Multitenant nature of SaaS solution, Understanding open SaaS Solutions. Understanding service oriented architecture.		
UNIT II: Platform as a Service and Infrastructure.		8 Hours
IT evolution leading to the cloud, Benefits of Paas Solution, Disadvantages of PaaS Solutions. Understanding IaaS, Improving performance through load balancing, system and storage redundancy, utilising cloud based NAS devices, Advantages of IaaS Solutions, Identify as a service, Understanding single sign on, Understanding open ID, mobile ID management.		
UNIT III: Data storage and Virtualisation		8 Hours
Examining the evaluation of network storage, understanding Cloud Based data storage, Advantages and Disadvantages of Cloud Based Storage, Cloud Based backup system, Industry specific Cloud Based data storage, Cloud Based database Solutions, Cloud based Block Storage. Understanding virtualisation, history of virtualisation, server virtualisation, Desktop virtualisation, Desktop solution on demand, virtual networks, data storage virtualisation.		
UNIT IV: Cloud security and management.		8 Hours
General security advantages of Cloud Based Solutions, Introducing business continuity and Disaster recovery, Know your service level Agreement, Ensure and audit system backup, Know your system's backup. Beware of Vender lock in. Determine technical support, Determine training Procedures, Monitor capacity planning and scaling capabilities, Monitor Audit Log use.		
UNIT V: Migration to the Cloud and Mobile Cloud Computing.		8 Hours
Define the system goals and requirements, Protect your existing data, Use an experienced		

cloud consultant, Know your application's current characteristics, Remember Vendor Lock-in, Define your Training requirements, Establish a Realistic Deployment Schedule, Identify IT governance issues, Understanding Cloud Bursting.

Understanding 3G and 4G, The mobile cloud ecosystem, Introducing mobile players- pages, apps and widgets, Mobile development considerations.

Total Hours: 40

BOOKS FOR REFERENCE:

1. Cloud Computing by Kris Jamsa.
2. Cloud Computing by NICK Antonopouls, Lee Gillam
3. Cloud Computing- A Practical Approach for Learning and Implementation by A. Srinivasan and J. Suresh.

CLOUD COMPUTING

TEACHING PLAN

Total hours: 40H

UNIT I

Introduction

8hrs

- Session 1: Web 2.0 and the cloud, Cloud types
- Session 2: Deployment models
- Session 3: Service models
- Session 4: Exploring the uses of cloud,
- Session 5: Scalability, Virtualization
- Session 6: Collecting processing power through Grid Computing, Getting starting with SaaS,
- Session 7: Understanding the Multitenant nature of SaaS solution
- Session 8: Understanding open SaaS Solutions, Understanding service-oriented architecture

UNIT – 2

Platform as a Service and Infrastructure

8 hrs

- Session 1: IT evolution leading to the cloud Benefits of Paas Solution
- Session 2: Disadvantages of Paas Solutions Understanding Iaas
- Session 3: Utilizing cloud based NAS devices
- Session 4: Advantages of IaaS Solutions
- Session 5: Identity as a service
- Session 6: Understanding single sign on
- Session 7: Understanding open ID
- Session 8: Mobile ID management

UNIT – 3

Cloud data storage and virtualization

8hrs

- Session 1: Examining the evaluation of network storage
- Session 2: Understanding Cloud Based data storage
- Session 3: Advantages and Disadvantages of Cloud Based Storage
- Session 4: Cloud Based backup system, Industry specific Cloud Based data storage
- Session 5: Cloud Based database Solutions
- Session 6: Cloud based Block Storage, Understanding virtualization
- Session 7: History of virtualization, Server virtualization, Desktop virtualisation
- Session 8: Desktop solution on demand, Virtual networks, Data storage virtualization

UNIT – 4

Cloud security and management

8hrs

- Session 1: General security advantages of Cloud Based Solutions
- Session 2: Introducing business continuity and Disaster recovery
- Session 3: Know your service level Agreement
- Session 4: Ensure and audit system backup ,Know your system's backup
- Session 5: Beware of Vender lock in
- Session 6: Determine technical support ,Determine training Procedures
- Session 7: Monitor capacity planning and scaling capabilities
- Session 8: Monitor Audit Log

UNIT – 5

Migration to the Cloud and Mobile Cloud Computing

8hrs

- Session 1: Define the system goals and requirements, Protect your existing data, Use an experienced cloud consultant
- Session 2: Know your application's current characteristics, Remember Vender Lock-in
- Session 3: Define your Training requirements, Establish a Realistic Deployment Schedule
- Session 4: Identify IT governance issues
- Session 5: Understanding Cloud Bursting
- Session 6: Understanding 3G and 4G
- Session 7: The mobile cloud ecosystem
- Session 8: Introducing mobile players- pages, apps and widgets, Mobile development considerations

TEXT BOOKS

1. Cloud Computing by Kris Jamsa.
2. Cloud Computing by NICK Antonopouls, Lee Gillam
3. Cloud Computing- A Practical Approach for Learning and Implementation by A. Srinivasan and J. Suresh.

UNIT - I
CHAPTER1
INTRODUCTION TO CLOUD COMPUTING

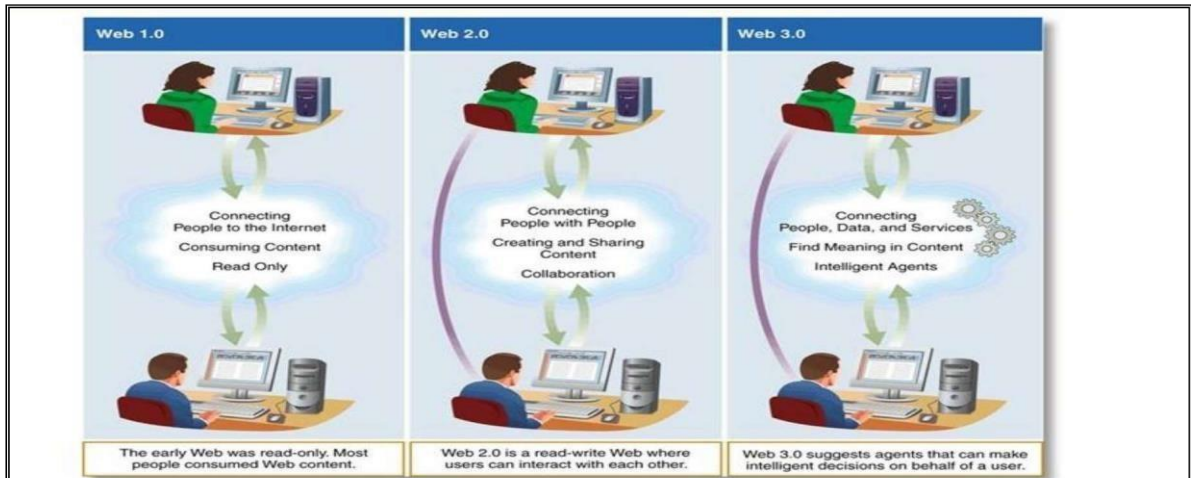
INTRODUCTION TO CLOUD COMPUTING

Cloud computing refers to applications and services that run on a distributed network using virtualized resources and accessed by common Internet protocols and networking standards. It is distinguished by the notion that resources are virtual and limitless and that details of the physical systems on which software runs are abstracted from the user. In an effort to better describe cloud computing, a number of cloud types have been defined.

There are two different classes of clouds: those based on the deployment model and those based on the service model. The deployment model tells you where the cloud is located and for what purpose. Public, private, community, and hybrid clouds are deployment models. Service models describe the type of service that the service provider is offering. The best-known service models are Software as a Service, Platform as a Service, and Infrastructure as a Service. The service models build on one another and define what a vendor must manage and what the client's responsibility is.

Web 2.0 and the Cloud

The web is the primary interface (medium) through which cloud computing delivers or provides its services to everyone. Basically Web 2.0 is the current state of online technology as it compares to early days of the Web, it comprises by greater user interactivity and collaboration and enhanced communication channels. The name Web 2.0 used to elaborate the second generation of www i.e. World Wide Web. It focused on the ability for user or people to collaborate and share information online via social media, web based communities, blogging and internet of course.



Benefits of Web 2.0

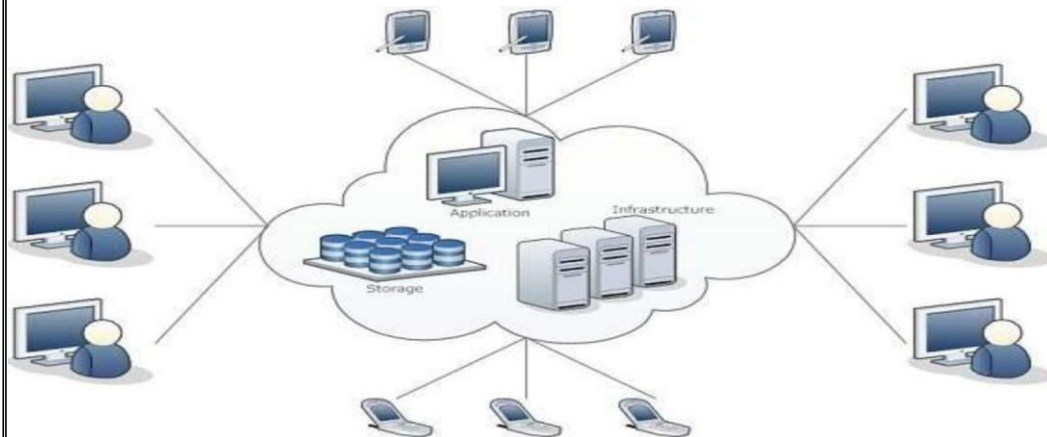
- Access all the services available at internet anytime or from anywhere.
- Web services are user friendly or has better user interface.
- Multiple kind or different type of media is available on web.
- It can build dynamic learning communities.
- Anyone can have real time discussion now.
- Success for search results is efficient.
- Updates of any information are easy.

Service running on web 2.0

Facebook, Flickr, linked in, YouTube, Wikipedia, BlogSpot, Word press, Twitter, Pinterest, Google Docs, Instagram and many more. Today every service available on internet is under Web 2.0.

CLOUD COMPUTING AND CLOUD TYPES

Cloud computing describes how applications are stored and deployed on network of internet servers. We need not to install a piece of software on our local PC and this is how, the cloud computing overcomes platform dependency issues. Hence, the Cloud Computing is making our business application mobile and collaborative.



A cloud represents internet, cloud can be classified

1. Based on deployment model
 - a. Private cloud
 - b. Public cloud
 - c. Hybrid cloud
 - d. Community cloud
2. Based on service model
 - Software as a service(SaaS)
 - Platform as a service(PaaS)
 - Infrastructure as a service(IaaS)

DEPLOYMENT MODELS

As cloud technology is providing users with so many benefits, these benefits must have to be categorized based on user's requirement. Cloud deployment model represents the exact category of cloud environment based on proprietorship, size, and access and also describes the nature and purpose of the cloud. Most organizations implement cloud infrastructure to minimize capital expenditure & regulate operating costs.

There are four types of deployment models:

1. Public Cloud
2. Private Cloud
3. Hybrid Cloud
4. Community Cloud

Public Cloud:

Public Cloud is a type of cloud hosting that allows the accessibility of systems & its services to its clients/users easily. Some of the examples of those companies which provide public cloud facilities are IBM, Google, Amazon, Microsoft, etc. This cloud service is open for use. This type of cloud computing is a true specimen of cloud hosting where the service providers render services to various clients. From the technical point of view, there is the least difference between private clouds and public clouds along with the structural design. Only the security level depends based on the service providers and the type of cloud clients use. Public cloud is better suited for business purposes for managing the load. This type of cloud is economical due to the decrease in capital overheads.

Advantages of the Public cloud are:

- Flexible
- Reliable
- High Scalable
- Low cost
- Place independence

Disadvantages of public cloud:

- Less Secured
- Poor Customizable

Private Cloud:

Private Cloud also termed as 'Internal Cloud'; which allows the accessibility of systems and services within a specific boundary or organization. The cloud platform is implemented in a cloud-based secure environment that is guarded by advanced firewalls under the surveillance of the IT department that belongs to a particular organization. Private clouds permit only authorized users, providing the organizations greater control over data and its security. Business organizations that have dynamic, critical, secured, management demand-based requirement should adopt Private Cloud.

Advantages of private cloud are:

Highly private and secured: Private cloud resource sharing is highly secured.

Control Oriented: Private clouds provide more control over its resources than public cloud as it can be accessed within the organization's boundary.

Disadvantages of private cloud:

- Poor scalability: Private type of clouds is scaled within internal limited hosted resources.
- Costly: As it provides secured and more features, so it's more expensive than a public cloud.
- Pricing: is inflexible; i.e., purchasing new hardware for up-gradation is more costly.
- Restriction: It can be accessed locally within an organization and is difficult to expose globally.

Hybrid Cloud

Hybrid Cloud is another cloud computing type, which is integrated, i.e., it can be a combination of two or more cloud servers, i.e., private, public or community combined as one architecture, but remain individual entities. Non-critical tasks such as development and test workloads can be done using public cloud whereas critical tasks that are sensitive such as organization data handling are done using a private cloud. Benefits of both deployment models, as well as a community deployment model, are possible in a hybrid cloud hosting. It can cross isolation and overcome boundaries by the provider; hence, it cannot be simply categorized into any of the three deployments - public, private or community cloud.

Advantages of Hybrid Cloud Computing are:

- Flexible
- Secure
- Cost Effective
- Rich Scalable

Disadvantages of Hybrid Cloud are:

- Complex networking problem
- Organization's security Compliance

Community Cloud

Community Cloud is another type of cloud computing in which the setup of the cloud is shared manually among different organizations that belong to the same community or area. Example of such a community is where organizations/firms are there along with the financial institutions/banks. A multi-tenant setup developed using cloud among different organizations that belong to a particular community or group having similar computing concern.

For joint business organizations, ventures, research organizations and tender's community cloud are the appropriate solution. Selection of the right type of cloud hosting is essential in this case. Thus, community-based cloud users need to know and analyse the business demand first.

SERVICE MODELS

- Infrastructure as a Service (IaaS): Earlier called "Hardware as a service"
 - Service provider delivers a virtual computing resources(storage, networking, servers...) over the internet

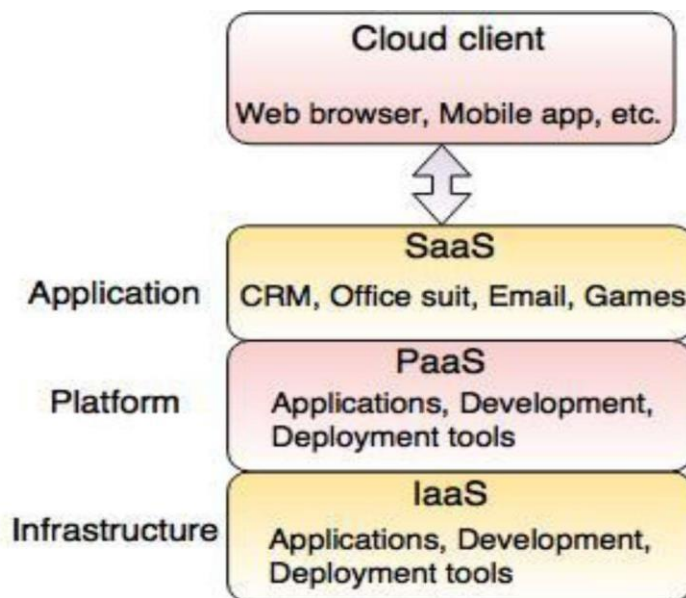
Example: AWS (Amazon Web Services), GCP (Google Cloud Platform), firewall, storage

- Platform as a Service (PaaS):
 - Service provider delivers a computing platform over the Internet
 - users can build and deliver applications.

Example: Salesforce.com, Google App Engine,

- Software as a Service (SaaS):
 - Service provider delivers software applications over the Internet
 - Users subscribe to the software and access it via internet

Example: Email, Games, Office...



EXPLORING USES OF CLOUD



1. Cloud Follows Pay for What you Use Model:

This means users will have to pay only for the amount of service used by them. This way, money spent on improving business through cloud doesn't prove to be expensive.

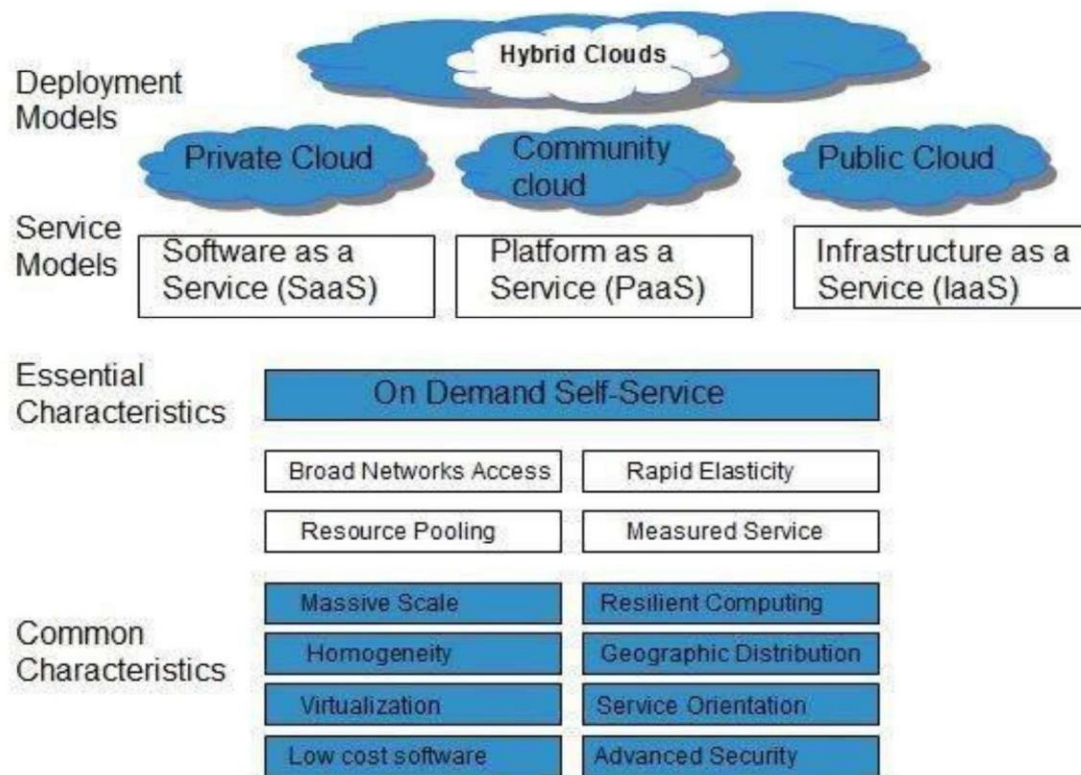
2. 24*7 Availability and Performance Oriented Infrastructure
3. Cloud Technology believes in Optimization of Existing Resources
4. Disaster Recovery Plan Help to Restore Data Quickly on Cloud
5. Agile Development on Cloud.
6. Cloud, Cost-Effective in Nature

The cloud promotes device independence, the cost of hardware and software automatically cuts short. Although, the initial expense of deploying cloud architecture is there, but that also focuses on pay for what you use model.

7. Cloud Architecture is Exceptionally Flexible
Cloud has various deployment models like private, public, hybrid, community which can control the access of control resources, hence flexible.
8. SRC (Security, Risk Management and Compliance Management)

CHARACTERISTICS OF CLOUD COMPUTING

There are four key characteristics of cloud computing. They are shown in the following diagram:



On Demand Self Service

Cloud Computing allows the users to use web services and resources on demand. One can login to a website at any time and use them.

Broad Network Access

Since cloud computing is completely web based, it can be accessed from anywhere and at any time.

Resource Pooling

Cloud computing allows multiple tenants to share a pool of resources. One can share single physical instance of hardware, database and basic infrastructure.

Rapid Elasticity

It is very easy to scale the resources vertically or horizontally at any time. Scaling of resources means the ability of resources to deal with increasing or decreasing demand.

The resources being used by customers at any given point of time are automatically monitored.

Measured Service

In this service cloud provider controls and monitors all the aspects of cloud service. Resource optimization, billing, and capacity planning etc. depend on it.

Scalability /elasticity

Cloud computing gives the illusion of infinite computing resources available on demand. Therefore, user expects clouds to rapidly provide resources in any quantity at any time. In particular, it is expected that the additional resources can be provisioned, possibly automatically, when an application load increases and when load decreases (scale up and scale down)

VIRTUALIZATION

Virtualization is the process of creating a virtual environment of something which may include hardware platforms, storage devices, OS, network resources, etc.

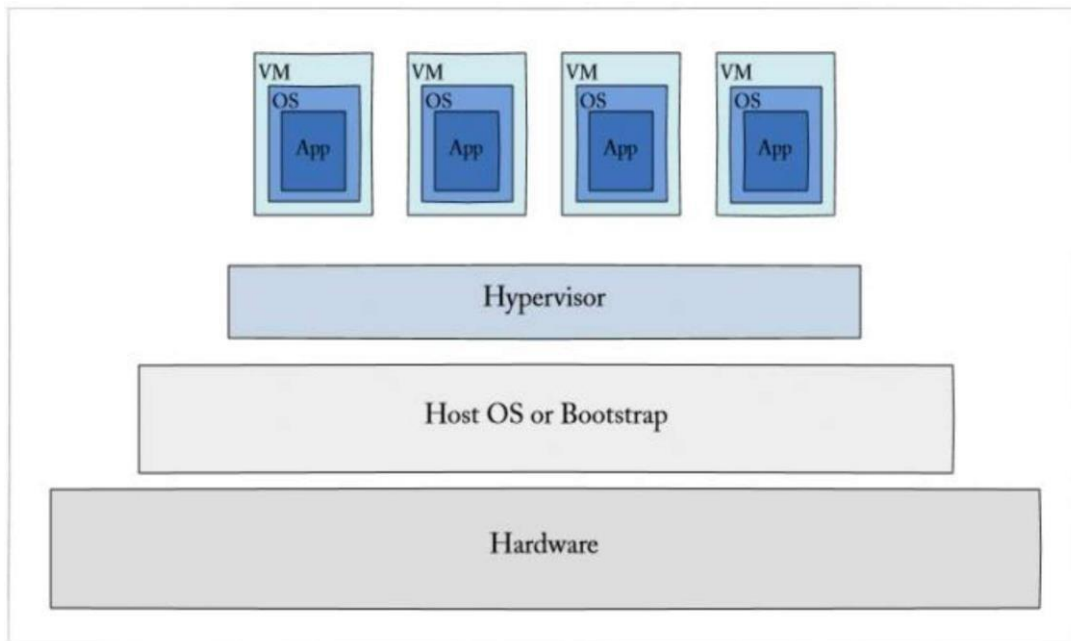
Defining Virtualization

Virtualization is the “ability which allows sharing the physical instance of a single application or resource among multiple organizations or users”. This technique is done by assigning a name logically to all those physical resources & provides a pointer to those physical resources based on demand.

Over an existing operating system & hardware, we generally create a virtual machine which and above it we run other operating systems or applications. This is called Hardware

Virtualization. The virtual machine provides a separate environment that is logically distinct from its underlying hardware. Here, the system or the machine is the host & virtual machine is the guest machine. This virtual environment is managed by a firmware which is termed as a hypervisor.

Figure - The Cloud's Virtualization:



Types of Virtualization

The virtualization of cloud has been categorized into different types based on their characteristics. These are:

- Hardware Virtualization
- Software Virtualization
- OS Virtualization
- Server Virtualization
- Storage Virtualization

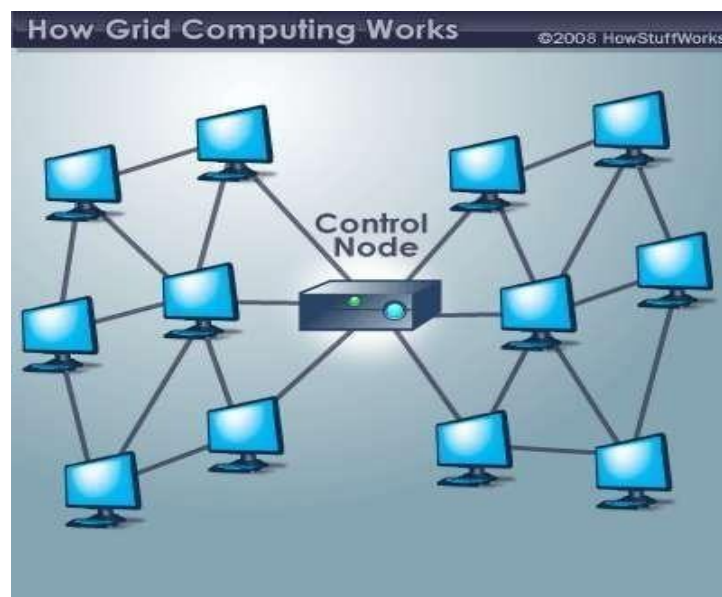
Virtualization is used mainly to provide applications with standard versions for the cloud customers & with the release of the latest version of an application the providers can efficiently provide that application to the cloud and its users and it is possible using virtualization only.

GRID COMPUTING

Grid computing is a computer network in which each computer's resources are shared with every other computer in the system.

Collecting processing power through grid computing

It's a special kind of distributed computing. In distributed computing, different computers within the same network share one or more resources. In the ideal grid computing system, every resource is shared, turning a computer network into a powerful supercomputer. With the right user interface, accessing a grid computing system would look no different than accessing a local machine's resources. Every authorized computer would have access to enormous processing power and storage capacity.



Grid computing is often confused with cloud computing, but they are quite different. Grid computing applies the resources of numerous computers in a network to work on a single problem at the same time. Grid computing necessitates the use of software that can divide and then send out pieces of the program to thousands of computers. It can be done throughout the computers of an organization, or it can be done as a form of public collaboration. *Sun Microsystems* offers Grid Engine software that allows engineers at companies to pool the computer cycles on up to 80 workstations at a time.

Benefits of Grid computing:

- It is a *cost-effective* way to use a given amount of computer resources.
- It is a way to solve problems that need a tremendous amount of computing power.
- The resources of several computers can be shared cooperatively, without one computer managing the other.
- In grid computing, a large project is divided among multiple computers to make use of their resources. Cloud computing does just the opposite.

CHAPTER 2

SOFTWARE AS A SERVICE

SOFTWARE AS A SERVICE

This model allows providing software application as a service to the end users. It refers to software that is deployed on a hosted service and is accessible via Internet.

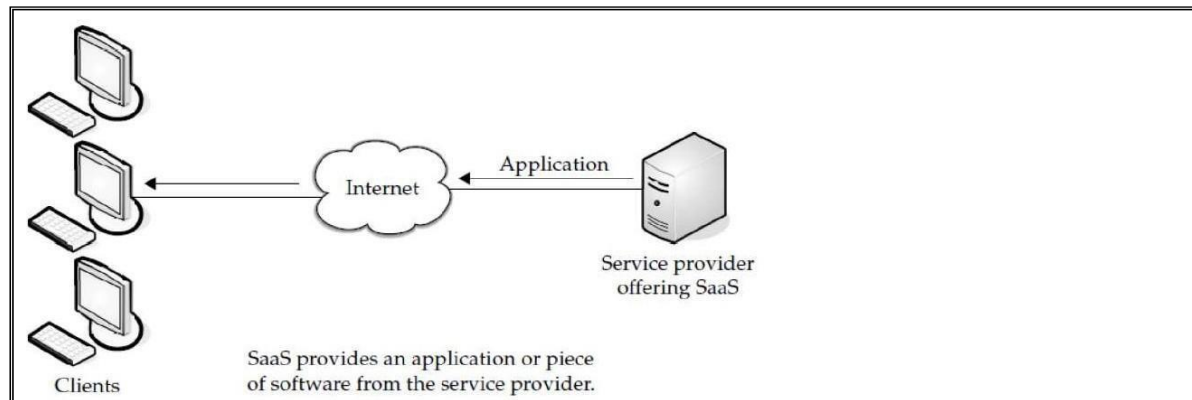
There are several SaaS applications; some of them are listed below:

- Billing and Invoicing System
- Customer Relationship Management (CRM) applications
- Help Desk Applications
- Human Resource (HR) Solutions

Some of the SaaS applications are not customizable such as an Office Suite. But SaaS provides us Application programming Interface (API), which allows the developer to develop a customized application.

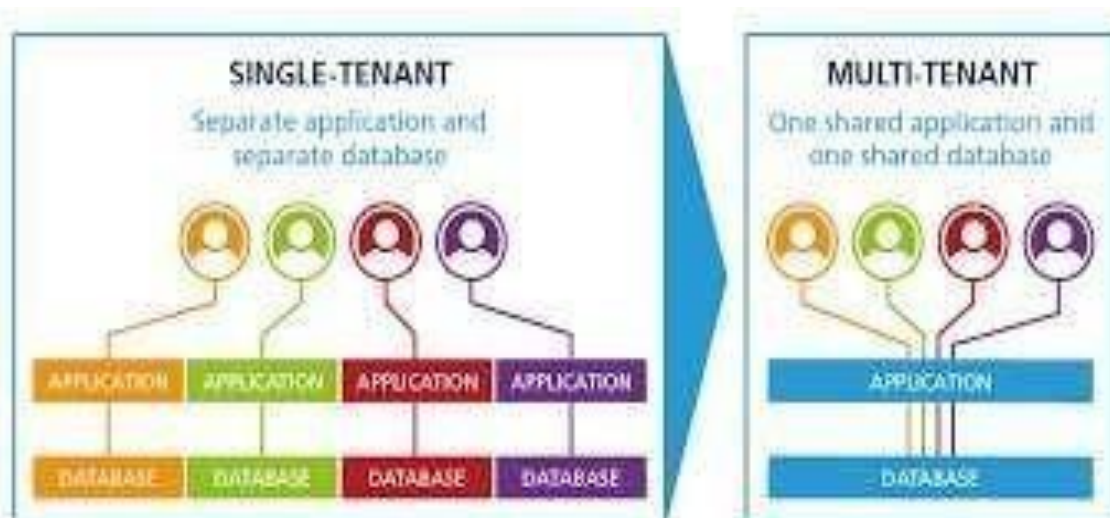
Characteristics of SaaS

- SaaS makes the software available over the Internet.
- The Software is maintained by the vendor rather than where they are running.
- The license to the software may be subscription based or usage based. And it is billed on recurring basis.
- SaaS applications are cost effective since they do not require any maintenance at end user side.
- They are available on demand.
- They can be scaled up or down on demand.
- They are automatically upgraded and updated.
- SaaS offers share data model. Therefore, multiple users can share single instance of infrastructure. It is not
- Required to hard code the functionality for individual users.
- All users are running same version of the software.



MULTI-TENANCY NATURE OF SAAS

Multi-tenancy is an architecture in which a single instance of a software application services multiple customers, or tenants. Multi-tenancy can be economical because software development and maintenance costs are shared. Additionally, with a multi-tenancy architecture, the provider only has to make updates once in order to share them with all of its 'tenants.' A software-as-a-service (SaaS) vendor, for example, can run one instance of its application on one instance of a database and provide web access to multiple customers. In such a scenario, each tenant's data is isolated and remains invisible to other tenants. In this way customers do not share or see each other's data



Benefits of SaaS multi-tenant architecture Shared

- infrastructure leads to lower costs

- Ongoing maintenance and updates: Customers don't need to pay costly maintenance fees to keep their software up to date. Vendors roll out new features and updates. These are often included with a SaaS subscription.
- Elasticity — the ability to scale up and down based on current requirements demanded by the business — is a benefit of multi-tenancy.
- Multitenancy feature distributes cost across a large pool of user via subscription

OPEN SAAS SOLUTION

Open SaaS refers to software as a service (SaaS) based on open source code. Similar to SaaS applications, Open SaaS is a web-based application that is hosted, supported and maintained by a service provider accessed via internet. Open SaaS is best for community software. Open SaaS uses those SaaS applications, which are developed using open source programming language. These SaaS applications can run on any open source operating system and database.

Open SaaS has several benefits listed below:

- No License Required
- Low Deployment Cost
- Less Vendor Lock-in
- More portable applications
- More Robust Solution

SaaS does not provide users with a copy of the executable file; they can't touch it or see it. Therefore, it's pretty impossible for users to see what it does or change it. Open source code, on the other hand, is openly available to anyone and is a huge accelerator of growth, since it encourages collaborative effort from thousands of developers.

Open SaaS basically provides customers with the best of both worlds:

1. It's open source so you can build your own platforms with it and customize all you want.
2. It's SaaS, which means you can choose not to manage the platform at all, you can simply pay your software provider.

Service-oriented architecture (SOA) is an application development methodology with which developers create solutions by integrating one or more web services. Think of a web service as a function or subroutine a program can call to accomplish a specific task. As shown in FIGURE 2-10, when a program running on computer calls a web service, a message, possibly containing

parameter values, is sent across the network (or Internet) to the computer housing the web service. That computer, in turn, performs its processing and normally returns a result to the caller.

Some developers refer to web services as remote-procedure calls. Further, developers refer to a set of web services as an application program interface (API). Amazon and eBay, for example, provide APIs that programmers can use to purchase products from across the web using the programs they create.

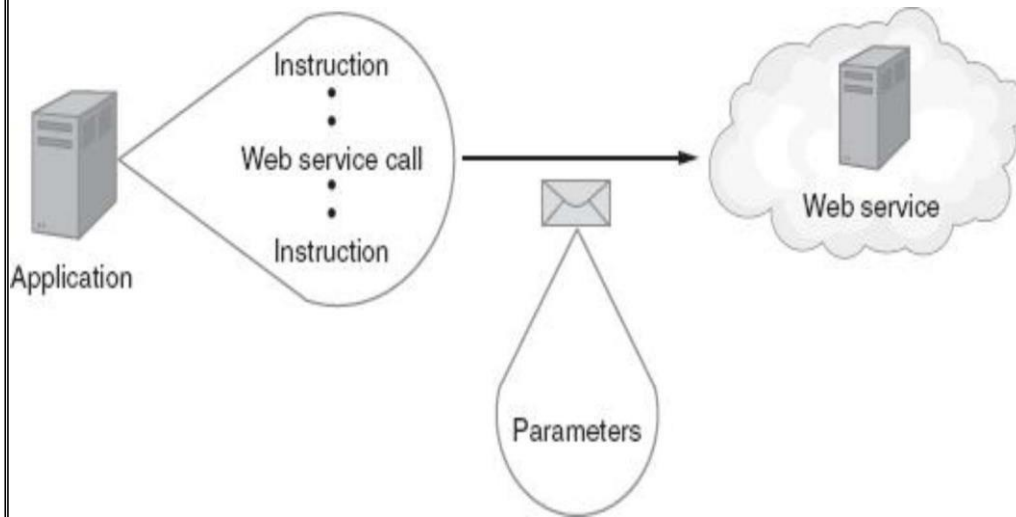


FIGURE 2-10 To call a web service, a program typically sends a message to the web service that resides on a remote computer and then waits for the web service to return a result.

SOA architecture is viewed as five horizontal layers. These are described below:

- Consumer Interface Layer: These are GUI based apps for end users accessing the applications.
- Business Process Layer: These are business-use cases in terms of application.
- Services Layer: These are whole-enterprise, in service inventory.
- Service Component Layer: are used to build the services, such as functional and technical libraries.
- Operational Systems Layer: It contains the data model.

Question Bank**Multiple Choice Questions****(Questions for Remembering)**

1. Which of the following is the second generation of WWW?
A. Cloud computing
B. Grid computing
C. Web 2.0
D. AWS
2. What is example for IaaS?
A. Email
B. Google App Engine
C. Salesforce.com
D. Amazon Web Services
3. Which cloud deployment model crosses isolation and overcomes boundaries?
A. Private
B. Public
C. Hybrid
D. Community
4. What is another name for “Internal cloud”?
A. Private
B. Public
C. Hybrid
D. Community
5. Which of the following deployment model is a issues as “Poor scalability “?
A. Private
B. Public
C. Hybrid
D. Community
6. Which of the following cloud has Complex networking issue?
A. Private
B. Public
C. Hybrid
D. Community
7. Which of the following cloud deployment model Organization having similar computing concern opt for?

- A. Private
 - B. Public
 - C. Hybrid
 - D. Community**
8. SOA stands for:
- A. Software Oriented Architecture
 - B. Service Oriented Architecture**
 - C. Software Originated Architecture
 - D. Service Originated Architecture
9. A Computer network where every computer shares resource of every other computer is?
- A. Cloud computing
 - B. Grid computing**
 - C. Utility computing
 - D. Virtualization
10. Which of the following can turn a computer network into powerful super computer.
- A. Cloud computing
 - B. Grid computing**
 - C. Utility computing
 - D. Virtualization
11. Which of the following is not a type of virtualization
- A. Hardware Virtualization
 - B. Server Virtualization
 - C. Storage Virtualization
 - D. Data virtualization**
12. Software as a Service (SaaS) based on open source code is?
- A. Web 2.0
 - B. SOA
 - C. Open SaaS**
 - D. AWS
13. Customer Relationship Management (CRM) is an example of
- A. IaaS application
 - B. SaaS Application**
 - C. PaaS Application
 - D. Utility Application
14. Which of the following offered Grid Engine software that allows engineers at companies to pool the computer cycles.

- A. Apple
- B. Microsoft
- C. Sales force
- D. Sun Microsystems**

15. Which layer of SOA is used to build the functional and technical libraries?

- A. Consumer Interface Layer
- B. Service Layer
- C. Service Component Layer**
- D. Operational Systems Layer

(Questions for Understanding)

16. Cloud in cloud computing refers to _____

- A. Service model
- B. Internet**
- C. Service Provider
- D. Software

17. Organizations are implementing cloud computing for _____

- A. reducing capital expenditure**
- B. increasing business
- C. reducing data redundancy
- D. for user interactivity

18. _____ is services on demand software.

- A. SaaS**
- B. PaaS
- C. IaaS
- D. NaaS

19. Which is also known to be hardware as a service.

- A. SaaS
- B. PaaS
- C. IaaS**
- D. NaaS

20. Which application is well suited for scientific and complex mathematical processing.

- A. Cloud computing
- B. Grid computing**
- C. Utility computing
- D. Open SaaS

21. Single instance of software application services multiple customers, this is seen in?

- A. Grid computing
 - B. Multitenancy architecture**
 - C. SOA
 - D. Open SaaS
22. The ability which allows sharing the physical instance of a single application or resource among multiple organizations or users is _____
- A. Cloud computing
 - B. Grid computing
 - C. Utility computing
 - D. Virtualization**
23. Which layer of SOA have GUI based apps for end users accessing the applications.
- A. Consumer Interface Layer**
 - B. Services Layer
 - C. Service Component Layer
 - D. Operational Systems Layer
24. Developers refer to a set of web services as?
- A. Web 2.0
 - B. Application package
 - C. SaaS solution
 - D. Application program interface (API)**
25. Less vendor lock-in and no licensing is a benefit of
- A. Multi tenant architecture
 - B. SOA
 - C. API
 - D. Open SaaS**

Questions carrying 4 marks

(Questions for Remembering)

1. Explain private cloud mention its advantage and disadvantages.
2. Explain public cloud mention its advantage and disadvantages.
3. Explain hybrid cloud mention its advantage and disadvantages.
4. Write the benefits of grid computing.
5. Write the characteristics of SaaS.
6. What are the benefits of Multitenant nature of SaaS.

(Questions for Understanding)

1. What do you mean by cloud computing? Write the uses of cloud computing?
2. Briefly explain about web 2.0 and its benefits.
3. Briefly explain all cloud deployment models.
4. Explain briefly the cloud service models.
5. Write a short note on open SaaS.
6. Briefly explain Service Oriented Architecture.

UNIT 2

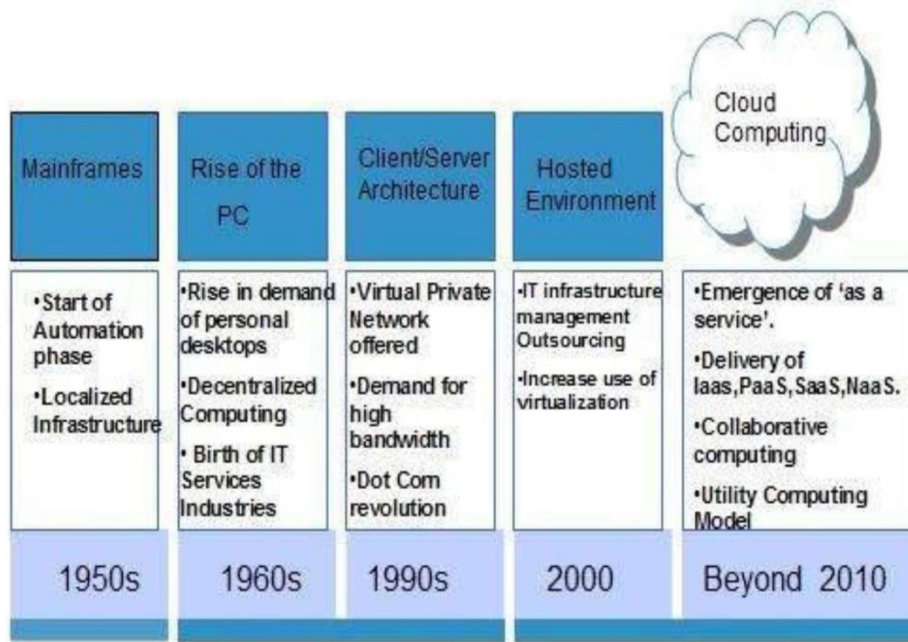
CHAPTER 3

PLATFORM AS A SERVICE AND INFRASTRUCTURE AS A SERVICE

2.1 Evolution of cloud computing

History of Cloud Computing surprisingly began almost 50 years ago. The father of this idea is considered to be John McCarthy, a professor at MIT University in US, who first in 1961 presented the idea of sharing the same computer technology as being the same as for example sharing electricity. Electrical power needs many households/firms that possess a variety of electrical appliances but do not possess power plant. Since that time, Cloud computing has evolved through a number of phases which include grid and utility computing, application service provision (ASP), and Software as a Service (SaaS). One of the first milestones was the arrival of Salesforce.com in 1999, which pioneered the concept of delivering enterprise applications via a simple website. The next development was Amazon Web Services in 2002, which provided a suite of cloud based services including storage, computation and even human intelligence. Another big milestone came in 2009 as Google and others started to offer browser based enterprise applications, though services such as Google Apps.

The concept of Cloud Computing came into existence in the year 1950 with implementation of mainframe computers, accessible via thin/static clients. Since then, cloud computing has been evolved from static clients to dynamic ones and from software to services. The following diagram explains the evolution of cloud computing:



3 Platform as a Service (PaaS)

Following on the heels of SaaS, Platform as a Service (PaaS) is another service model which provides access to computing platform over cloud computing solution. PaaS supplies all the resources required to build applications and services completely from the Internet, without having to download or install software.

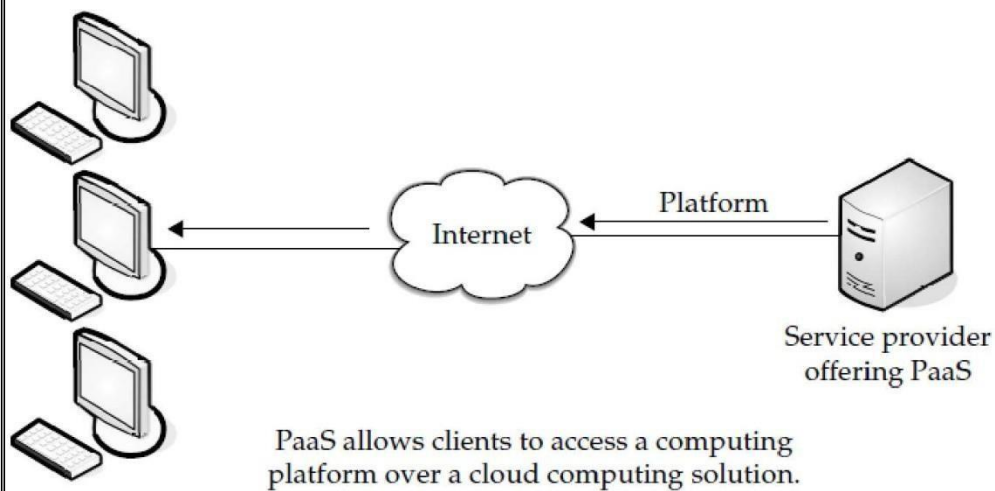


Figure 2.1: Platform as Service

Characteristics:

Here are the characteristics of PaaS service model:

- PaaS offers browser-based development environment. It allows the developer to create database and edit the application code either via Application Programming Interface or point-and-click tools.
- PaaS provides built-in security, scalability, and web service interfaces.
- PaaS provides built-in tools for defining workflow and approval processes and defining business rules.
- It is easy to integrate with other applications on the same platform.
- PaaS also provides web services interfaces that allow us to connect the applications outside the platform.

Benefits

Following are the benefits of PaaS model:

Lower administrative overhead

Customer need not bother about the administration because it is the responsibility of cloud provider.

Lower total cost of ownership

Customer need not purchase expensive hardware, servers, power, and data storage.

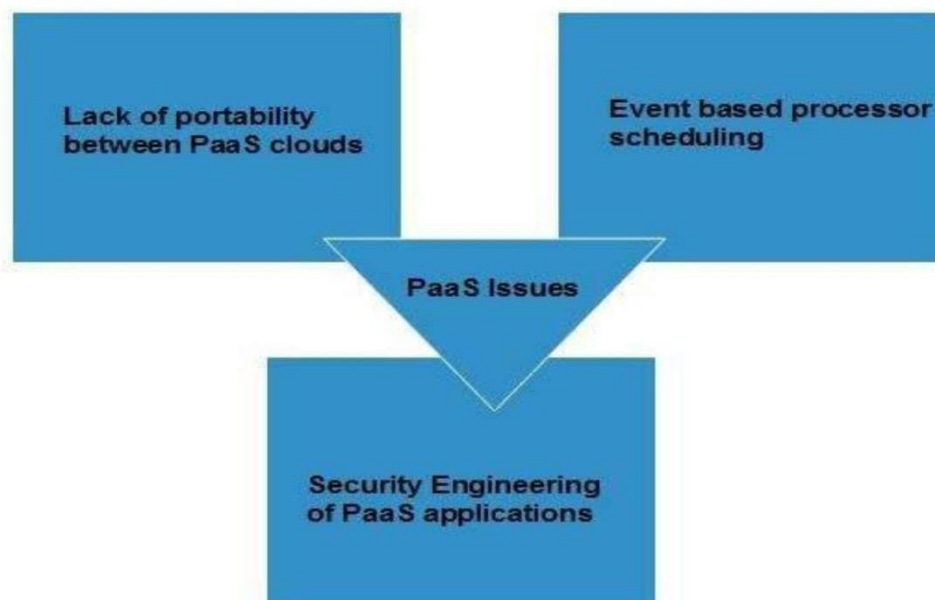
Scalable solutions

It is very easy to scale the resources up or down automatically, based on their demand.

More current system software

It is the responsibility of the cloud provider to maintain software versions and patch installations.

Challenges



Like SaaS, PaaS also place significant burdens on consumer's browsers to maintain reliable and secure connections to the provider systems. Therefore, PaaS shares many of the issues of SaaS. However, there are some specific issues associated with PaaS.

- Lack of portability between PaaS clouds

Although standard languages are used yet the implementations of platforms services may vary. For example, file, Queue, or hash table interfaces of one platform may differ from another, making it difficult to transfer workloads from one platform to another.

- Event based processor scheduling

The PaaS applications are event oriented which poses resource constraints on applications, i.e., they have to Answer a request in a given interval of time.

- Security engineering of PaaS applications

Since the PaaS applications are dependent

In this scenario, we're using the cloud provider's machines. Another term for this type of Infrastructure as a Service (IaaS)

computing is everything as a Service. Infrastructure as a service (IaaS) is a form of cloud computing that provides virtualized computing resources (such as virtual machines, storage drives, servers, operating systems & networks) on demand over the internet. Amazon Web Services (AWS) and Google Cloud Platform (GCP) are examples of independent IaaS providers

IaaS, as the name suggests, is a way of providing Cloud computing infrastructure such as virtual machines, storage drives, servers, operating systems & networks, which is also an on-demand service like that of SaaS. Rather than purchasing servers or developing software, clients buy those resources as a fully outsourced service based on their requirement. "Public cloud" is considered as an infrastructure that consists of shared resources, based on a self-service over the Internet. In one word, it is the only layer of the cloud where the customer gets the platform for their organization to outsource IT infrastructure on a pay-per-use basis.

IaaS provides users with:

- Load balancers
- Disk storage via virtual machines
- Software Packages
- IP address
- VLANs

Characteristics

Here are the characteristics of IaaS service model:

- Virtual machines with pre-installed software.
- Virtual machines with pre-installed Operating Systems such as Windows, Linux, and Solaris.
- On-demand availability of resources.

- Allows storing copies of particular data in different locations.
- The computing resources can be easily scaled up and down.

Advantages of IaaS are:

Dynamic: Users can dynamically opt & configure devices such as CPU, storage drive, etc.

- Easy Access: Users can easily access the vast cloud computing power.
- Renting: Flexible and efficient while renting IT infrastructure.

IaaS resources such as virtual machines, storages, bandwidth, ip addresses, monitoring services, firewalls, etc., all are made available to the consumers on rent. The consumer has to pay based the length of time a consumer, retains a resource. Also, with administrative access to virtual machines, the consumer can also run any software, even a custom operating system.

- Full control of computer resources through administrative access to VM's

IaaS allows the consumer to access computing resources through administrative access.

- Consumer issues administrative command to cloud provider to run the virtual machine or to save data on cloud's server.
- Consumer issues administrative command to virtual machines they owned to start web server or installing new applications.
- Portability, Interoperability with Legacy Applications
- It is possible to maintain legacy between applications and workloads between IaaS clouds. For example, network Applications such as web server, e-mail server that normally runs on consumer-owned server hardware can also be run from VMs in IaaS cloud.

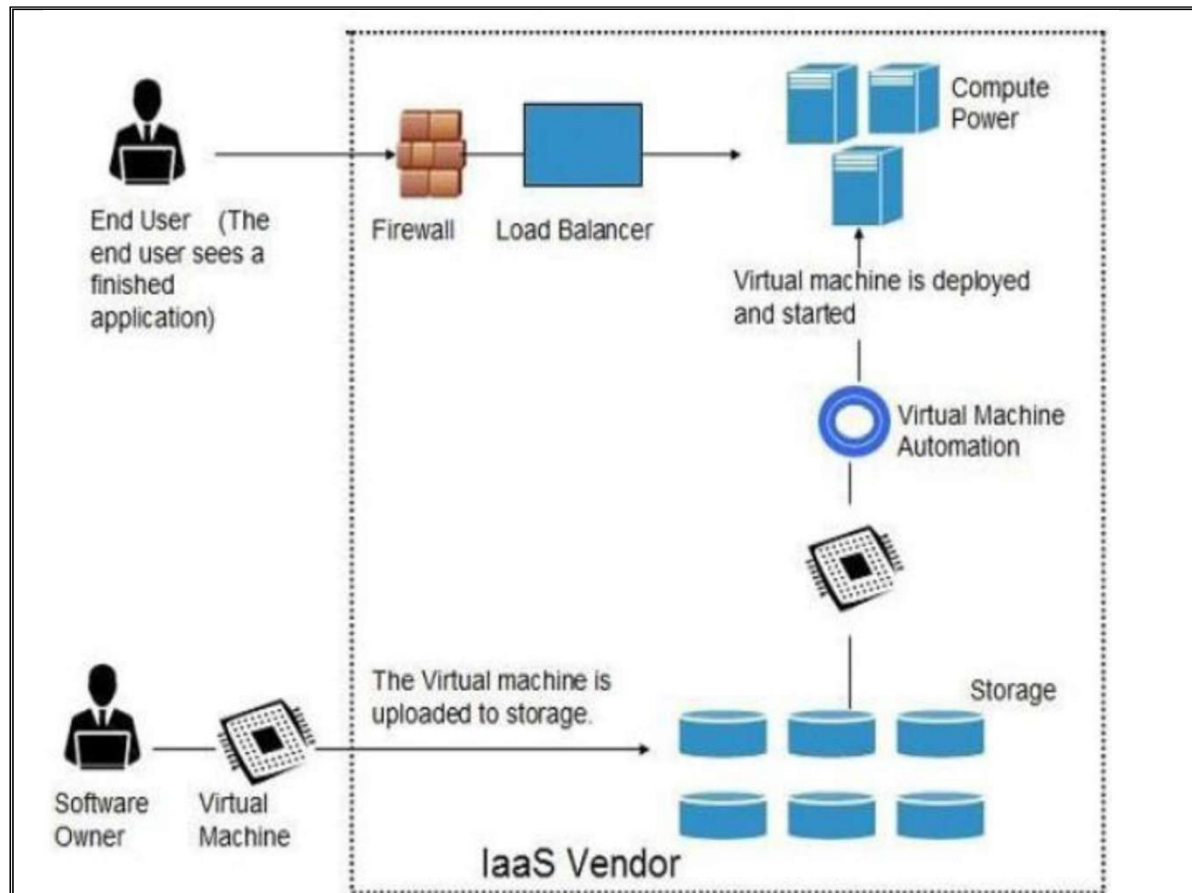


Figure 2.2: Infrastructure as a Service

Benefits

IaaS allows the cloud provider to freely locate the infrastructure over the Internet in a cost-effective manner. Some of the key benefits of IaaS are listed below:

- Full Control of the computing resources through Administrative Access to VMs. Flexible and efficient renting of Computer Hardware.
- Portability, Interoperability with Legacy Applications.

Full control over computing resources through administrative access to VMS

IaaS allows the consumer to access computing resources through administrative access to virtual machines in the following manner:

- Consumer issues administrative command to cloud provider to run the virtual machine or to save data on cloud's server.
- Consumer issues administrative command to virtual machines they owned to start web server or installing new applications.

Flexible and efficient renting of computer hardware

IaaS resources such as virtual machines, storages, bandwidth, IP addresses, monitoring services, firewalls, etc., all are made available to the consumers on rent. The consumer has to pay based

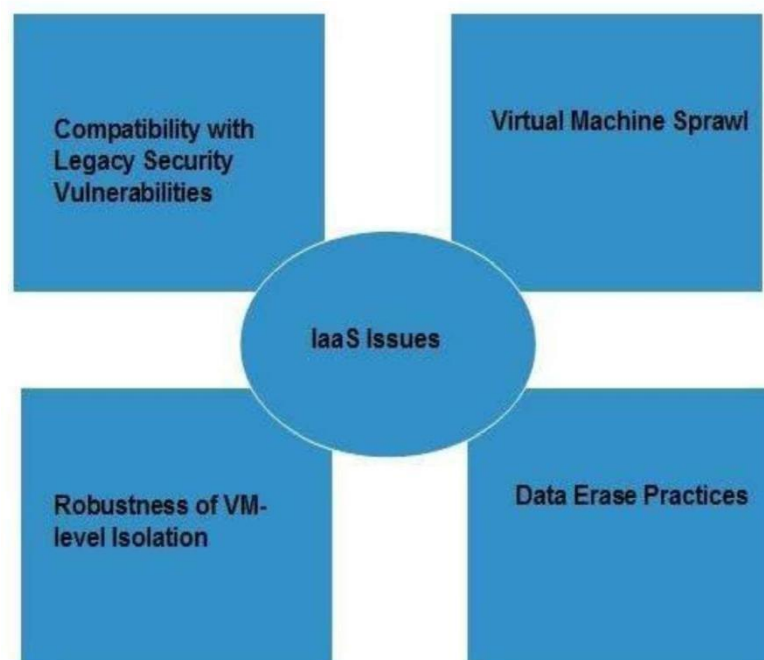
the length of time a consumer, retains a resource. Also, with administrative access to virtual machines, the consumer can also run any software, even a custom operating system.

Portability, interoperability with legacy applications

It is possible to maintain legacy between applications and workloads between IaaS clouds. For example, network Applications such as web server, e-mail server that normally runs on consumer-owned server hardware can also be run from VMs in IaaS cloud.

Challenges

IaaS shares issues with PaaS and SaaS, such as Network dependence and browser-based risks. It also has some specific issues associated with it.



Compatibility with legacy security vulnerabilities

Because IaaS offers the consumer to run legacy software in provider's infrastructure, therefore it exposes consumers to all of the security vulnerabilities of such legacy software.

Virtual machine sprawl

The VM can become out of date with respect to security updates because IaaS allows the consumer to operate the Virtual machines in running, suspended and off state. However, the provider can automatically update such VMs, but this mechanism is hard and complex.

Robustness of VM-level isolation

IaaS offers an isolated environment to individual consumers through hypervisor. Hypervisor is a software layer that includes hardware support for virtualization to split a physical computer into multiple virtual machines.

Data erase practices

The consumer uses virtual machine that in turn uses the common disk resources provided by the cloud provider. When the consumer releases the resource, the cloud provider must ensure that next consumer to rent the resource, does not observe data residue from previous consumer.

Improving performance through Load balancing

Rapid spikes in online traffic can challenge the best websites also. During online sales events, even a little interruption can cost lots of damage. Cloud manages online traffic by distributing workloads across multiple servers & resources. One characteristics of cloud technology is virtualized network access to a service - no matter where you access the service, it will direct you to the available resources.

Cloud load-balancing is the process of distributing workloads & computing resources within a cloud technology's environment. It also helps organizations & enterprises to manage workload demands by allocating resources among multiple systems or servers. Cloud load balancing also involves hosting the distribution of workload traffic that resides over the internet.

Application of Load Balancing

High performance level of tasks can be achieved using this load balancing technique on a lower cost than traditional on-premises load-balancing technology. In addition to workload and traffic distribution, cloud technology's load balancing can also provide health check for Cloud applications.

Load balancing is an optimization technique which can be used to enhance utilization & throughput, lower latency, reduce response time & avoid overloading of systems.

Without load balancing, it would have been very difficult to manage the cloud technology. It provides necessary redundancy procedures to make unreliable system reliable via managed redirection.

Load balancing system can use a different mechanism to assign direction to various services. In a simple case, load balancers are used to listen to network ports of the service request. When the request from a client arrives, load-balancers use the scheduling algorithm (round robin & weighted round-robin are mostly used) to assign where the request is sent.

Cloud storage system and storage redundancy

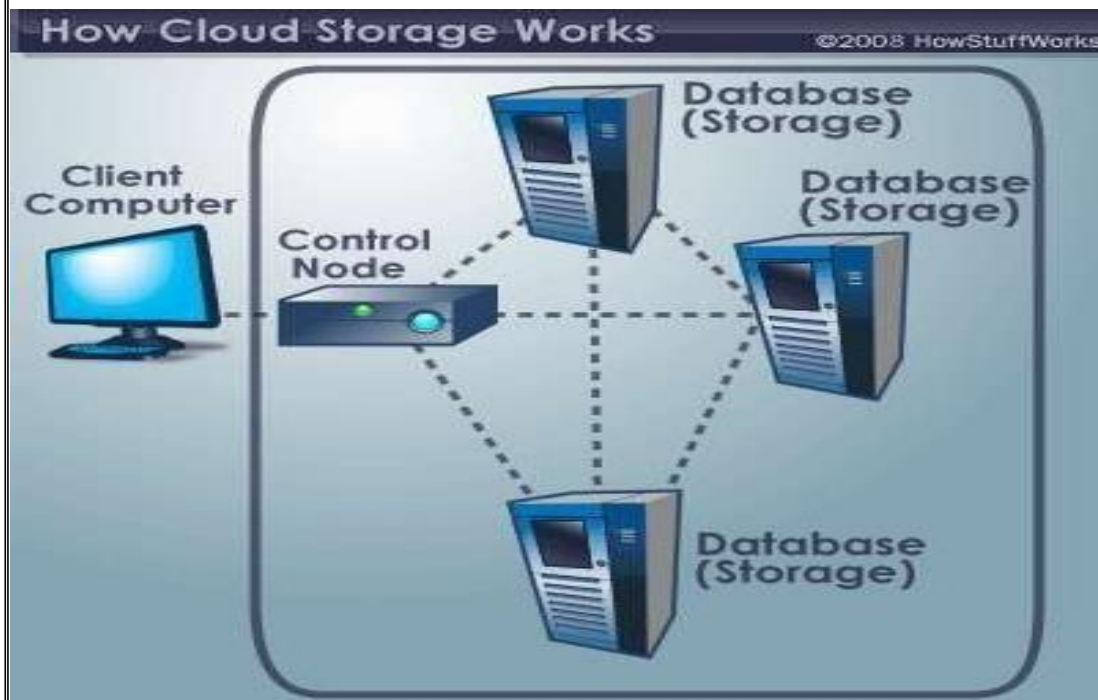
There are hundreds of different cloud storage systems. Some have a very specific focus, such as storing Web e-mail messages or digital pictures. Others are available to store all forms of

digital data. Some cloud storage systems are small operations, while others are so large that the physical equipment can fill up an entire warehouse. The facilities that house cloud storage systems are called data centers.

At its most basic level, a cloud storage system needs just one data server connected to the Internet. A client (e.g., a computer user subscribing to a cloud storage service) sends copies of files over the Internet to the data server, which then records the information. When the client wishes to retrieve the information, he or she accesses the data server through a Web-based interface. The server then either sends the files back to the client or allows the client to access and manipulate the files on the server itself.

Cloud storage systems generally rely on hundreds of data servers. Because computers occasionally require maintenance or repair, it's important to store the same information on multiple machines. This is called redundancy. Without redundancy, a cloud storage system couldn't ensure clients that they could access their information at any given time. Most systems store the same data on servers that use different power supplies. That way, clients can access their data even if one power supply fails.

Not all cloud storage clients are worried about running out of storage space. They use cloud storage as a way to create backups of data. If something happens to the client's computer system, the data survives off-site. With 1000's jobs to manage, network jam network interruptions can be expected. A WAN optimization services, reduces network latency, helps avoid blockage in network traffic and keeps the speed at optimum level.



A typical cloud storage system architecture includes a master control server and several storage servers.

Network Attached Storage

Whereas most data storage solutions are either local (i.e. within physical reach, like an external hard drive) or cloud-based (i.e. stored by another company, like Dropbox or Google Drive), a NAS device is like a mix of the two.

NAS devices are hard drives that are connected to a network.

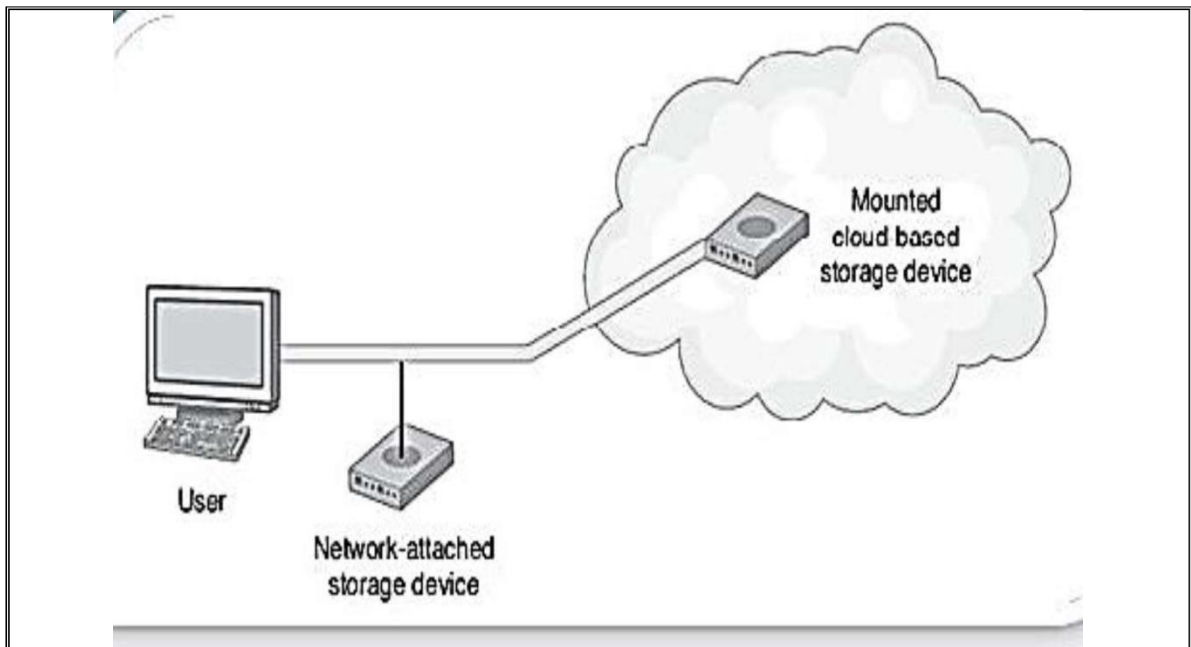
While adding an external hard disk to a PC to increase storage is not a bad option, it can only be connected to and accessed by one PC at a time. On the other hand, a NAS is connected to an entire network, meaning that it can be accessed by any number of PCs as long as they are also on the same network.



Unlike traditional external hard drives, NAS devices generally have some kind of built-in operating system which adds software functions like native media streaming, printer streaming, or remote access. But otherwise, they're functionally similar to any other kind of external hard drive you may have used before.

Cloud storage is incredibly convenient — that's why so many of us use it. But in the backs of our minds, I'm sure we all have worries about who can truly access our documents, especially for those of us storing confidential information like bank accounts or ID certificates.

Cloud based NAS device which is present in cloud-based storage as a mounted device, may be replicated in the cloud to meet a company's data redundancy needs



Most NAS devices have software options that allow you to configure remote access (i.e. access from beyond your local network) so that you can get to your documents and files wherever you are in the world. In other words, your own private cloud storage.

Cloud-based storage services like Dropbox effectively let you rent space on someone else's network-connected drives, enabling you to access your data from anywhere with an internet connection. A NAS gives you this same functionality without handing your data over to a host company, thus alleviating any concerns about privacy or cost.

IDaaS

Identity as a service (IDaaS) comprises cloud-based solutions for identity and access management (IAM) functions, such as single sign-on (SSO). These methods allow all users (customers, employees, and third parties) to more securely access sensitive information both on and off-premises.

Why IDAAS

Example: Employees in a company require to login to system to perform various tasks. These systems may be based on local server or cloud based. Following are the problems that an employee might face:

- Remembering different username and password combinations for accessing multiple servers.
- If an employee leaves the company, it is required to ensure that each account of that user is disabled. This increases workload on IT staff.

To solve above problems, a new technique emerged which is known as Identity-as a-Service (IDaaS).

IDaaS offers management of identity information as a digital entity. This identity can be used during electronic transactions.

Identity

Identity refers to set of attributes associated with something to make it recognizable. All objects may have same attributes, but their identities cannot be the same. A unique identity is assigned through unique identification attribute.

There are several identity services that are deployed to validate services such as validating web sites, transactions, transaction participants, client, etc.

Identity-as-a-Service may include the following:

- Directory services
- Federated services
- Registration
- Authentication services
- Risk and event monitoring
- Single sign-on services
- Identity and profile management

IDaaS also means collecting intelligence (i.e., logging events and reporting on which users accessed what information and when) to better understand, monitor, and improve their behaviors.

Multi-factor authentication (MFA), including biometrics, are core components of IDaaS.

Multi-factor authentication (MFA)

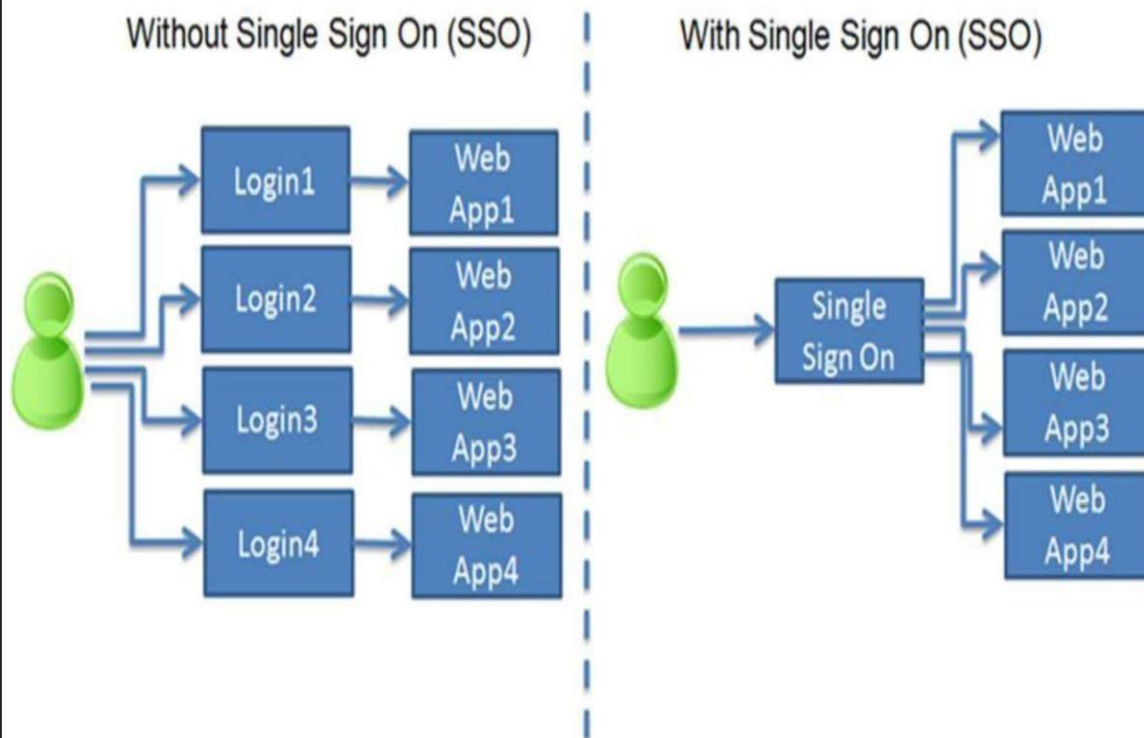
Multi-factor authentication (MFA) is an increasingly popular way of verifying a user's identity. MFA requires more than one piece of identifying information (i.e., a password).

Typically, two or more of the following criteria are used:

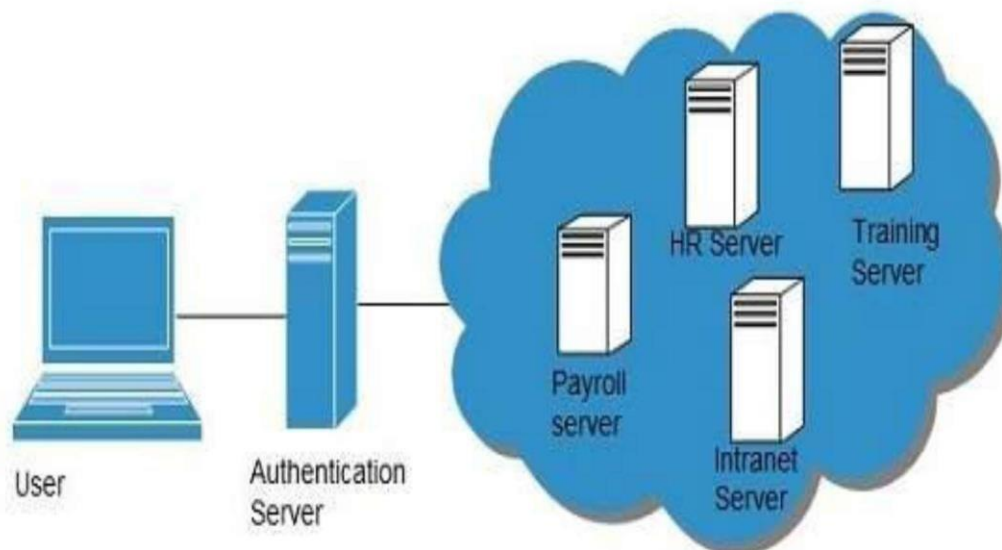
- Knowledge: Something the user knows (such as a password)
- Possession: Something the user has (such as a cell phone)
- Inheritance: Something the user is (such as a fingerprint or retina scan)

Single Sign-On (SSO)

To solve the problem of using different username and password combinations for different servers, companies now employ Single Sign-On software, which allows the user to login only one time and manage the access to other systems.

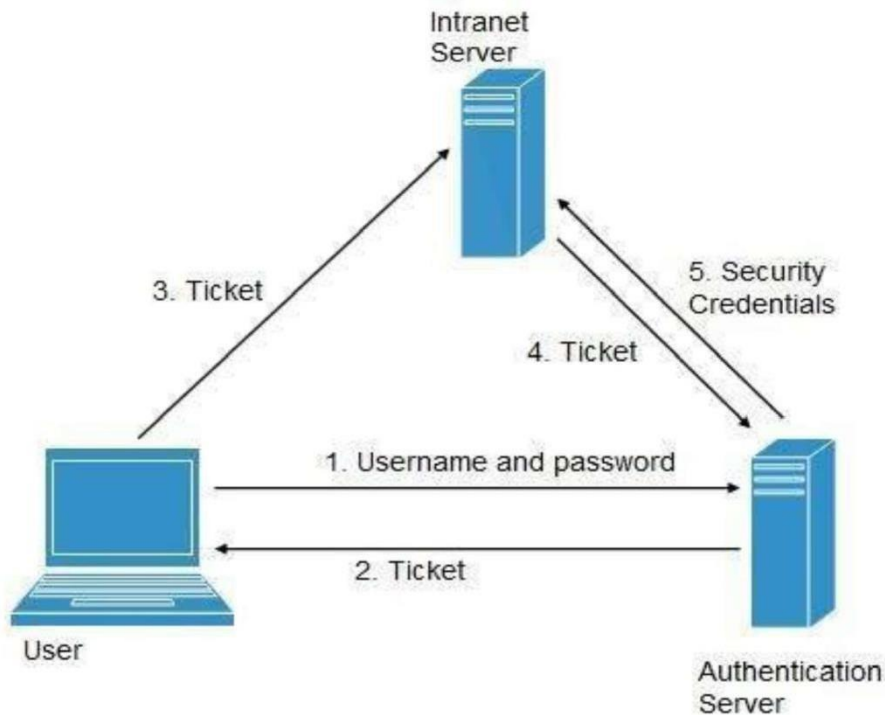


SSO has single authentication server, managing multiple accesses to other systems, as shown in the following diagram:



SSO Working

There are several implementations of SSO. Here, we discuss the common ones:



Following steps explain the working of Single Sign-On software:

- User logs into the authentication server using a username and password.
- The authentication server returns the user's ticket.
- User sends the ticket to intranet server.
- Intranet server sends the ticket to the authentication server.
- Authentication server sends the user's security credentials for that server back to the intranet server.

If an employee leaves the company, then disabling the user account at the authentication server prohibits the user's access to all the systems.

OpenID- identity in the cloud

OpenID is a unified user identification method released as an open standard that essentially acts as a single user identification system that can be used across multiple websites.

OpenID is a way to eliminate multiple user accounts across different websites, which often leads to confusion on the part of the user, especially when trying to remember all the different usernames and password combinations that have come and gone. OpenID allows users to log on to virtually any website that supports the standard with a single ID, eliminating the agony of the sign-up process and simplifying signing in to any affiliate website.

It offers users to login into multiple websites with single account. As of 2012, OpenID is supported by at least 27,000 sites, including Google, Yahoo, PayPal, Flickr, MySpace, WordPress.com etc.

The original authentication protocol for OpenID was developed by Brad Fitzpatrick in May of 2005. OpenID allows users to sign in to thousands of websites without having to create new usernames and passwords.

Benefits Of Open ID

- Increased site conversation rates
- Access to greater user profile content
- Fewer problems with lost passwords
- Ease of content integration into social networking site

Also, increases security as user information and passwords are stored centrally. This means if one account is hacked, the hacker won't be able to retrieve a password that can be used to hack the same user's account on other sites.

Mobile identity management

Identity management is essential to authentication, encryption and user activity policy. Mobile identity management is the natural evolution of what's historically been called AAA: a uthentication, authorization and accounting, three of the core components of any effective IT security strategy.

Recent years have seen a broad range of new identity and access management products and services introduced, driven in large part by the wireless LAN industry. This is perhaps to be expected; after all, Wi-Fi has become the primary or default network access for almost everyone today, and smartphones and tablets don't provide any mechanisms to connect to wired networks.

But identity management isn't just about wireless networking. The most important thing to remember when it comes to mobile identity management best practices is that the technology needs to be deployed across the entire network, no matter how a user connects.

Two-factor authentication: Because it's impossible to guarantee absolute security, each identity management element must be as reliable as possible. Mobile identity management systems that make use of the "something you have plus something you know" ethic of two-factor authentication have a major advantage. The second factor might use a specific device, biometrics, hardware token or other element.

With mobile identity and access management features, enterprises can control and manage users activity on company network and devices. MIM also helps businesses avoid the hassle of managing multiple usernames and passwords while improving end user experience.

Question Bank**Multiple Choice Questions****(Questions for understanding)**

1. _____ is a form of cloud computing that provides virtualized computing resources .
A. PaaS
B. SaaS
C. IaaS
D. IDaaS
2. _____ offers browser-based development environment.
A. PaaS
B. SaaS
C. IaaS
D. IDaaS
3. Which service model provides access to computing platform over cloud computing solution.
A. PaaS
B. SaaS
C. IaaS
D. IDaaS
4. Event based processor scheduling is an issue of _____
A. SaaS
B. PaaS
C. IaaS
D. IDaaS
5. _____ offers management of identity information as a digital entity
A. IDaaS
B. SaaS
C. PaaS
D. IaaS
6. The facilities that house cloud storage systems are called _____
A. Load balancing
B. Virtualization
C. Data centers
D. NAS
7. Expand SSO
A. Single sign-On

- B. Server sign-On
 - C. Single sign-Of
 - D. Single server-On
8. Single sign-on is a service of ____
- A. SaaS
 - B. IaaS
 - C. PaaS
 - D. IDaaS**
- 9.comprises cloud-based solutions for identity and access management (IAM) functions.
- A. SaaS
 - B. IaaS
 - C. PaaS
 - D. IDaaS**
10. Cloud provider maintaining software versions and patch installation is a benefit of ____ solution.
- A. Nas
 - B. PaaS**
 - C. IaaS
 - D. IDaaS
11. Knowledge ,Possession and inheritance is a criteria used in
- A. MIM(Mobile Id Management)**
 - B. MFA(Multi-factor Authentication)
 - C. SSO(Single Sign On)
 - D. Open Id
12. _____refers to set of attributes associated with something to make things recognizable.
- A. Object
 - B. Instance
 - C. Identity**
 - D. Class
13. AAA in Mobile Id Management is _____
- A. Access, Authorization and Accounting
 - B. Authentication, Available and Access
 - C. Authentication, Authorization and Available
 - D. Authentication, Authorization and Accounting**

14. Expand MFA
 - A. Multi-factor authorization
 - B. Multi-factor authentication**
 - C. Multi-factor accounting
 - D. Multi -factor application

15. Dropbox is an example for _____
 - A. IAAS solution
 - B. Cloud storage solution**
 - C. IDaaS solution
 - D. SAAS solution

16. MFA(Multi factor authentication) and biometric are the core components of
 - A. SSO
 - B. open ID
 - C. IDaaS**
 - D. cloud based NAS

17. An open standard that acts as a single user identification system which can be used across multiple websites.
 - A. Multi-Factor Authentication
 - B. Open ID**
 - C. Single Sign On
 - D. Mobile ID Management MIM

18. Two factor authentication is used in
 - A. Multi Factor Authentication
 - B. Open ID
 - C. Single Sign On
 - D. Mobile ID Management MIM**

19. Amazon Web Services (AWS) and Google Cloud Platform (GCP) are examples of
 - A. IDaaS providers
 - B. SaaS providers
 - C. PaaS providers
 - D. IaaS providers**

20. Which of the following is the benefit of Infrastructure as a service
 - A. Reduced hardware cost**
 - B. Event based processor scheduling
 - C. Virtual machine sprawl
 - D. browser-based development environment

21. The process of distributing workloads & computing resources within a cloud technology's environment.
A. Cloud load-balancing
B. Multi-factor authentication
C. Single Sign On
D. Network Attached Storage
22. _____ is an optimization technique which can be used to enhance utilization & throughput, lower latency, reduce response time & avoid overloading of systems.
A. Redundancy
B. Load balancing
C. Virtualization
D. Scalability
23. In which system, a user log in to a system one time and then move freely among related servers and applications without having to authenticate him- or herself each time.
A. MIM
B. SSO
C. MFA
D. PaaS
24. The scheduling algorithm ,round robin & weighted round-robin are mostly used to assign resource requested, by a_____
A. load balancers
B. virtual machine
C. NAS
D. SSO service
25. Which Algorithm is used by load balancer to assign requested resource
A. Shortest job first
B. Priority based
C. Round robin
D. First come first serve

Questions carrying 4 marks

(Questions for understanding)

1. What do you mean by PaaS? Write the characteristics of PaaS.
2. What do you mean by IaaS? Write the characteristics of IaaS.
3. Write about disadvantage of PaaS.
4. Brief on cloud based NAS device.

5. Write a short note on mobile ID management.
6. Write a short note on cloud storage system.

(Questions for Application)

1. What are the advantages of IaaS. Explain?
2. What do you mean load balancing? Write its application.
3. What do you mean by SSO? Explain the working of SSO with a neat diagram.
4. What do you mean open ID? Mention the benefits of Open ID.
5. Explain benefits of Platform as a service.
6. Summarize how storage redundancy is beneficial in cloud computing.

UNIT 3

CHAPTER 4

DATA STORAGE

Computer Network Storage

Network storage is a term used to describe a storage device (usually many devices grouped together) that is available to a network.

This kind of storage maintains copies of data across high-speed local area network (LAN) connections and is designed to back up files, databases, and other data to a central location that can be easily accessed via standard network protocols and tools.

Why Network Storage Is Important

Storage is an essential aspect of any computer. Hard drives and USB keys, for example, are designed to hold personal data in a place that's close to where they need to access the information, like directly inside or next to their computer.

However, when these types of *local storage* fail, and especially when they aren't backed up online, the data is lost. Additionally, the process of sharing local data with other computers can be time-consuming, and sometimes the amount of local storage available is insufficient to store everything desired.

Network storage addresses these problems by providing a reliable, external data repository for all computers on the LAN to share efficiently. Freeing up local storage space, network storage systems also typically support automated backup programs to prevent critical data loss.

For example, a network with 250 computers spanning a large building with multiple floors would benefit from network storage. With network access and proper permissions, users could access folders on the network storage device without worrying that those files are affecting their local storage capacity.

Without a network storage solution, a file that needs accessing by multiple users who are not physically close would have to be emailed, moved manually with something like a flash drive, or uploaded online only to be downloaded again on the destination side. All of those alternative solutions pose time, storage, and privacy concerns that are alleviated with central storage.

In basic terms, network storage is simply about storing data using a method by which it can be made available to clients on the network. Over the years, the storage of data has evolved through various phases. This evolution has been driven partly by the changing ways in which we use technology, and in part by the exponential increase in the volume of data we need to store. It has

also been driven by new technologies, which allow us to store and manage data in a more effective manner.

In the days of mainframes, data was stored physically separate from the actual processing unit, but was still only accessible through the processing units. As PC based servers became more commonplace, storage devices went 'inside the box' or in external boxes that were connected directly to the system. Each of these approaches was valid in its time, but as our need to store increasing volumes of data and our need to make it more accessible grew, other alternatives were needed. Enter network storage.

Network storage is a generic term used to describe network based data storage, but there are many technologies within it which all go to make the magic happen. Here is a rundown of some of the basic terminology that you might happen across when reading about network storage.

Direct Attached Storage (DAS)

Direct attached storage is the term used to describe a storage device that is directly attached to a host system. The simplest example of DAS is the internal hard drive of a server computer. DAS is still, by far, the most common method of storing data for computer systems.

SAN and NAS Network Storage

Two standard types of network storage are called Storage Area Network (SAN) and Network Attached Storage (NAS).

Storage Area Network (SAN)

A SAN is a network of storage devices that are connected to each other and to a server, or cluster of servers, which act as an access point to the SAN. In some configurations a SAN is also connected to the network. SAN's use special switches as a mechanism to connect the devices. These switches, which look a lot like a normal Ethernet networking switch, act as the connectivity point for SAN's. Making it possible for devices to communicate with each other on a separate network brings with it many advantages. Consider, for instance, the ability to back up every piece of data on your network without having to 'pollute' the standard network infrastructure with gigabytes of data. This is just one of the advantages of a SAN which is making it a popular choice

SAN is typically used on business networks and utilizes high-end servers, high-capacity disk arrays, and Fibre Channel interconnection technology.

Network Attached Storage (NAS)

Network Attached Storage, or NAS, is a data storage mechanism that uses special devices connected directly to the network media. These devices are assigned an IP address and can then be accessed by clients via a server that acts as a gateway to the data, or in some cases allows the

device to be accessed directly by the clients without an intermediary. Home networks usually use NAS, which involves installing hardware called NAS devices onto the LAN via TCP/IP.

The beauty of the NAS structure is that it means that in an environment with many servers running different operating systems, storage of data can be centralized, as can the security, management, and backup of the data. An increasing number of companies already make use of NAS technology, if only with devices such as CD-ROM towers (stand-alone boxes that contain multiple CD-ROM drives) that are connected directly to the network.

Some of the big advantages of NAS include the expandability; need more storage space, add another NAS device and expand the available storage. NAS also bring an extra level of fault tolerance to the network. In a DAS environment, a server going down means that the data that that server holds is no longer available. With NAS, the data is still available on the network and accessible by clients.

High speed, reliable method for data storage involves technologies like SCSI(Small Computer System Interface)and RAID (Redundant Array of Independent Disks).In addition to these mainstays of storage technology, other technologies feature in our network storage picture. One of the most significant of these technologies is Fibre channel .

Fibre Channel is a technology used to interconnect storage devices allowing them to communicate at very high speeds being faster than more traditional storage technologies like SCSI, Fibre Channel also allows for devices to be connected over a much greater distance. In fact, Fibre Channel can be used up to six miles. This allows devices in a SAN to be placed in the most appropriate physical location.

Network Storage Pros and Cons

PROS:

- Provides an easy way for networked devices to access the same files
- It's easier to back up data for many users when they store their files in the same area
- Network storage saves local storage devices from reaching their capacity
- Less local storage needs to be purchased for users if they store most of it on the network

CONS

- Devices not connected to the network cannot access the data (e.g. a down network prevents users from reaching their files)
- Serving shared files to users requires proper permissions to be set up, which can be time-consuming when lots of new users are configured at once
- More network storage requires more *backup storage* space

Storage Devices

Storage devices can be broadly classified into two categories:

- Block Storage Devices

- File Storage Devices

Block Storage Devices

The block storage devices offer raw storage to the clients. These raw storage are partitioned to create volumes.

File Storage Devices

The file Storage Devices offer storage to clients in the form of files, maintaining its own file system. This storage is in the form of Network Attached Storage (NAS).

Cloud based data storage

Cloud Storage is a service that allows to save data on offsite storage system managed by third-party and is made accessible by a web services API.

The cloud storage system stores multiple copies of data on multiple servers, at multiple locations. If one system fails, then it is required only to change the pointer to the location, where the object is stored.

Cloud Storage Classes

Cloud storage can be broadly classified into two categories:

- Unmanaged Cloud Storage
- Managed Cloud Storage

Unmanaged Cloud Storage

Unmanaged cloud storage means the storage is preconfigured for the customer. The customer can neither format, nor install his own file system or change drive properties.

Managed Cloud Storage

Managed cloud storage offers online storage space on-demand. The managed cloud storage system appears to the user to be a raw disk that the user can partition and format.

Advantages of cloud based data storage

Cloud storage, or storage that occurs online so users can access data remotely, is now very common. Already, it's the preferred method of storing digital data, especially for small to medium sized organizations. Storing remotely versus locally offers huge advantages to both consumers and businesses.

1. Accessibility and Synchronization

With local file storage, you only receive access to your data from a certain location. With cloud storage, on the other hand, the devices we now use every day, all day, become access points.

This includes tablets, smartphones, desktops, laptops and more. With these devices, you're able to access your data in any location with Internet access. This removes the stress of transferring files back and forth between devices, which is both annoying and sometimes complicated.

Additionally, files remain the same across all devices, as they are automatically updated when you make changes. You will always have the latest version of a file, no matter how you're retrieving it.

2. Collaboration Tools

Just as it's annoying to transfer files back and forth between devices, it's a pain to send dozens of emails just to share files. With cloud storage, there's a better solution. Whether you want to share a single file, or a folder full of documents, cloud storage makes it pretty simple, eliminating time wasted.

Additionally, many storage services are more than just places to put your data. They also allow you to collaborate with peers, making editing capabilities available to authorized users. And again, the changes made are saved automatically and shared with all collaborators.

3. Cost Efficiency

Cloud storage eliminates the costs of using external hard drives or backup tapes, and the cost of time it takes to complete routing backups manually. Organizations don't need to spend extra money on additional servers that are expensive to maintain or hire extra IT staff to do the work. Online storage solutions offer plenty of space for your data in the cloud, often for pretty low, affordable prices.

4. Space Efficiency

And the efficiency doesn't end with costs. When you move your data into the cloud, it suddenly doesn't have a physical presence on your devices and hardware. It will no longer take up valuable space at home or in your office because the provider is the one offering the space to store it virtually.

5. Security and Recovery

Of course it can be frightening to store sensitive information in a new place, but the cloud has proven to be more secure than local business storage in most cases. With good storage solutions, data is encrypted in flight and at rest, unauthorized users are unable to access the files, and the cloud provider offers enterprise-grade security measures that businesses would otherwise not have.

Additionally, if disaster strikes, you won't face catastrophic data loss. Instead, you'll be able to retrieve backups of your original files and restore them with minimal downtime.

Challenges

Storing the data in cloud is not that simple task. Apart from its flexibility and convenience, it also has several challenges faced by the customers.

Accessibility: If you have no internet connection, you have no access to your data.

-The network connection has to be very reliable - if the connection is slow or unreliable you might not be able to access your files

-If the data storage company goes out of business then all of the data stored on their servers may be lost (or not available to access)

Bandwidth & Cost issue: Several cloud storage services have a specific bandwidth allowance. If an organization surpasses the given allowance, the additional charges could be significant. However, some providers allow unlimited bandwidth.

It usually costs money to store data in the cloud. Organisations might be charged for making large data backups or for retrieving large files from the cloud back to their network.

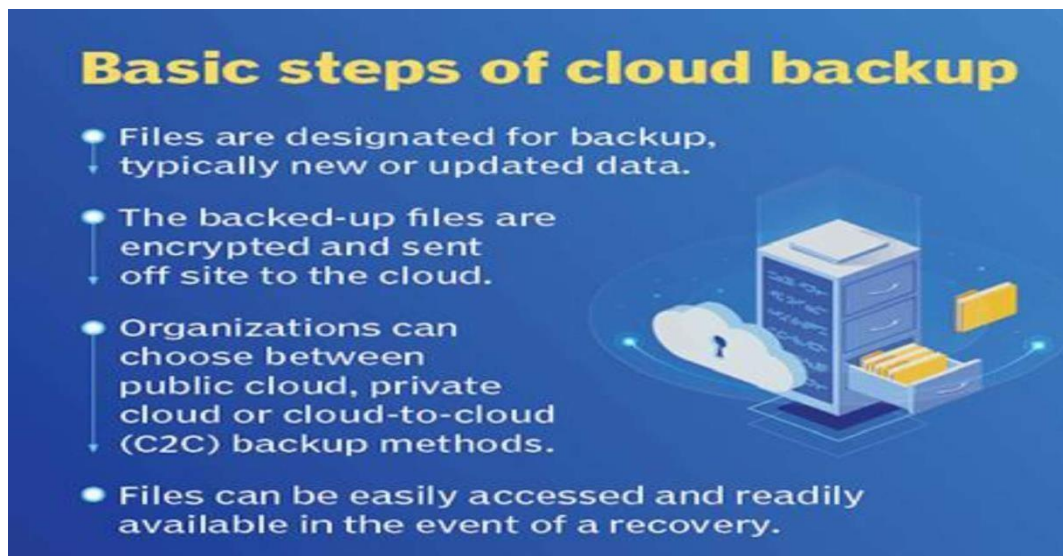
Data Security: There are concerns with the safety and privacy of important data stored remotely. You need to be certain that the company is reputable i.e. that they will perform regular backups of your data and that they have good security in place so that no one can access your data

- You will need to remember a login and user name in order to access your data remotely

Cloud based backup system

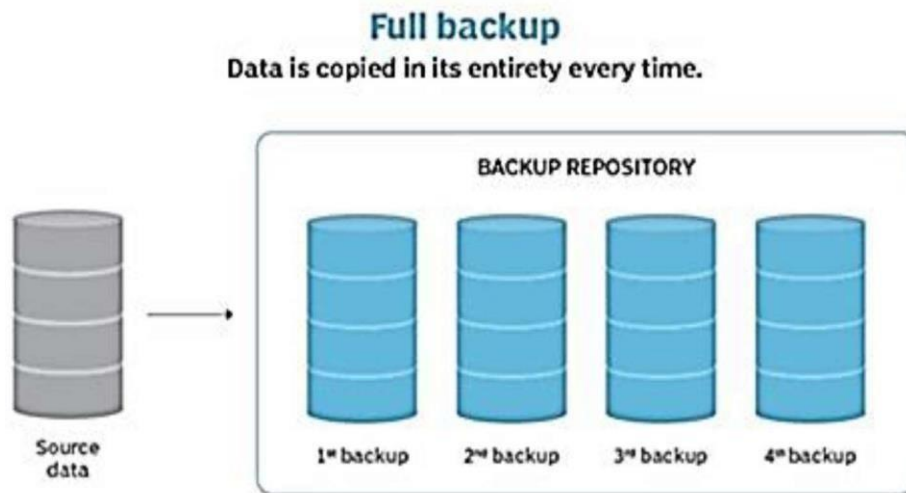
Cloud backup, also known as online backup or remote backup, is a strategy for sending a copy of a physical or virtual file or database to a secondary, off-site location for preservation in case of equipment failure or catastrophe. The secondary server and storage systems are usually hosted by a third-party service provider, who charges the backup customer a fee based on storage space or capacity used, data transmission bandwidth, number of users, number of servers or number of times data is accessed. Cloud Backup Storage Provides high encryption.

While there are multiple options and approaches to off-site backup, cloud backup serves as the off-site facility for many organizations. In an enterprise, the company might own the off-site server if it hosts its own cloud service, but the chargeback method would be similar if the company uses a service provider to manage the cloud backup environment.

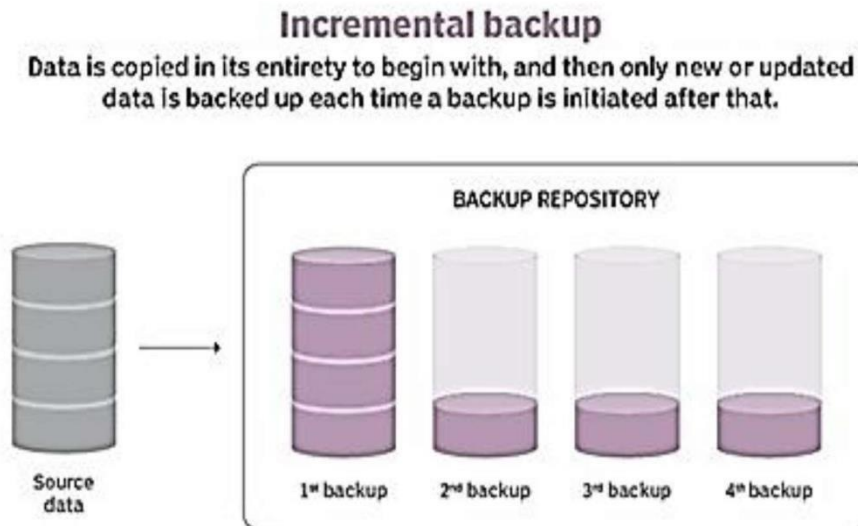


Types of backup:

1. Full backups copy the entire data set every time a backup is initiated. As a result, they provide the highest level of protection. However, most organizations cannot perform full backups frequently because they can be time-consuming and take up too much storage capacity.

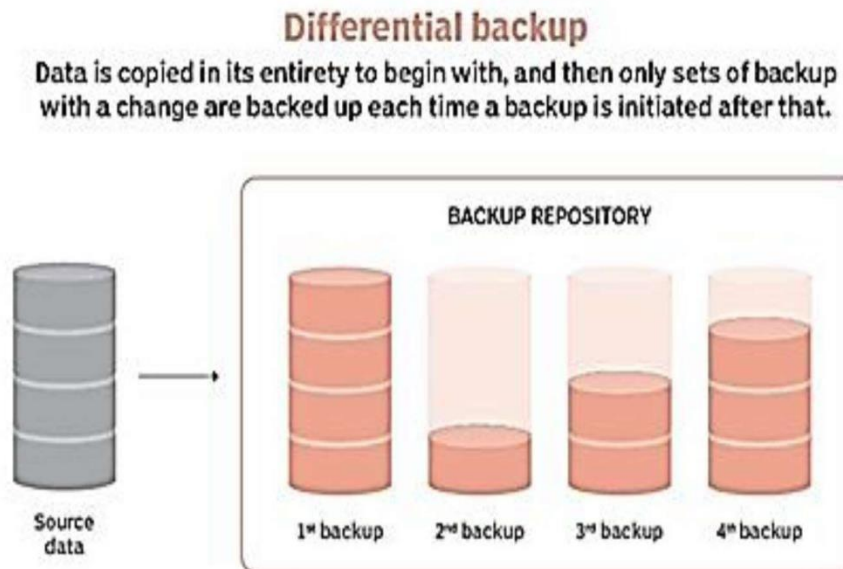


2. Incremental backups only back up the data that has been changed or updated since the last backup. This method saves time and storage space, but can make it more difficult to perform a complete restore. Incremental is a common form of cloud backup because it tends to use fewer resources.



3. Differential backups are similar to incremental backups because they only contain data that has been altered. However, differential backups back up data that has changed since the

last *full* backup, rather than the last backup in general. This method solves the problem of difficult restores that can arise with incremental backups.



Varieties of cloud backup approaches include:

- Backing up directly to the public cloud. One way to store organizational resources is by duplicating resources in the public cloud. This method entails writing data directly to cloud providers, such as AWS or Microsoft Azure. The organization uses its own backup software to create the data copy to send to the cloud storage service. The cloud storage service then provides the destination and safekeeping for the data, but it does not specifically provide a backup application. In this scenario, it is important that the backup software is capable of interfacing with the cloud's storage service. Additionally, with public cloud options, IT professionals may need to look into supplemental data protection procedures.
- Backing up to a service provider. In this scenario, an organization writes data to a cloud service provider that offers backup services in a managed data center. The backup software that the company uses to send its data to the service may be provided as part of the service, or the service may support specific commercially-available backup applications.
- Choosing a cloud-to-cloud (C2C) backup. These services are among the newest offerings in the cloud backup arena. They specialize in backing up data that already lives in the cloud, either as data created using a software as a service (SaaS) application or as data stored in a cloud backup service. As its name suggests, a cloud-to-cloud backup service copies data from one cloud to another cloud. The cloud-to-cloud backup service typically hosts the software that handles this process.
- Using online cloud backup systems. There are also hardware alternatives that facilitate backing up data to a cloud backup service. These appliances are all-in-one backup machines that include backup software and disk capacity along with the backup server. The appliances

are about as close to plug-and-play as backup gets, and most of them also provide a seamless (or nearly so) link to one or more cloud backup services or cloud providers. The list of vendors that offer backup appliances that include cloud interfaces is long, with Quantum, Unitrends, Arcserve, Rubrik, Cohesity, Dell EMC, StorageCraft and Asigra active in this arena. These appliances typically retain the most recent backup locally, in addition to shipping it to the cloud backup provider, so that any required recoveries can be made from the local backup copy, saving time and transmission costs.

When an organization engages a cloud backup service, the first step is to complete a full backup of the data that needs to be protected. This initial backup can sometimes take days to finish uploading over a network as a result of the large volume of data that is being transferred. In a 3-2-1 backup strategy, where an organization has three copies of data on two different media, at least one copy of the backed up data should be sent to an off-site backup facility so that it is accessible even if on-site systems are unavailable.

Using a technique called cloud seeding, a cloud backup vendor sends a storage device -- such as a disk drive or tape cartridge -- to its new customer, which then backs up the data locally onto the device and returns it to the provider. This process removes the need to send the initial data over the network to the backup provider. High Encryption is provided by Cloud Backup Storage.

If the amount of data in the initial backup is substantial, the cloud backup service may provide a full storage array for the seeding process. These arrays are typically small network-attached storage (NAS) devices that can be shipped back and forth relatively easily. After the initial seeding, only changed data is backed up over the network.

Hybrid cloud backup

Hybrid cloud backup providers connect traditional backups to the cloud. This strategy is useful for organizations that produce a large volume of data and need quick restore access. With one approach, a network-attached storage (NAS) appliance serves as a local backup target and syncs backup data to the cloud. When an organization needs a quick restore, the data is available in the on-site NAS. If an organization loses its primary site, the cloud backup is still available. This method can also be referred to as D2D2C backup. In another hybrid approach, an organization uses both the public and private cloud for backup.

It is difficult to get data consistency with hybrid cloud backup, especially if the data transfer takes a long time. Point-in-time snapshots and continuous backups help, but costs rise as backup frequency increases.

Comparatively, in a pure cloud backup scenario, backups go directly to the service provider's cloud.

Industry specific cloud based data storage

A growing number of industries appreciate the advantages of cloud computing, but also recognize the barriers that stand in the way of making a meaningful transition. For instance; finance, healthcare and government operations are areas where sensitive and confidential data must be carefully handled and protected at all times from unauthorized access and other malicious activity. As a result, a variety of national and international authorities have created regulations and compliance measures that govern not only how a number of industries handle critical data, but also how cloud service providers are expected to safely host said data.

Regulatory and compliance requirements are a major concern for organizations interested in cloud-based computing. For example, U.S. government agencies are required to meet a variety of regulatory and compliance guidelines when utilizing data and application processing and storage solutions. For example, the International Traffic in Arms Regulations (ITAR) places heavy restrictions on both data access and location, requiring data storage facilities to be located specifically within the U.S. and restriction of data access solely to U.S. citizens.

The healthcare industry has its own regulations to contend with when it comes to handling sensitive and confidential data. The Health Insurance Portability and Accountability (HIPAA) Act and Health Information Technology for Economic and Clinical Health (HITECH) Act place heavy restrictions on data access, processing and dissemination in the interest of patient confidentiality. Nevertheless, many healthcare organizations are aware of the benefits cloud computing offers, making it imperative for them to find innovative ways of overcoming these regulatory and compliance obstacles.

Maintenance of security protocol as well as compliance with federal and international regulations is a driving force behind the reluctance of many industries to fully utilizing the cloud. As a result, many of the emerging cloud providers are taking a hard look at ways to accommodate these concerns, as well as how to approach the issue of compliance and certification.

Amazon and Nirvana solution are top industry for industry specific cloud storage service. AWS offers "Amazon S3"(simple storage serviced),IBM offers "blue cloud ",Google offers "gdrive".

Cloud database

A cloud database is a collection of informational content, either structured or unstructured, that resides on a private, public or hybrid cloud computing infrastructure platform. From a structural and design perspective, a cloud database is no different than one that operates on a business's own on-premises servers. The critical difference lies in where the database resides.

Where an on-premises database is connected to local users through a corporation's internal local area network (LAN), a cloud database resides on servers and storage furnished by a cloud or database as a service (DBaaS) provider and it is accessed solely through the internet. To a software application, for example, a SQL database residing on-premises or in the cloud should appear identical.

The behavior of the database should be the same whether accessed through direct queries, such as SQL statements, or through API calls. However, it may be possible to discern small differences in response time. An on-premises database, accessed with a LAN, is likely to provide a slightly faster response than a cloud-based database, which requires a round trip on the internet for each interaction with the database.

How Cloud Databases Work

Cloud databases, like their traditional ancestors, can be divided into two broad categories: relational and nonrelational.

A relational database, typically written in structured query language (SQL), is composed of a set of interrelated tables that are organized into rows and columns. The relationship between tables and columns (fields) is specified in a schema. SQL databases, by design, rely on data that is highly consistent in its format, such as banking transactions or a telephone directory. Popular cloud platforms and cloud providers include MySQL, Oracle, IBM DB2 and Microsoft SQL Server. Some cloud platforms such as MySQL are open sourced.

Non-relational databases, sometimes called NoSQL, do not employ a table model. Instead, they store content, regardless of its structure, as a single document. This technology is well-suited for unstructured data, such as social media content, photos and videos.

Types of Cloud Databases

Two cloud database environment models exist: traditional and database as a service (DBaaS).

In a traditional cloud model, a database runs on an IT department's infrastructure with a virtual machine. Tasks of database oversight and management fall upon IT staffers of the organization.

The DBaaS model is a fee-based subscription service in which the database runs on the service provider's physical infrastructure. Different service levels are usually available. In a classic DBaaS arrangement, the provider maintains the physical infrastructure and database, leaving the customer to manage the database's contents and operation.

Alternatively, a customer can set up a managed hosting arrangement, in which the provider handles database maintenance and management. This latter option may be especially attractive to small businesses that have database needs but lack adequate IT expertise.

Cloud database benefits

Compared with operating a traditional database on an on-site physical server and storage architecture, a cloud database offers the following distinct advantages:

- **Elimination of physical infrastructure-** In a cloud database environment, the cloud computing provider of servers, storage and other infrastructure is responsible for maintenance and keeping high availability. The organization that owns and operates the database is only responsible for supporting and maintaining the database software and its contents. In a DBaaS environment, the service provider is responsible for managing and operating the database software, leaving the DBaaS users responsible only for their own data.
- **Cost savings-** Through the elimination of a physical infrastructure owned and operated by an IT department, significant savings can be achieved from reduced capital expenditures, less staff, decreased electrical and HVAC operating costs and a smaller amount of needed physical space.
- DBaaS benefits also include instantaneous scalability, performance guarantees, failover support, declining pricing and specialized expertise.

Cloud based Block Storage

Block Storage Systems

Block storage systems are used to host databases, support random read/write operations, and keep system files of the running virtual machines. Data is stored in volumes and blocks where files are split into evenly-sized blocks. Each block has its own address, but unlike objects they do not have metadata. When storing large amounts of data, files are split into smaller units of a fixed size, the “blocks,” which are distributed amongst the storage nodes. This also supports the volume IO performance.

Block storage is the most commonly used storage type for most applications. The most common examples of Block Storage are SAN, iSCSI, and local disks.

It can be either locally or network-attached and are typically formatted with a file system like FAT32, NTFS, EXT3, and EXT4.

Public cloud providers allow creating various file systems on their block storage systems to enable their users to store persistent data like a database. In addition, an Amazon EBS volume is accessed from an Amazon EC2 instance over an AWS dedicated or shared network. Another advantage of using block storage in the cloud is its backup mechanism. For example, AWS EBS offers a snapshot feature that is essentially an incremental point in time backup of your volume.

Block storage options in the cloud

1. **AWS Elastic Block Storage (EBS):** Amazon EBS provides raw storage –just like a hard disk which you can attach to your Elastic Cloud Compute (EC2) instances. Once attached, you create a file system and get immediate access to your storage. You can create EBS General Purpose (SSD) and Provisioned IOPS (SSD) volumes up to 16 TB in size, and slower, legacy magnetic volumes.
2. **Rackspace Cloud Block Storage:** Rackspace provides raw storage devices capable of delivering super fast 10GbE internal connections.

3. Azure Premium Storage: Premium Storage delivers high-performance, low-latency disk support for I/O intensive workloads running on Azure Virtual Machines. Volumes allow up to 32 TB of storage.
4. Google Persistent Disks: Compute Engine Persistent Disks provide network-attached block storage, much like a high speed and highly reliable SAN, for Compute Engine instances. You can remove a disk from one server and attach it to another server, or attach one volume to multiple nodes in read-only mode. Two types of block storage are available: Standard Persistent Disk and Solid-State Persistent Disks.

Use cases

Ideal for databases, since a DB requires consistent I/O performance and low-latency connectivity.

Use block storage for RAID Volumes, where you combine multiple disks organized through stripping or mirroring.

Any application which requires service side processing, like Java, PHP, and .Net will require block storage.

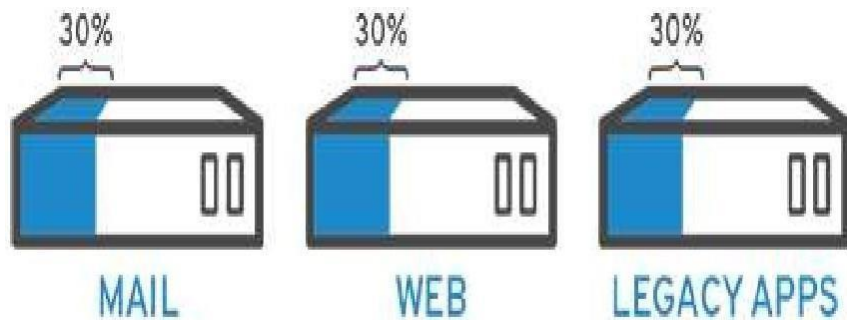
Running mission-critical applications like Oracle, SAP, Microsoft Exchange, and Microsoft SharePoint.

Chapter 5

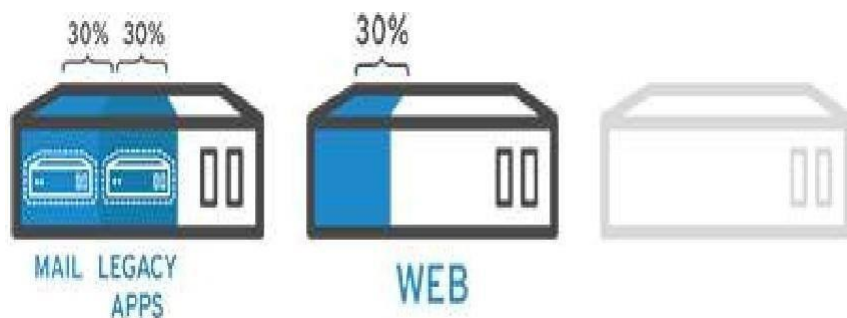
Virtualization

Virtualization is technology that lets you create useful IT services using resources that are traditionally bound to hardware. It allows you to use a physical machine's full capacity by distributing its capabilities among many users or environments.

In more practical terms, imagine you have 3 physical servers with individual dedicated purposes. One is a mail server, another is a web server, and the last one runs internal legacy applications. Each server is being used at about 30% capacity just a fraction of their running potential. But since the legacy apps remain important to your internal operations, you have to keep them and the third server that hosts them, right?



Traditionally, yes. It was often easier and more reliable to run individual tasks on individual servers: 1 server, 1 operating system, 1 task. It wasn't easy to give 1 server multiple brains. But with virtualization, you can split the mail server into 2 unique ones that can handle independent more efficiently.



Keeping security in mind, you could split the first server again so it could handle another task increasing its use from 30%, to 60%, to 90%. Once you do that, the now empty servers could be reused for other tasks or retired altogether to reduce cooling and maintenance costs.

Features of virtualization

1. Partitioning: Multiple virtual servers can run on a physical server at the same time
2. Encapsulation of data: All data on the virtual server including boot disks is encapsulated in a file format
3. Isolation: The Virtual server running on the physical server are safely separated & don't affect each other
4. Hardware Independence: When the virtual server runs, it can migrate to the different hardware platform

A brief history of virtualization

While virtualization technology can be sourced back to the 1960s, it wasn't widely adopted until the early 2000s. The technologies that enabled virtualization like hypervisors were developed decades ago to give multiple users simultaneous access to computers that performed batch processing. Batch processing was a popular computing style in the business sector that ran routine tasks thousands of times very quickly (like payroll).

But, over the next few decades, other solutions to the many users/single machine problem grew in popularity while virtualization didn't. One of those other solutions was time sharing, which isolated users within operating systems—~~inadvertently~~ leading to other operating systems like UNIX, which eventually gave way to Linux®. All the while, virtualization remained a largely unadopted, niche technology.

Fast forward to the the 1990s. Most enterprises had physical servers and single-vendor IT stacks, which didn't allow legacy apps to run on a different vendor's hardware. As companies updated their IT environments with less-expensive commodity servers, operating systems, and applications from a variety of vendors, they were bound to underused physical hardware each server could only run 1 vendor-specific task.

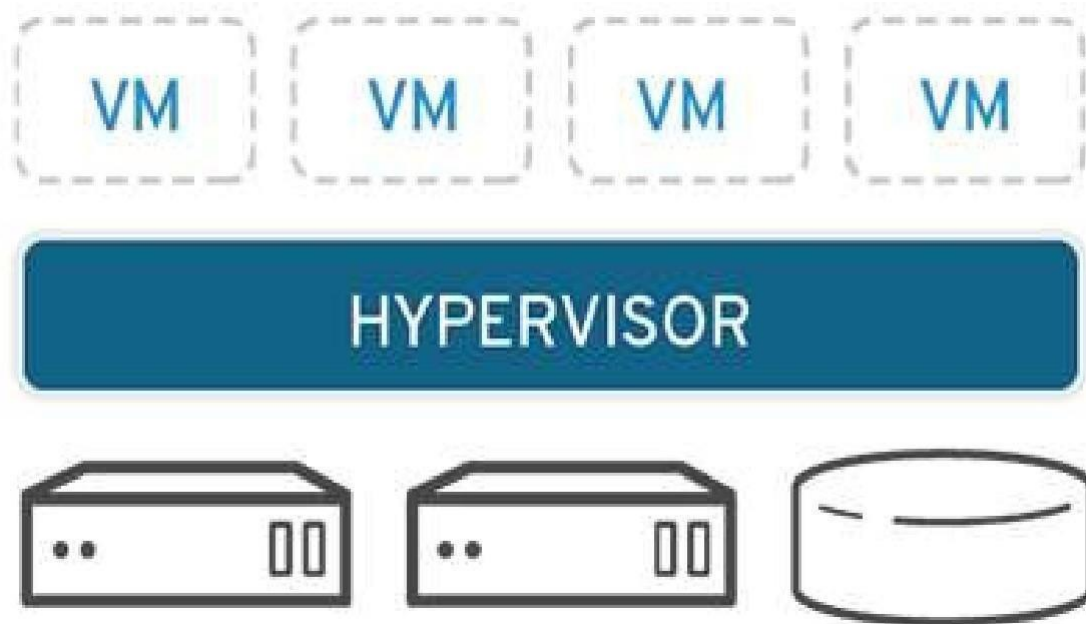
This is where virtualization really took off. It was the natural solution to 2 problems: companies could partition their servers *and* run legacy apps on multiple operating system types and versions. Servers started being used more efficiently (or not at all), thereby reducing the costs associated with purchase, set up, cooling, and maintenance.

Virtualization's widespread applicability helped reduce vendor ~~lock~~ and made it the foundation of cloud computing. It's so prevalent across enterprises today that specialized virtualization management software is often needed to help keep track of it all.

How does virtualization work?

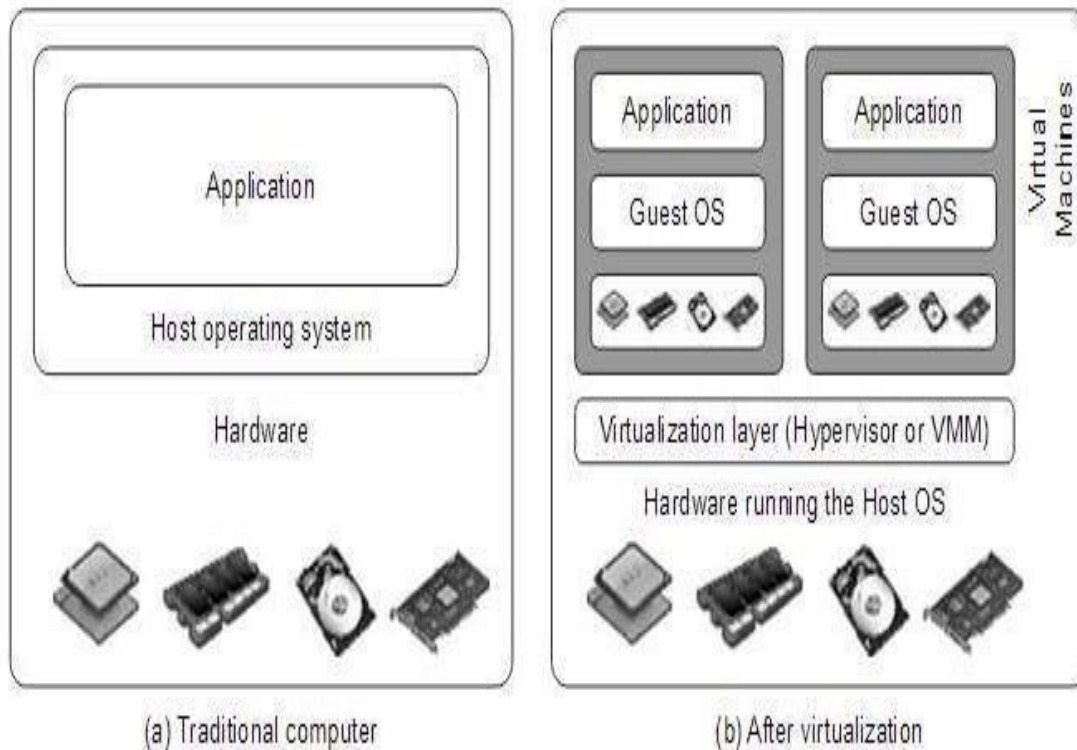
Software called hypervisors separate the physical resources from the virtual environments ~~the~~ things that need those resources. Hypervisors can sit on top of an operating system (like on a

laptop) or be installed directly onto hardware (like a server), which is how most enterprises virtualize. Hypervisors take your physical resources and divide them up so that virtual environments can use them.



Resources are partitioned as needed from the physical environment to the many virtual environments. Users interact with and run computations within the virtual environment (typically called a guest machine or virtual machine). The virtual machine functions as a single data file. And like any digital file, it can be moved from one computer to another, opened in either one, and be expected to work the same.

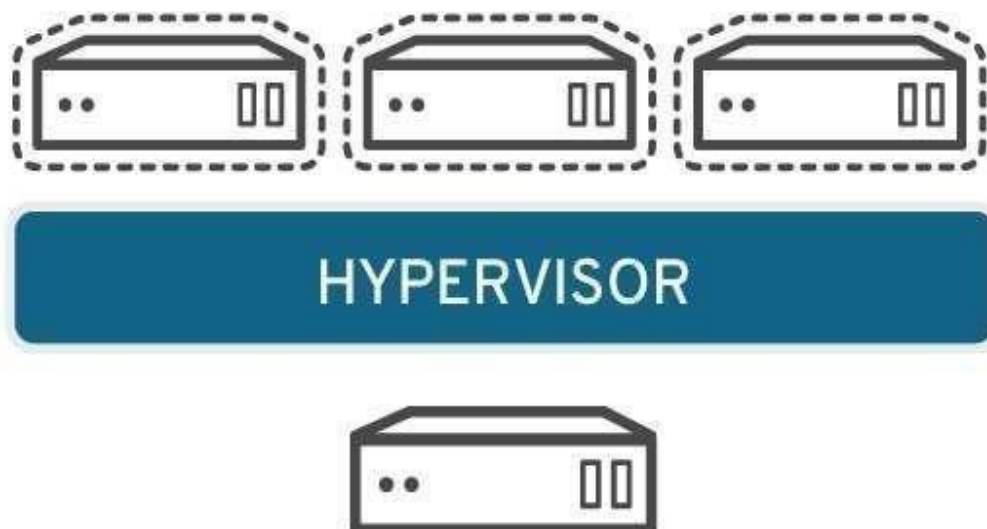
When the virtual environment is running and a user or program issues an instruction that requires additional resources from the physical environment, the hypervisor relays the request to the physical system and caches the changes—which all happens at close to native speed (particularly if the request is sent through an open source hypervisor based on KVM, the Kernel-based Virtual Machine).



The architecture of a computer system before and after virtualization

Server Virtualization

Servers are computers designed to process a high volume of specific tasks really well so other computers like laptops and desktops can do a variety of other tasks. Virtualizing a server lets it to do more of those specific functions and involves partitioning it so that the components can be used to serve multiple functions.



Server virtualization is the partitioning of physical server into several virtual servers and this division is mainly done to improvise the utility of server resource. In other word it is the masking of resources that are located in server which includes the number & identity of processors, physical servers & the operating system. This division of one physical server into multiple isolated virtual servers is done by server administrator using software. The virtual environment is sometimes called the virtual private-servers.

In this process, the server resources are kept hidden from the user. This partitioning of physical server into several virtual environments; result in the dedication of one server to perform a single application or task.

This technique is mainly used in web-servers which reduces the cost of web-hosting services. Instead of having separate system for each web-server, multiple virtual servers can run on the same system/computer.

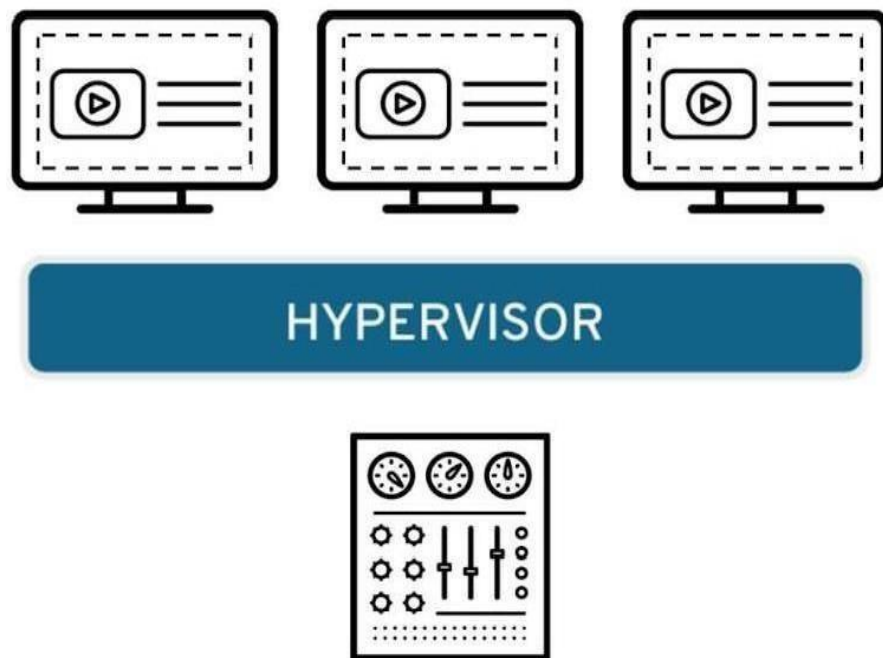
The primary uses of server virtualization are:

- To centralize the server administration
- Improve the availability of server
- Helps in disaster recovery
- Ease in development & testing
- Make efficient use of server resources.

Advantages of Server Virtualization

- Cost Reduction: Server virtualization reduces cost because less hardware is required.
- Independent Restart: Each server can be rebooted independently and that reboot won't affect the working of other virtual servers.
- Server virtualization eliminates server sprawl (Server sprawl is a situation in which many under-utilized servers utilize more space or consume more resources than can be justified by their workload) & uses server resources efficiently.

Desktop virtualization



Easily confused with operating system virtualization—which allows you to deploy multiple operating systems on a single machine—desktop virtualization allows a central administrator (or automated administration tool) to deploy simulated desktop environments to hundreds of physical machines at once. Unlike traditional desktop environments that are physically installed, configured, and updated on each machine, desktop virtualization allows admins to perform mass configurations, updates, and security checks on all virtual desktops.

Desktop solutions on demand

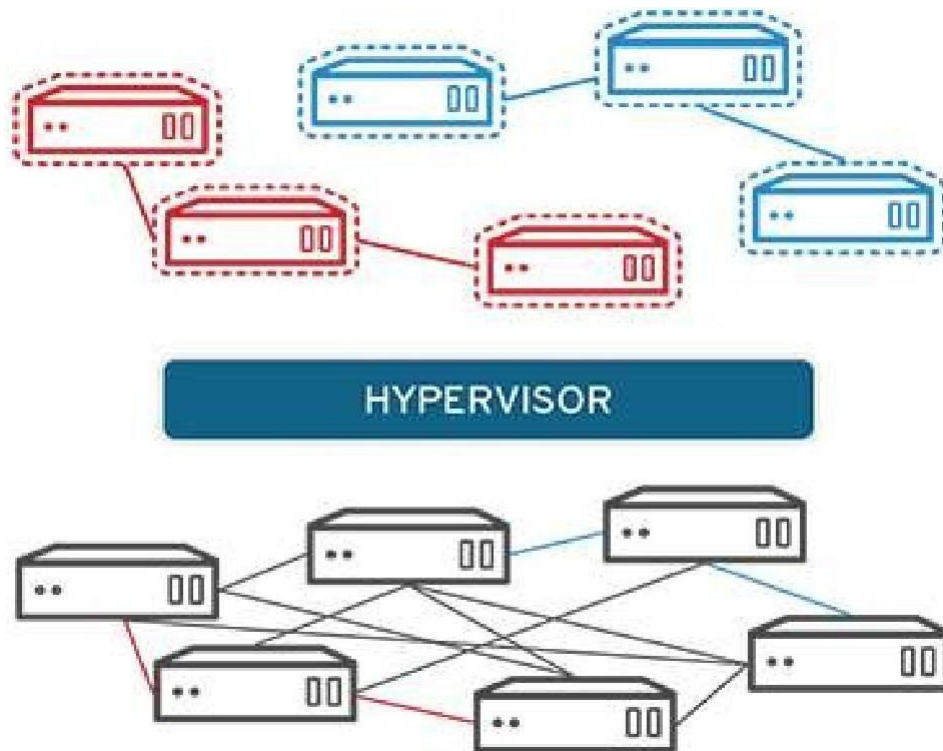
As the presence of virtual desktop infrastructure grew, VMware -- which came up with the term - joined Citrix and Microsoft as the leaders in the desktop virtualization market. Other vendors that offer various types of desktop virtualization technology, focusing mostly on client virtualization and remote desktop virtualization, include Parallels International, LogMeIn Inc. and Splashtop Inc.

There are also many third-party vendors whose products and services aim to improve the management, security and usability of virtual desktops. These offerings range from monitoring and application management tools to user environment management and bandwidth optimization software.

Virtual Desktops: Multinational flexibility provides seamless transitions between different operating systems on a single machine reducing desktop footprint and hardware expenditure. Parallels Desktop for Mac, a virtual machine application, instead of Boot Camp's dual-boot

approach, Parallels Desktop runs Windows XP directly on the Mac OS desktop (in what Parallels calls "near-native performance")--allowing you to run both OSs simultaneously and switch back and forth seamlessly."

Virtual networking



Virtual networking is a technology that facilitates the control of one or more remotely located computers or servers over the Internet. Data can be stored and retrieved, software can be run and peripherals can be operated through a Web browser as if the distant hardware were onsite.

Virtual networking facilitates consolidation of diverse services and devices on a single hardware platform called a virtual services switch. The centralization of control reduces the cost and complexity of operating and maintaining hardware and software compared with administering numerous separate devices in widely separated geographical locations. Maintenance personnel and administrators can install device drivers, perform tests and resolve problems on the remote machines from a single location.

The chief limitation of virtual networking is the fact that certain problems can be resolved only by direct physical contact with the hardware involved. Examples include broken wires, frozen drives and defective chips. In some cases, virtual networking may make it possible to circumvent

a damaged component by taking advantage of alternative resources until a technician can perform onsite repair or replacement. It may be necessary to install virtual networking software on the remote computers or servers to take advantage of this technology.

Virtualizing networks reduces the number of physical components —like switches, routers, servers, cables, and hubs —that are needed to create multiple, independent networks, and it's particularly popular in the telecommunications industry.

Several vendors, including Microsoft and VMware, offer virtual networking software. Some vendors offer comprehensive virtual networking services, allowing business network administrators to outsource labor and resources to the vendor. Virtual networking capability is a standard feature of Windows XP and Vista.

Data storage Virtualization

It pools the physical storage from different network storage devices and makes it appear to be a single storage unit that is handled from a single console. As we all know there has been a strong bond between physical host & locally installed storage device; and with the change in paradigm, local storage is no longer needed. More advanced storage has come to the market with an increase in functionality.

Storage virtualization helps the storage administrator to backup, archive and recovery data more efficiently, in less amount of time by masking the actual complexity of SAN (Storage Area Network). Through the use of software hybrid appliances, the storage administrator can implement virtualization.

Importance of Storage Virtualization

Storage virtualization is becoming more and more important in different forms such as:

- **Storage Tiering:** Using the storage technique as a bridge or as a stepping stone, this technique analyzes and select out the most commonly used data & place it on its highest performing storage pool and the least used data in the weakest performance storage pool.
- **WAN Environment:** Instead of sending multiple copies of the same data over WAN, WAN accelerator is used to locally cache the data and present it in a LAN speed, and not impacting the WAN performance.
- **SAN Storage:** SAN technology present the storage as block-level storage & the storage is presented over the Ethernet network of OS.
- **File Server:** OS writes the data to a remote server location to keep it separate and secure from local users.

Benefits of Storage Virtualization

- Data is stored in a very convenient location. This is because if the host failure data don't get compromised necessarily.
- By using storage level abstraction, it becomes flexible how storage is provided, protected, partitioned and used.
- Storage Devices are capable of performing advanced functions such as disaster recovery, duplication, replication of data & re-duplication of data.
- Storage virtualization facilitates management and monitoring of storage in a virtualized environment.

Question Bank

Multiple Choice Questions

(Questions for Application)

1. Which of the following storage devices exposes its storage to clients as Raw storage that can be partitioned to create volumes?
A. block
B. file
C. disk
D. cloud
2. _____ means the storage is preconfigured for the customer. The customer can neither format, nor install his own file system or change drive properties.
A. Configured cloud storage
B. Managed Cloud Storage
C. Unformatted cloud storage
D. Unmanaged Cloud Storage
3. _____ offers online storage space on-demand.
A. Managed cloud storage
B. Partitioned cloud storage
C. Unmanaged cloud storage
D. Open cloud Storage
4. _____ is a strategy for sending a copy of a physical or virtual file or database to a secondary, off-site location for preservation in case of equipment failure or catastrophe.
A. Cloud Backup
B. Storage virtualization
C. Block Storage
D. Cloud database
5. _____ backups copy the entire data set every time a backup is initiated.
A. Incremental Backup
B. Full Backup
C. Differential Backup
D. hybrid backup
6. Which Back up service copy data from one cloud to another.
A. Point-in-time snapshots
B. C2C back up
C. Differential back up
D. Cloud seeding
7. _____ and continuous backups help, but costs rise as backup frequency

increases.

A. Point-in-time snapshots

B. C2C back up

C. Differential back up

D. Full back Up

8. _____ backups only back up the data that has been changed or updated since the last backup.

A. Incremental Backup

B. Full Backup

C. Differential Backup

D. hybrid back up

9. Which is the common form of cloud backup used because of less resource use.

A. Incremental Backup

B. Full Backup

C. Differential Backup

D. hybrid back up

10. Which of the following database is well-suited for unstructured data, such as social media content, photos and videos.

A. MySQL

B. Oracle

C. IBM DB2

D. NoSQL

11. Which of the following is not a block storage options in the cloud.

A. AWS Elastic Backup Storage

B. Azure Premium Storage

C. AWS Elastic Block Storage

D. Google Persistent Disks

12. Any application which requires service side processing like Java, PHP, and .Net will require _____

A. Block storage

B. Network storage

C. File storage

D. Disk storage

13. RAID Volumes, where you combine multiple disks organized through stripping or mirroring, uses which storage?

A. Block

B. Network

C. File

D. Disk Drive

14. MySQL, Oracle, IBM DB2 and Microsoft SQL Server are Popular cloud platforms for _____
- A. Cloud Storage
 - B. **Cloud database**
 - C. Cloud block storage
 - D. Cloud Provider
15. How many categories of storage devices broadly exist in cloud?
- A. one
 - B. **two**
 - C. three
 - D. four
16. _____ is a situation in which many under-utilized servers utilize more space or consume more resources than can be justified by their workload.
- A. **Server sprawl**
 - B. Server utilization
 - C. Server scheduling
 - D. Server Virtualization
17. A technique in which, cloud backup vendor sends a storage device such as a disk drive to its new customer, which then backs up the data locally onto the device and returns it to the provider.
- A. Incremental Back Up
 - B. Database virtualization
 - C. mirroring
 - D. **Cloud seeding**
18. _____ is technology that lets you create useful IT services using resources that are traditionally bound to hardware.
- A. Mirroring
 - B. Cloud back up
 - C. **Virtualization**
 - D. Cloud Storage
19. Multiple virtual servers can run on a physical server at the same time is called _____
- A. Mirroring
 - B. Sprawl
 - C. **Partitioning**
 - D. Isolation
20. The Virtual server running on the physical server are safely separated & don't affect each other is called _____

- A. Partitioning
 - B. **Isolation**
 - C. Encapsulation
 - D. Mirroring
21. ____ is the partitioning of physical server into several virtual servers and this division is mainly done to improve the utility of server resources.
- A. Mirroring
 - B. Resource Virtualization
 - C. Server Partitioning
 - D. **Server virtualization**
22. ____ pools the physical storage from different network storage devices and makes it appear to be a single storage unit that is handled from a single console.
- A. **Data storage Virtualization**
 - B. Network storage virtualisation
 - C. Server virtualization
 - D. database virtualization
23. Which of the following is a cloud storage service offered by IBM.
- A. **Blue cloud**
 - B. Nirvanax
 - C. Gdrive
 - D. simple storage service(S3)
24. Which of the following is a open source cloud database platform.
- A. NoSQL
 - B. **MySQL**
 - C. Oracle
 - D. MSAccess
25. Which of the following do not employ table model.
- A. **NoSQL**
 - B. MySQL
 - C. Oracle
 - D. MSAccess

Questions carrying 4 marks

(Questions for understanding)

1. Discuss the Pros and Cons of network storage.
2. What is cloud based data storage. Explain its types.
3. Explain the types of back up.
4. Explain the various cloud backup approaches.
5. Discuss the benefits of cloud database.
6. Identify the features of Virtualization.
7. What is storage virtualization explain its importance.

(Questions for Application)

1. Write a short note on network storage.
2. Write the advantages of cloud based data storage.
3. Write the disadvantages of cloud based data storage.
4. Write a brief note on cloud back up.
5. Write a brief note on the concept of virtualization.
6. Write a note on server virtualization and its uses.
7. Write a note on virtual networking.

UNIT 4

CLOUD SECURITY AND MANAGEMENT

For years, IT data centers physically secured resources to prevent users who do not have a need to physically touch computers, servers, and storage devices from doing so. IT security professionals know that anyone who can physically touch a device can more easily break into it, and so much of IT security is based on preventing access. The cloud, however, changes the security model significantly by making data available to users anywhere, anytime, with virtually any device. Securing cloud-based applications takes planning and resources. As you examine cloud security issues, think in terms of two types of threats: those that are common to both cloud-based and on-site solutions, and threats that are specific to the cloud.

Security Advantages of Cloud-Based Systems

The incidents of cybercrime have affected many of the top service providers due to this reason, smaller businesses remain at risk of possible intellectual property rights violations. However, the cloud-based storage systems offer a practical algorithm to process and manage a vast workflow securely. In addition to the numerous advantages such as ease of access and secure storage of data, security features of cloud-based solutions make the Cloud one of the best possible option. Let's have a look at some of the most prominent security advantages of cloud-based systems.

Data Encryption

Robust data encryptions within cloud-based security systems have substantially reduced the possibilities of data breaches; these solutions offer a layered approach that consists of security intelligence, key management, and secure access controls. Cloud-based systems give the required freedom to companies to choose their users who will be accessing the data that has been outsourced to the cloud. This way, any attempts to tamper with personal or profession data can be thwarted.

Most companies face the threat of internal data theft by their employees, and stronger access controls can nip these threats in the bud. The multi-layered security features weed out the possibilities of a breach of data to a great extent. Data, irrespective of its type, needs to be protected at all times. Any violations can be hazardous to the goodwill and the functioning of an enterprise.

Avoid DDoS Attacks

Distributed Denial of Service (DDoS) attacks can result in hefty losses. Hackers target the

website by directing traffic from several sources to the end website, and, as a result, the system gets overwhelmed. These DDoS attacks may tarnish the image of the company, as clients begin to lose trust.

A computer or networked device under the control of an intruder is known as a zombie, or bot. The attacker creates what is called a command-and-control server to command the network of bots, also called a botnet. The person in control of a botnet is sometimes referred to as the botmaster.

There are three types of DDoS attacks. Network-centric or volumetric attacks overload a targeted resource by consuming available bandwidth with packet floods. Protocol attacks target network layer or transport layer protocols using flaws in the protocols to overwhelm targeted resources. And application layer attacks overload application services or databases with a high volume of application calls.

Cloud-based security systems guard this imminent threat with real-time scanning of potential risks; this function is further used as a warning tool for various systems which allows for the tracking of incoming threats and attacks instantly this enables website admins to divert the traffic to several different locations.

Regulatory Compliance

Cloud computing security solutions usually provide reliable SOC1 and SOC2 certifications to the entertainment businesses. These certifications ensure periodic scrutiny of data and all types of possible glitches. Cloud-based solutions manage the requisite infrastructure for regulatory compliance and the protection of data. Detailed AWS reports about management of security controls ensure all organizations focus on their business operations, without worrying about compliance requirements.

Secure Storage

Traditional storage solutions don't provide any protection against possible disasters that have the potential to erase required data from devices. Cloud computing allows the users to store their data safely, thereby negating any mishaps that may affect the equipment. Cloud storage solutions offer private, public, and hybrid solutions which the businesses may choose as per their requirements. The hybrid cloud storage systems allow the users to keep their data secure in the most effective manner.

Patch Management

The vulnerabilities of a website are often exploited by hackers to breach the security system of a company. Cloud service providers keep their sites up to date; further on, they ensure that no vulnerabilities exist. Moreover, cloud solutions offer real-time assistance to clients by providing

companies with the option to scale cloud solutions during high traffic situations. This flexibility allows companies to reduce their cost of services substantially.

These large number of security features are quite flexible, agile, and affordable. Enhanced security features offer sufficient protection to the private and financial data of both media and entertainment companies and help to thwart data and intellectual property breaches. In this era of digitalization, where cybercrime has emerged as a norm, cloud-based solutions seem to be the best alternative to traditional security systems.

Benefits of proper patch management include:

- ❖ Security: The most critical and obvious benefit of patch management is heightened network security. Patches are often created after a company has experienced a data breach to ensure other businesses' data remains safe, and applying a patch as quickly as possible lessens the risk of your business becoming affected. By securing your network before hackers exploit a program's security vulnerabilities, you can avoid the data theft, legal issues and lasting reputation damage that often follows a security breach.
- ❖ Productivity: A single patch can lead to a considerable boost in network productivity, and an entire patch management system can transform the efficiency of your entire company.
- ❖ Remote protection: Patch management programs can be implemented as part of a remote workforce support solution to protect all of your company's devices, no matter where in the world they're located.
- ❖ Innovation: Technology is moving quickly. It seems that just as soon as developers introduce one feature, a newer, better one comes along to take its place. patches— which usually contain new features and functionality—allow you to keep up with the latest technological innovations and improve your services without the need to buy entirely new software.

Hardware and software redundancy: Most cloud-based solution providers have redundant hardware and software resources they can quickly deploy in an emergency.

CASE study : MCAFEE SECURITY AS A SERVICE

To stay current with virus and spyware threats, most antivirus solutions perform constant updates to their virus signature dictionary/database. When you license an antivirus solution, you normally receive at least one year of automatic security system updates. As you work, the antivirus software updates your system against new threats behind the scenes. In general, the antivirus software, as shown in FIGURE 1, pulls in the threat signatures from across the cloud.

McAfee now offers a range of security solutions that deploy from the cloud. The solutions protect e-mail (spam, phishing, redirection, and virus elimination), websites, desktop computers, mobile devices, and more. FIGURE 2 shows the McAfee security as a service web page.

Exercise Assume that you must provide desktop antivirus and antispam support for 1,000 desktop computers. Visit the McAfee website and create a proposal that includes your solution features and cost.

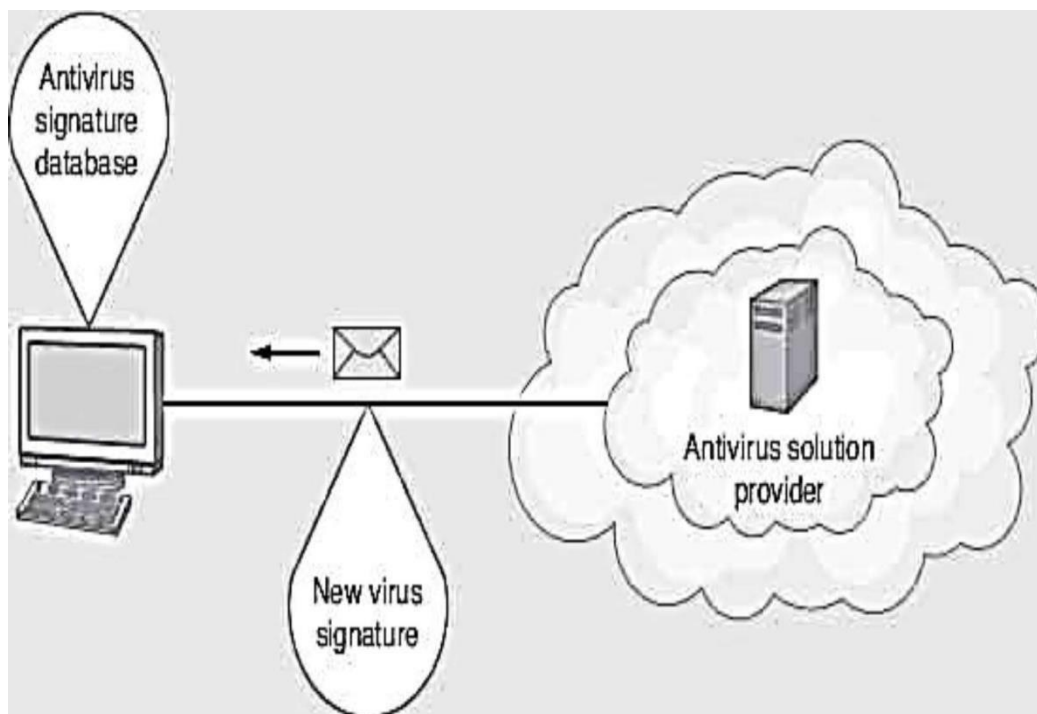


FIGURE 1 Antivirus systems retrieve cloud-based updates regarding new threats in near real time.

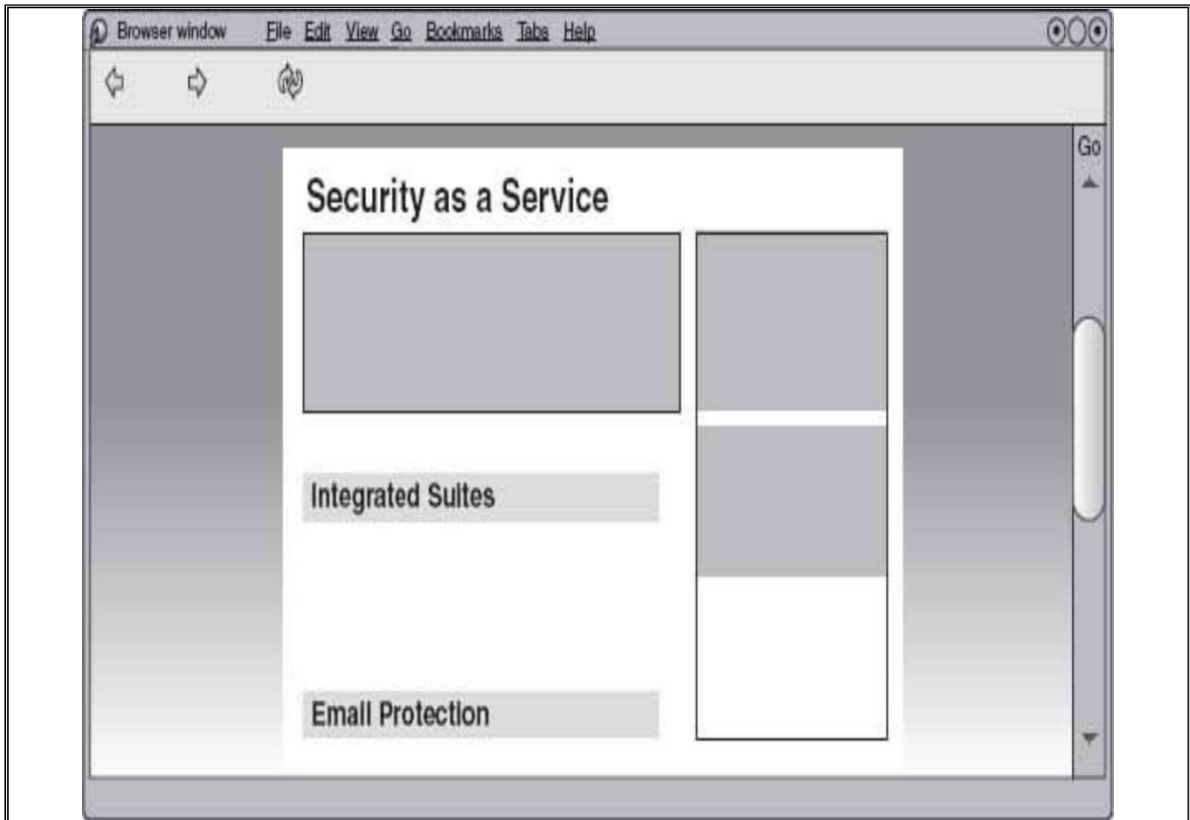


FIGURE 2 McAfee, like many security providers, now offers a range of cloud-based security solutions.

Introducing business continuity

The most basic business continuity requirement is to keep essential functions up and running during a disaster and to recover with as little downtime as possible. A business continuity plan considers various unpredictable events, such as natural disasters, fires, disease outbreaks, cyberattacks and other external threats.

Business continuity is important for organizations of any size, but it may not be practical for any but the largest enterprises to maintain all functions for the duration of a disaster. According to many experts, the first step in business continuity planning is deciding which of an organization's functions are essential and allocating the available budget accordingly. Once crucial components have been identified, failover mechanisms can be put in place.

Technologies such as disk mirroring allow an organization to maintain up-to-date copies of data in geographically dispersed locations. This enables data access to continue uninterrupted if one location is disabled.

A business continuity plan has three key elements:

- Resilience

- Recovery
 - Contingency
-
- A company can increase resilience by designing critical functions and infrastructures with various disaster possibilities in mind; this can include staffing rotations, data redundancy and maintaining a surplus of capacity. Ensuring resiliency against different scenarios can also help businesses maintain essential services on location and off-site without interruption.
 - Rapid recovery to restore business functions after a disaster is crucial. Setting recovery time objectives for different systems, networks or applications can help prioritize which elements need to be recovered first. Other recovery strategies include resource inventories, agreements with third parties to take on company activity and using converted spaces for mission-critical functions.
 - A contingency plan has procedures in place for a variety of external scenarios and can include a chain of command that distributes responsibilities within the organization. These responsibilities can include hardware replacement, leasing emergency office spaces, damage assessment and contracting third-party vendors for assistance.

DISASTER RECOVERY

Disaster recovery is mainly data focused, concentrating on storing data in a way that can be more easily accessed following a disaster.

Similar to a business continuity plan, disaster recovery planning specifies an organization's planned strategies for post-failure procedures. However, a disaster recovery plan is just a subset of business continuity planning.

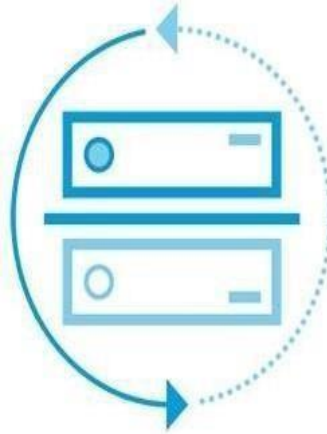
DRaaS: Disaster recovery as a service

At the most basic level, DRaaS is the process of replicating and hosting servers and their data via a third party provider with the purpose of enabling a failover option in the case of an unexpected interruption or event.

To help create a more complete picture of the DRaaS process, let's take a closer look at its major components:

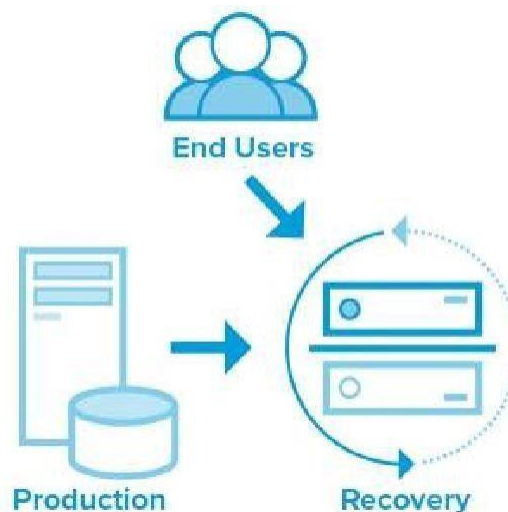
- Replication is the process of duplicating data from a production environment and transmitting it to the third party host. Since most organizations have not fully-virtualized their infrastructure, the replication technology should have the capability to frequently capture both virtualized and non-virtualized (physical) servers. This is often referred to as a "Hybrid" solution. In instances where data is constantly added and changed, frequent data snapshots are essential in order to prevent data loss during failover.

- Replication



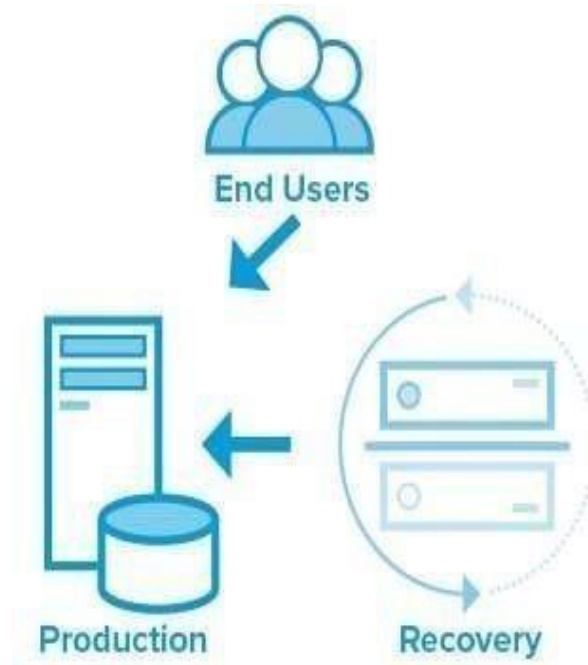
- Failover is the transitioning of end-user access during an event to the third party hosting environment. A DRaaS solution should provide a simple, straight-forward process for declaring and failing over to the replicated environment. The speed in which the solution can “stand up” the replicated environment corresponds directly to the impact on business operations. Integrated tools and support should be in place so that failover can occur, even if your IT is unavailable.

- Failover



- Failback is the process of transitioning user access from the third party failover site back to the original production environment. Once the impacted environment is fully up and running again, the DRaaS solution should have processes and tools in place that allow for a seamless switch back to production. Immediately following failback, replication processes should be re-engaged to ensure continuous protection.

- Failback



Service level agreement

A service-level agreement (SLA) is a contract between a service provider and its internal or external customers that documents what services the provider will furnish and defines the service standards the provider is obligated to meet.

SLA will typically include

a statement of objectives,

Scope of services to be covered: a list of the services to be covered by the agreement

define the responsibilities of the service provider and customer under the SLA.

Performance metrics (response time , resolution time

Penalties for contract breach /exclusions



The customer, for example, will be responsible for making a representative available to resolve issues with the service provider in connection with the SLA. The service provider will be responsible for meeting the level of service as defined by the SLA. The service provider's performance is judged according to a set of metrics. Response time and resolution time are among the key metrics included in an SLA, since they relate to how the service provider deals with a service interruption.

Performance metrics

SLAs establish customer expectations with regard to the service provider's performance and quality in a number of ways. Some metrics that SLAs may specify include the following:

- Availability and uptime percentage -- the amount of time services are running and accessible to the customer. Uptime is generally tracked and reported on per calendar month.
- Service provider response time -- the time it takes the service provider to respond to a customer's issue or request. A larger service provider may operate a service desk to respond to customer inquiries.
- Resolution time -- the time it takes for an issue to be resolved once logged by the service provider.
- The schedule for notification in advance of network changes that may affect users.
- Usage statistics that will be provided.

- In addition to establishing performance metrics, an SLA may include a plan for addressing downtime and documentation for how the service provider will compensate customers in the event of a contract breach. Service credits are a typical remedy. Here, the service provider issues credits to the customer based on an SLA-specified calculation. Service providers, for example, might provide credits commensurate with the amount of time it exceeded the SLA's performance guarantee. A service provider may cap performance penalties at a maximum dollar amount to limit exposure.
- The SLA will also include a section detailing exclusions, that is, situations in which an SLA's guarantees -- and penalties for failing to meet them -- don't apply. The list might include events such as natural disasters or terrorist acts. This section is sometimes referred to as a "force majeure clause", which aims to excuse the service provider from events beyond its control.

Ensure and Audit System Backups

Under Data Storage in the Clouds, we have observed different cloud-based backups in detail.

Managers should consider different forms of backups. First, your company may back up user files from on-site computers to disks that reside within the cloud. Hopefully you will never require these backups; but regardless, you should periodically audit the backups, perhaps by checking that you can successfully restore randomly selected files of different users.

Second, if the cloud provider stores some or all of your company data, you must understand the provider's backup process (and include it in the SLA). For governance purposes, you should know if the data is encrypted, who has access to it, and if it is replicated to a remote facility. If it is backed up to another location, you must know where and how often.

Additionally, if the provider uses a database to store your company data, you need to know if and how the data is replicated and whether your company's information is stored in a private or multitenant database. You should also know the system's guaranteed uptime.

Know Your System's Data Flow

Often, developers and managers think of a cloud solution as a black box, which means that they know what the system does but not how it does it.

Managers should create a detailed process-flow diagram that shows the movement of company data throughout the cloud solution. They should also identify within the dataflow various points for the placement of internal controls or auditing.

▪ Auditing

The company should examine the sources from which the values are derived to ensure that each is accurate and free from fraud. This inspection process is known as auditing.

The auditing process can be internal (done by the company) or external (done by a third party), as shown in figure below

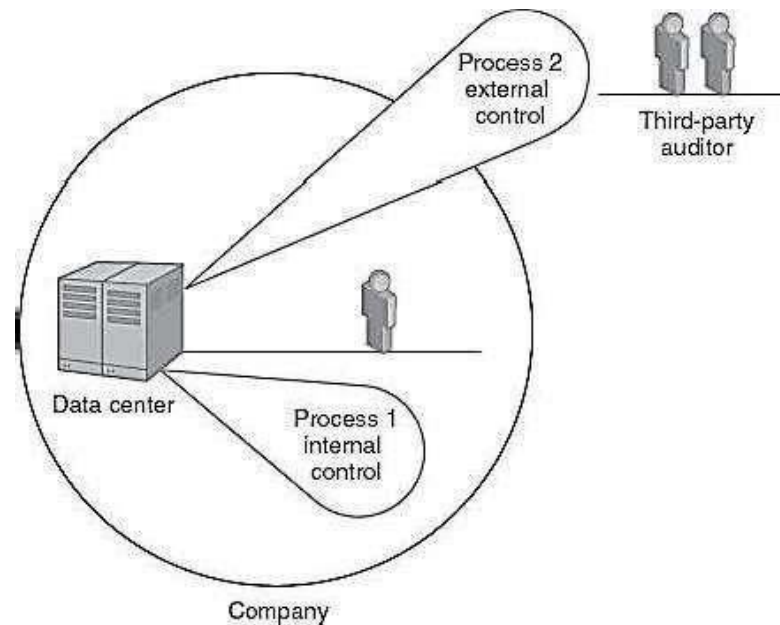


FIGURE Companies must audit the source of the values they measure and report using internal or external auditors.

- E. **Internal audit:** Conducted within an organization, they are voluntary. To be used as quality control checks, scrutinizing business operations against set standards. These standards can be governmentally set by regulatory bodies, or by the organization itself.
- F. **External audit:** Conducted externally to a business, by an independent, qualified auditor. These are **NOT** voluntary. They aim to make sure records kept by a business are complete, accurate and adhere to legal standards.

Role of a Cloud Auditor

The role of an auditor is to provide an objective opinion based on facts and evidence that a company has controls in place to meet a certain objective, criteria, or requirement. Additionally, in many cases, the auditor will also provide an opinion on whether or not those controls operated over a period of time.

The auditing process will identify key stages within processes that the auditor should inspect such as enquiry, physical inspection, observation, confirmation, analytics procedures, and/or re-performance to collect evidence. To support the process, the company should put in place its own internal controls (policies and procedures) at each of these key stages, as shown in figure below.

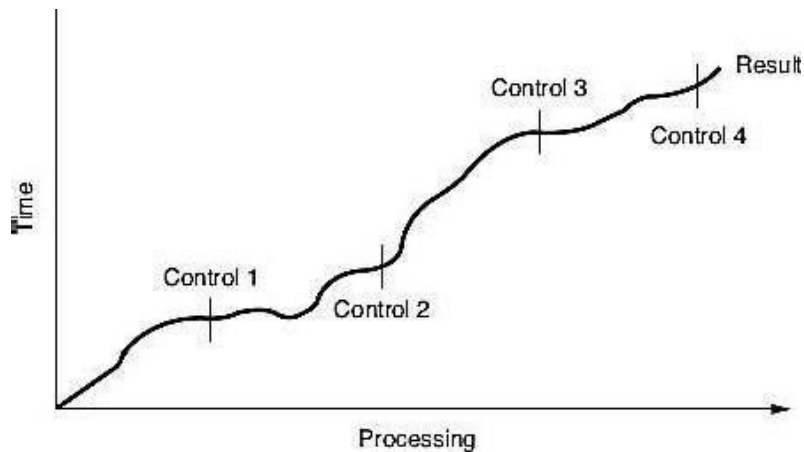


FIGURE Internal controls allow a company auditor to inspect data values at key stages.

Auditing can be done based on different control areas like Organization and Administration, Communication, Risk Assessment, Monitoring Activities, Logical and Physical Access, Systems Operations and Change Management etc.

Beware of Vendor Lock-In

Relationships can go bad—even those with a cloud-solution provider. The agreement you sign with a cloud provider should stipulate exit procedures in case the provider fails to meet the service levels or breaches any other aspect of the contract.

IT managers must be able to control their company's data. In the event of a worst-case scenario,

a manager must be able to export the company data, ideally to a file that can be imported by another provider. Managers should test this capability before a problem arises.

Vendor lock-in occurs when a provider does not support data export or when a provider's service is unavailable through others. Thus, the customer is "locked in" to the relationship with the vendor. If this happens, a company may have to put up with breaches of its SLA because it has no other place to move its data. Managers should consider the risk of vendor lock-in before they enter into a provider agreement.

Source-Code Escrow

Companies fail. Therefore, managers, should perform due diligence on a cloud-solution provider before they enter into an agreement. The manager may want to arrange a source code escrow agreement, which places a copy of the provider's programming language source code with a Third-party Escrow Company. Then, if the solution provider fails, the company can acquire and deploy the source code, put it on its own system, and implement the provider's solution.

Determine Technical Support and Help Desk Procedures

Depending on the solutions a company places in the cloud, a company may have various help desk support requirements. For example, in the case of a Solution as a Service (SaaS) solution, the cloud-solution provider may provide software technical support. For Platform as a Service (Paas) or Infrastructure as a Service (IaaS) solutions, however, the company may provide software technical support. There may also be shared support responsibilities. In all cases, an IT manager should ensure that the support specifics are defined within the SLA.

Determine Training Procedures

To be successful, large-scale cloud applications often require user training before, during, and after the integration. For SaaS solutions, the cloud-service provider normally provides user training. Depending on the application's processing, the company may need to augment/increase the training with in-house instruction. The IT manager should stipulate the training responsibilities within the SLA.

CASE study: NETUITIVE PREDICTIVE ANALYTICS AND CLOUD MANAGEMENT

Predictive analytics tools perform statistical analysis to predict future behavior. Netuitive integrates predictive analytics to provide IT managers with insights into how a solution will work under different conditions. Netuitive software can monitor a group of integrated or stand-alone cloud-based solutions. The software's self-learning capabilities allow the software to identify demand trends and more. As shown in FIGURE 12-5, Netuitive provides a dashboard and drill-down reporting.

Exercise Discuss factors a company might want to consider before using predictive analytics of a cloud-based solution.



FIGURE : Netuitive provides predictive analytics that an IT manager can use to model a site's future behavior.

Know the Provider's Security Policies and Procedures

Many clients are apprehensive about storing their data in the cloud. To reduce these concerns, IT managers should thoroughly understand the provider's security plans, policies, and procedures. Specifically, a manager should be

- Aware of the provider's multitenant use,
- E-commerce processing,
- Employee screening, and
- Encryption policy.
- Also examine the provider's use of firewalls, intrusion detection, and security mechanisms. These security factors should be defined in the sla.
- Define the data privacy requirements:
If a company has specific data-privacy requirements, such as the Health Insurance Portability and Accountability Act (HIPAA) for health care or the Family Educational Rights and Privacy Act (FERPA) for student data, it should ensure that the SLA details the specific requirements.

***CASE study* NEW RELIC CLOUD-PERFORMANCE MONITORING**

When it comes to cloud-performance monitoring, most managers spend 80 percent of their time monitor ing 20 percent of a solution's code (see Chapter 19 and the Pareto Principle). New Relic, shown in FIGURE 12-6, provides monitoring software that will examine system performance to identify potential bottlenecks. New Relic software supports most common programming languages and can be easily integrated into a site.

Exercise Discuss common bottleneck locations within cloud-based solutions.

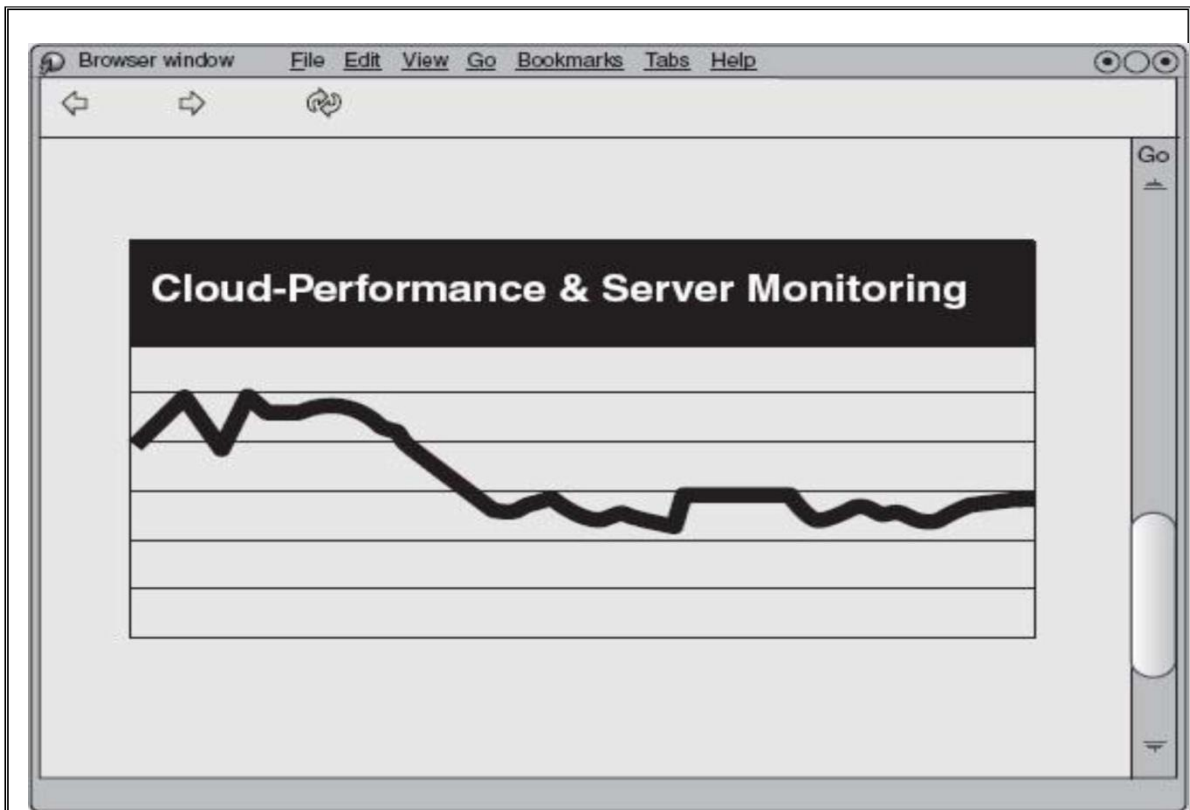
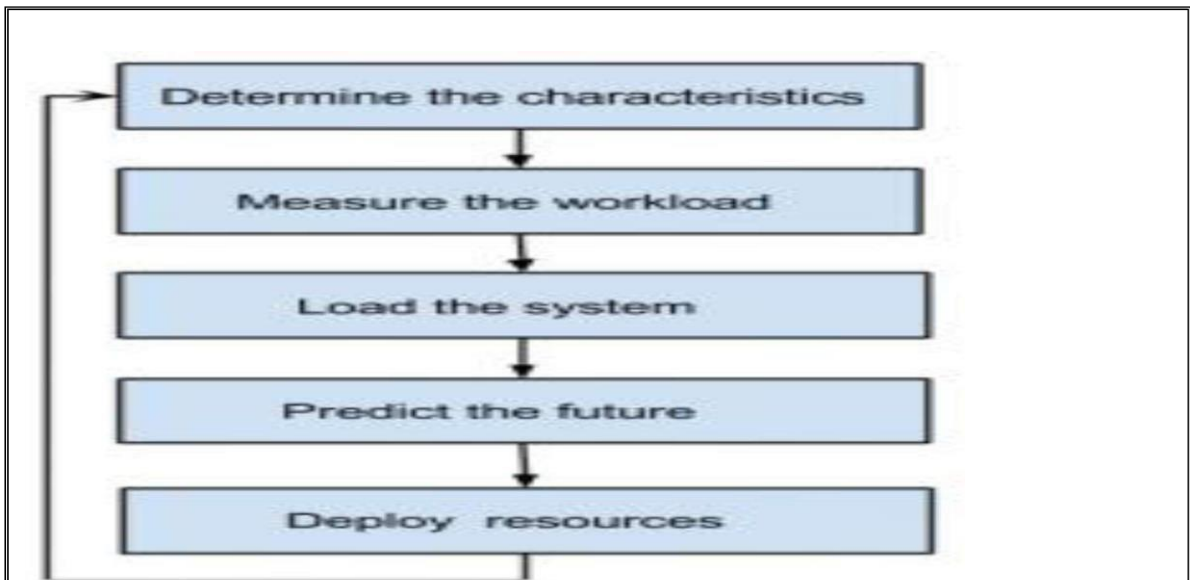


FIGURE : New Relic provides cloud-performance monitoring and bottleneck identification.

Monitor Capacity Planning and Scaling Capabilities

For SaaS solutions, the cloud-solution provider will scale the site to match user demand. An IT manager, however, must define in advance key response-time metrics the solution must provide and then include those measures within the SLA.

For PaaS and IaaS solutions, the IT manager must initially estimate the solution's capacity plan, which defines the resources the solution will need to operate satisfactorily. The IT manager should also estimate the site's potential growth and define, with the help of the solution provider, the plan for scaling the site resources as well as the related costs.



Capacity planning measures the maximum amount of work that can be done using the current technology and then adds resources to do more work as needed

Several sites within the cloud provide system-performance reports that managers can use to measure current performance and the potential system benefit from scaling specific resources.

Monitor Audit-Log Use

To identify potential system bottlenecks, detect errors within the system, and identify system-resource use, the IT manager may examine various system log files. In a PaaS or IaaS solution, the manager can likely turn on the log file reporting that meets his or her needs. For a SaaS solution, the manager should discuss in advance with the cloud service provider the various logs that will be available and the costs of running them, both in terms of dollars and system performance.

CASE study: UPTIME SOFTWARE

Too often, cloud-solution managers do not know that a system error has occurred until a user reports one. With Uptime, IT managers can easily monitor a wide range of servers and produce resource utilization reports similar to that shown in FIGURE 12-9. Companies can download, install, and use a trial version of the Uptime software from the company website.



FIGURE: The Uptime site provides system monitoring and utilization reporting capabilities a site manager can easily implement.

Solution Testing and Validation

Just because a company provides a solution does not mean that the solution is error free. An IT staff using a cloud-based solution must test the solution and periodically audit key processing to confirm that the application is providing correct results. In particular, a cloud-service provider will often perform patch management and version updates. The IT staff should be aware of all system modifications and test accordingly.

Question Bank**Multiple Choice Questions****(Questions for understanding)**

1. Cloud service providers keep their sites up to date to ensure that no vulnerabilities exist. This is possible because of?
A. Network performance
B. Patch management
C. Resilience
D. Failback process
2. Which of the following is hardest factor to determine?
A. Network performance
B. Network capacity
C. Network delay
D. Network latency
3. Which of the following is critical step in capacity planning?
A. Predict the future based on historical trends and other factors
B. Load the system until it is overloaded
C. Deploy or tear down resources to meet your predictions
D. Measuring the work load
4. _____measures the maximum amount of work that can be done using the current technology and then adds resources to do more work as needed.
A. Capacity planning
B. SLA metric
C. Vendor lock in
D. Training procedure
5. The auditing process of an cloud based solution can be of _____types
A. two
B. three
C. four
D. five
6. Which of the following model type is not trusted in terms of security?
A. Public
B. Private
C. Hybrid
D. Community
7. To keep essential functions up and running during a disaster and to recover with as little downtime as possible is the basic aim of _____
A. Capacity planning
B. Business continuity planning

- C. Disk mirroring
 - D. Cloud seeding
8. What about vendor lock in not true in the following.
- A. When the Provide doesn't support data export of the customer
 - B. When the Provider service is not available elsewhere
 - C. **the company can acquire and deploy the source code, put it on its own system and run it**
 - D. The agreement you sign with a cloud provider doesn't stipulate exit procedures
9. Which of the following is not correct about measures to avoid vendor lock in
- A. Negotiate both an entry and exit strategy upfront with your vendor
 - B. Have a backup vendor
 - C. Design portable applications
 - D. **Agreement must not have exit procedures**
10. Which of the following is not a part of Providers security plan must not include
- A. Encryption policy.
 - B. Provider's multitenant use,
 - C. E-commerce processing
 - D. **Un-authorization for data access**
11. Technology to maintain up-to-date copies of data in geographically dispersed locations the technology is ____
- A. **disk mirroring**
 - B. Capacity planning
 - C. Business continuity planning
 - D. Cloud seeding
12. _____ enables data access to continue uninterrupted if one location is disabled.
- A. **disk mirroring**
 - B. Capacity planning
 - C. Disaster recovery
 - D. resilience
13. Business continuity has _____key elements
- A. four
 - B. two
 - C. five
 - D. **three**
14. Which of the following is not true about source code escrow agreement
- A. It places a copy of the provider's programming language source code with a third-party escrow company
 - B. It is a way to avoid the risk of vendor lock in
 - C. **excuse the service provider from events beyond its control.**

D. if the solution provider fails, the company can acquire and deploy the source code, put it on its own system

15. SLA including a section detailing exclusions where agreement breach is not liable for punishments
- A. **force majeure clause**
 - B. source code escrow
 - C. statement of objective
 - D. performance metrics

(Questions for skills)

16. What is the relation of capacity attribute to performance?
- A. same
 - B. **different**
 - C. similar
 - D. Equal
17. Targeting the website by directing traffic from several sources to the end website is _____
- A. **DDOS attack**
 - B. Vendor lock in
 - C. Ethical hacking
 - D. Web traffic
18. "Force majeure clause", which aims to
- A. **excuse the service provider from events beyond its control.**
 - B. Punish the service provider from events beyond its control.
 - C. Force an action against customer
 - D. Force an action against provider
19. Which of the following is a security provider offering a range of cloud-based security solutions.
- A. **McAfee**
 - B. MySQL
 - C. Rackspace
 - D. Cloud switch
20. Which mechanism refers to the process of transitioning user access from the third party failover site back to the original production environment.
- A. Resilience
 - B. **Fail back**
 - C. Contingency plan
 - D. Replication
21. What is the time taken for an issue to be resolved once logged by the service provider.

- A. Resilience time
 - B. **Resolution time**
 - C. Response time
 - D. Reference time
22. What does internal controls of cloud solution refer to in auditing
- A. **Policies and Procedure**
 - B. Auditing tools
 - C. IT staff
 - D. Bottlenecks
23. Resilience, Recovery and Contingency are the key elements of _____
- A. Capacity planning
 - B. **Business continuity**
 - C. Cloud seeding
 - D. Disaster Recovery
24. The process of replicating and hosting servers and their data via a third party provider to enable a failover option in the case of an unexpected interruption.
- A. Disk mirroring
 - B. **DRaaS**
 - C. DDOS
 - D. IDaaS
25. Replication, failover and fail back are the components of
- A. **DRaaS Process**
 - B. Business continuity plan
 - C. Disk mirroring
 - D. Capacity planning

Questions carrying 4 marks

(Questions for understanding)

1. What do you mean by DDOS attack? How to avoid it.
2. What do you mean by Patch management? Why is it required.
3. What is Business continuity? Explain its Key elements.
4. What is SLA? What does it include.
5. What are the procedure to follow to ensure and audit system back up.
6. What is vendor lock in? How to avoid it.

(Questions for Application)

1. Write the security advantages of cloud Based solution.
2. Explain Disaster Recovery and components of DRaaS.
3. Summarize capacity Planning in cloud management.
4. Summarize Auditing/inspection of a cloud based solution.
5. What is the use of monitoring the log files.
6. Write a note on SLA performance metrics.

UNIT V

MIGRATION TO THE CLOUD

Define the System Goals and Requirements

All IT projects should begin with specific requirements. The process of taking an application to the cloud, known as cloud migration, is no exception. The following are the common issues to

- Data security and privacy requirements
- Site capacity plan the resources that the application initially needs to operate.
- Scalability requirements the measurable factors that should drive scaling events.
- Business continuity and disaster requirements.
- Budget requirements
- Type of cloud public, private, or hybrid
- Single tenant or multitenant solution requirements
- Data backup requirements
- Client device requirements, such as computer, tablet, or smartphone support Training requirements
- Help desk and support requirements Governance and auditing requirements
- Open source software requirements (open source-based cloud solutions reduce the risk of vendor lock-in)
- Programming API requirements Dashboard and reporting requirements Client access requirements
- Data export requirements

After you define your application requirements, discuss each in detail with potential cloud-solution providers. Make sure you define all of your system requirements clearly within the Service-level agreement (SLA).

Cloud Migration Benefits

Cloud migrating benefits your application by opening a horizon of new possibilities for high-level architecture applications that gather and process data in realtime. Properly deployed cloud migration strategies ensure a smooth cloud migration experience and hassle-free post-migration performance.

i. Scalable Resources

The most visible benefit of cloud migration is the inherent feature of flexible server resources. This is the main reason why applications that require varying levels of server resources opt for the scalable server solutions offered by cloud hosting solutions.

ii. Cost Reduction

In an era when the competition has become fierce, every business is looking for cost-effective solutions to remain profitable. Cloud migration helps companies to reduce both operational and capital expenses simply because of the Pay-As-You-Go billing model that is ideal for businesses who now have to pay only for the resources their server consumes.

iii. Mobility

Since a cloud server is accessible from anywhere, users experience the versatility of access and ease of use like none other. IP whitelisting feature ensures the authorization of the user to access the server from any part of the world regardless of device. This function is commonly available in all PaaS cloud services, such as Cloudways.

iv. Automation

A known cloud migration problem is to keep the backend server API's updated. When it comes to (managed) cloud migration, application management team do not have to worry about API and application update.

v. Flexible Operations

Cloud Solution allows organizations to deploy and test applications quickly over the cloud. Instead of manually using the application, the development teams can deploy the applications through automated deployment options without worrying about compatibility and performance issues.

vi. Secure Storage Options

Organization prefers to go for cloud migration because of the very secure storage that is available at a very reasonable operational cost. Besides, when the need arises, the room can be easily scaled up or down without disrupting the application usage. This is an excellent feature for businesses with seasonal or periodic data storage requirements.

CASE study : CLOUDSWITCH CLOUD MIGRATION

Many companies have enterprise-based applications that are widely used by their employees. These applications, therefore, are mission critical. CloudSwitch provides a downloadable

on-site applications to a cloud-based solution in a matter of minutes. In other words, application that companies can install within their data center that securely maps the company's CloudSwitch, shown in FIGURE 1 provides a way for companies running Windows or Linux solutions to migrate quickly to the cloud. The CloudSwitch migration program requires no programming or development. Additionally, CloudSwitch provides a suite of cloud-management tools that the IT staff can use to manage the solution after it moves to the cloud.

Exercise Most IT projects fail or come in over budget. Discuss steps you would take to reduce the risk of failure of a cloud-migration project.



FIGURE 1 CloudSwitch provides tools to migrate applications to the cloud and then to manage the application within the cloud.

CASE study 3TERA CLOUD SOLUTIONS

Formerly known as Computer Associates, CA Technologies provides a wide range of services and solutions to companies migrating to the cloud. To drive its cloud offerings, CA Technologies

acquired 3Tera, a company that helps businesses move solutions to public and private clouds using a graphical user interface (GUI) tool. As shown in FIGURE 2, the 3Tera website offers a turnkey cloud-computing platform, driven by the company's AppLogic software, which allows large (enterprise) or small companies to migrate to the cloud through the use of the 3Tera virtual appliance, a device that behaves very much like a virtual machine. As user demand for an application grows, 3Tera can easily scale solutions.

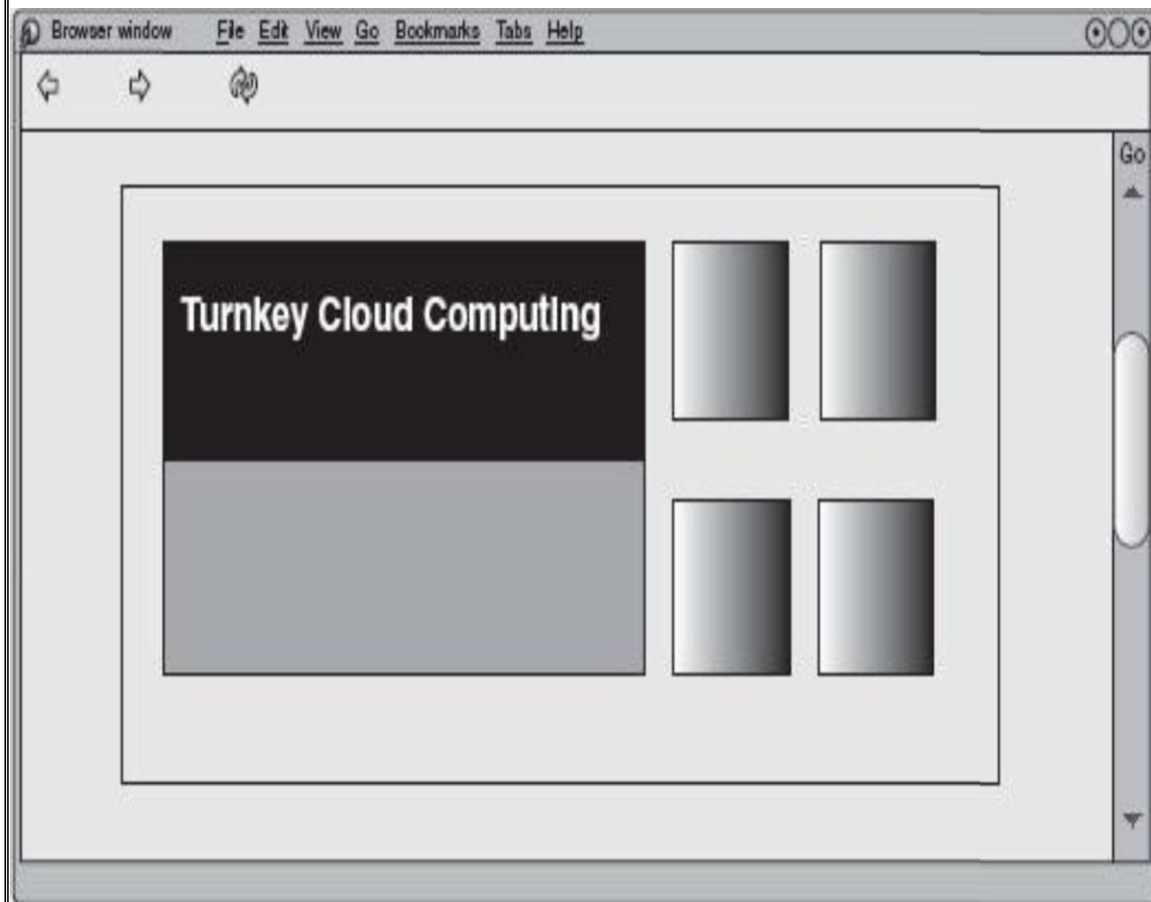


FIGURE 2 3Tera supports cloud migration through the use of its AppLogic software.

Protect Your Existing Data

Before you begin your application migration to a cloud provider, make sure that you back up your data so that you can revert, if necessary, to a known restore point. Then, be sure that you and the provider agree to the backup procedures that will be employed after they take control of your data. As discussed, it is easy to move a solution to a cloud provider. You need to ensure that it is equally easy to move out of the cloud if necessary.

Furthermore, if your company has specific privacy requirements, such as those of the Health Insurance Privacy and Portability Act (HIPAA) for health care, or of the Family Educational

Rights and Privacy Act (FERPA) for education, have your provider state explicitly, in writing, its data privacy policies and procedures.

Use an Experienced Cloud Consultant

Moving a solution to the cloud is a learning experience. The process has many options and a wide range of potential pitfalls. Many companies provide consultants who are experienced in the cloud migration process. Before you begin your application's migration to the cloud, you should consider hiring a consultant. In most cases, you will find that this saves you money in the end by helping you avoid costly mistakes.

Know Your Application's Current Characteristics

Before you move your application to the cloud where you might face scaling issues, be sure that you monitor your application to identify its key performance indicators:

- Demand periods: Does the application have periods of high or low demand, such as 8 A.M. to 5 P.M.?
- Average users: How many users typically use the system simultaneously?
- Disk-storage requirements: What are the application's typical disk-storage needs? Are the files permanent or temporary? Are most operations read or write operations?
- Database-storage requirements: What are the application's database requirements? Is the database replicated in real time? What is the application's database read/write ratio?
- RAM use: What is the application's range of physical and virtual memory use?
- Bandwidth consumption: What is the application's bandwidth requirement?
- Caching: How does the application currently cache data?

Remember Vendor Lock-In

Vendor lock-in occurs when a vendor makes it difficult for a company to switch to another provider, even if the vendor has failed to fulfill the SLA. This lock-in may occur because the vendor is unable to export data completely or because the vendor provides services its competitors do not. A cloud-service provider should make it easy for clients to move to another provider in the event that the provider fails to meet one or more of the SLA requirements.

In the case of a Platform as a Service (PaaS) or Infrastructure as a Service (IaaS) provider, moving a company's applications and data should be relatively straightforward. Moving from a Software as a Service (SaaS) provider, however, may prove to be more difficult because of the specific capabilities of the vendor, upon which the company relies. To reduce the risk of vendor

lock-in, many companies seek providers who support “open” solutions, which use open source software such as Linux, PHP, and MySQL.

Define Your Training Requirements

To reduce employee stress during an application’s migration to the cloud and to increase employee productivity with the cloud-based tools, you should consider training before, during, and after the cloud migration. As you define your training requirements, consider the following:

- Employee preparedness for the SaaS solution
- Developer training on the solution application program interfaces (APIs)
- Administrator training for cloud-based operations
- IT-audit group training for corporate governance issues and internal controls
- Help desk support preparedness training
- Business continuity and disaster preparedness training

Given the cloud’s cost-effective ability to deploy solutions, training may prove to be one of the most expensive aspects of the company’s cloud migration.

CASE study RIGHTSCALE CLOUD APPLICATION MANAGEMENT

RightScale, shown in FIGURE 4, provides a fully automated cloud-management platform that lets companies deploy cloud-based solutions across one or more clouds. RightScale provides its cloud-management software as a SaaS solution that lets customers deploy and manage their solutions quickly. The RightScale website also features valuable videos, white papers, and forums that focus on cloud computing.



FIGURE 4 RightScale provides support for application deployment to one or more clouds.

Establish a Realistic Deployment Schedule

Cloud-solution providers can quickly deploy solutions. Despite that, you should set a deployment schedule that provides sufficient time for training, testing, and benchmarking. Many organizations, when moving a new application to the cloud for the first time, will establish a beta-like release schedule that offers employees a pre-release opportunity to interact with the software and provide feedback. Furthermore, the testing period may provide time for the company to establish early system-performance benchmarks.

Identify IT Governance Issues

Before you migrate an application to the cloud, consider the following governance requirements:

- Identify how the cloud solution aligns with the company's business strategy.
- Identify and define the internal and external controls the company will need within the application, and at what control points, in order to validate that the application is performing correctly and is free from possible external modification.
- Describe risks the IT staff is trying to mitigate and ways the cloud can help.

- Describe who within the company will have access to data within the cloud and how they will get it.
- Determine who within the cloud provider's organization will have access to data within the cloud and how they will get it.
- Discover how the cloud provider logs errors and system events and how you can access them.
- Determine how and when the cloud provider performs system updates and patches.
- Discover which performance-monitoring tools are available for your use.

Understanding Cloud Bursting

One of the cloud's biggest advantages is its ability to scale on the fly to meet user demand. Some companies that run on-site applications have started to use the cloud as a way to scale their applications on demand. When the on-site application encounters increased user demand, the application expands into the cloud through a process called cloud bursting, as shown in FIGURE 5. When the user demand declines, the application leaves the cloud. Cloud bursting is most common for seasonal demand, or event-driven demand, such as the load on Google Maps when an earthquake or other natural disaster occurs.

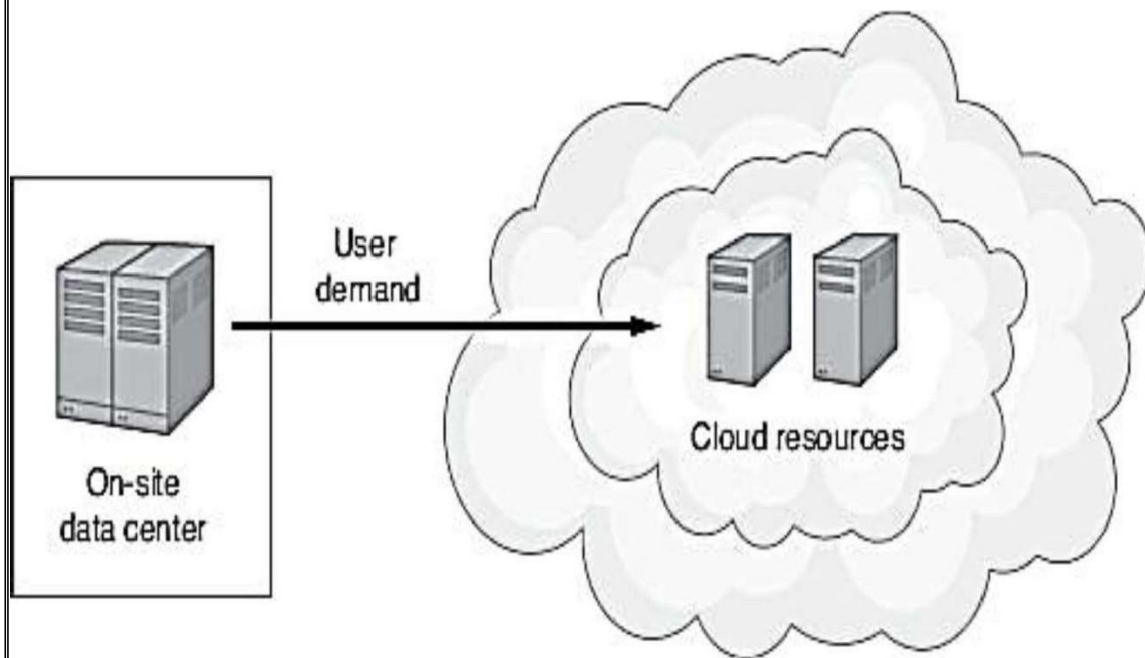


FIGURE 5 Cloud bursting lets a company extend an on-site application into the cloud to scale temporarily to meet user demand

Mobile Cloud Computing

The Evolution of Mobile Computing

If you list inventions that have had the biggest impact on society, from the automobile to airplanes to television and the telephone, the cell phone may (or likely eventually will) top the list. It is important to note that in less than 30 years the cell phone has evolved from a heavy cumbersome device, as shown in FIGURE 6, to a small handheld device with more than 1 billion users. Today, the liveliest sector of the mobile phone market is “smartphones,” which integrate computing capabilities, and often a web browser, which allows the phones to provide a wide range of solutions. Beyond traditional phone calls, users use smartphones to do the following:



FIGURE 6 In less than 30 years, the cell phone has gone from a cumbersome device used by early adopters to a handheld device used worldwide.

- Browse websites, including Google, Facebook, eBay, and more
- Place face-to-face video calls to phones and computers
- Perform GPS-based navigational operations
- Exchange text messages
- Perform e-commerce operations
- Run a myriad of applications (apps)

Understanding the G in 3G and 4G

When you discuss phone capabilities, you will hear terms such as 3G and 4G. In the simplest sense, the G stands for generation. A 4G phone, therefore, is a fourth-generation phone. Although there are standards with respect to the potential speeds associated with each generation, the speed of most devices today depends upon the location (some cities support faster speeds), the provider network, and the phone technology, as well as whether the user is moving or stationary. TABLE 1 describes the key generational attributes.

TABLE -1 THE COMMON GENERATION OF CELL PHONES

Generation	Capability
1G	First-generation phones supported analog communication.
2G	Second-generation phones introduced digital communication.
3G	Third-generation phones supported faster speeds, which, in turn, made web browsing and e-mail readily available.
4G	Fourth-generation phones support near Wi-Fi speed, which enables rich media and video streaming.

The Mobile Cloud Ecosystem

An ecosystem is an environment that consists of living and non-living things with which one interacts. Many cloud-based companies use the term ecosystem to describe the user's environment. To that end, you might describe the mobile-cloud ecosystem as consisting of the following:

- Phone class, which may be voice or face-to-face
- Web browsing
- Apps and widgets

- Voice commands and voice recognition
- Display screens
- Transmission speeds for upload and download operations
- Keyboard interface
- Touchscreens

The mobile cloud consists of apps and web pages that originate from sites within the cloud from which users download, or with which they interact via a mobile device. If you are creating a mobile solution, you should consider how your solution interacts with or supports the various mobile-device ecosystem components.

Introducing the Mobile Players

To understand mobile data communication, you should understand the players. To start, the operator, or network, is the company that makes the mobile network available. Within the United States, mobile operators include Verizon and AT&T. In the simplest sense, the operator owns the cell tower through which the data communication occurs. Most operators will provide internal groups that support developers in bringing mobile solutions to the market.

Within most mobile networks, operators will place special servers, called transcoders, which examine the content a mobile device is downloading from a website. The transcoder, in turn, may change the document content, such as changing a PNG graphic to a GIF graphic to improve the download performance or device display. If you are developing mobile content, you must be aware of how the transcoder may change the content. FIGURE 7, for example, shows how a transcoder (in this case a Google transcoder) might modify the content of a website.

Unfortunately, standards for transcoders are still being developed. As a result, you may see differences, in some cases considerable ones, among the content produced by different transcoders.

Many operators may include proxy servers (proxies) which perform operations on behalf of a device. By serving as an intermediary, a proxy provides a level of security that separates the device from the web server with which the device is interacting.



FIGURE 7 Within a mobile network, a transcoder may modify web content to a form and layout more suitable for a mobile device.

Pages, Apps, and Widgets

When developers build mobile solutions, they can approach their solution in one of three ways. First, they can build a web page, ideally targeted for a mobile display. As users browse the Web using mobile browsers built into their phones, the contents of the web page appear.

Second, developers can build an app, which typically is a device-specific program that users download and install (either free or for a price) onto a device. The app, in turn, displays an icon on the device, which users click to start the app. FIGURE 8 illustrates a page from which users can access hundreds of thousands of apps that have been created for the iPhone.

Third, developers can create a widget, which is much like an app that the user downloads and installs to his or her mobile device. A widget differs from an app in that the widget is always active. A clock widget's icon, for example, might constantly display the current time, a weather widget's icon, in turn, might display changing weather conditions, and a map widget may constantly update its icon to show the user's location. In other words, a widget is always running and may possibly be communicating. Having many active widgets at one time might affect a device's performance.



FIGURE 8 From business apps, to games, to education, there are hundreds of thousands of apps available for the iPhone.

CASE 14-1 W3C MOBILE PAGE CHECKER

Web developers should be familiar with the W3C website and its outstanding development tutorials and tools. When it comes to malformed HTML tags, mobile web browsers, because of their compact size (code size), tend not to be as forgiving as desktop browsers. Before you launch your mobile pages, you should validate your code using the S3C Mobile Checker at www.w3c.org/mobile, as shown in FIGURE 9.

Within the W3C Mobile Checker site, you simply enter the URL (web address) of the page you want to examine. The W3C Mobile Checker, in turn, analyzes the page's HTML and displays a summary similar to the one shown in FIGURE 10.

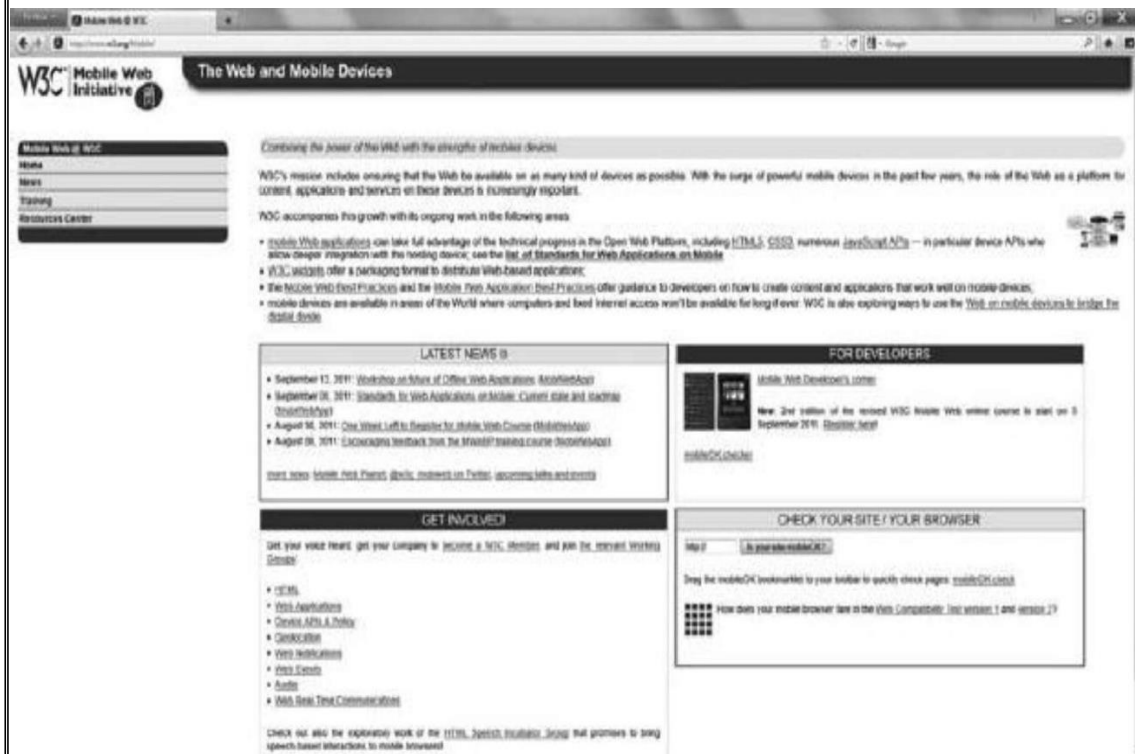


FIGURE 9 The W3C Mobile Checker provides warnings and error messages that developers can use to improve the quality of their mobile web pages.

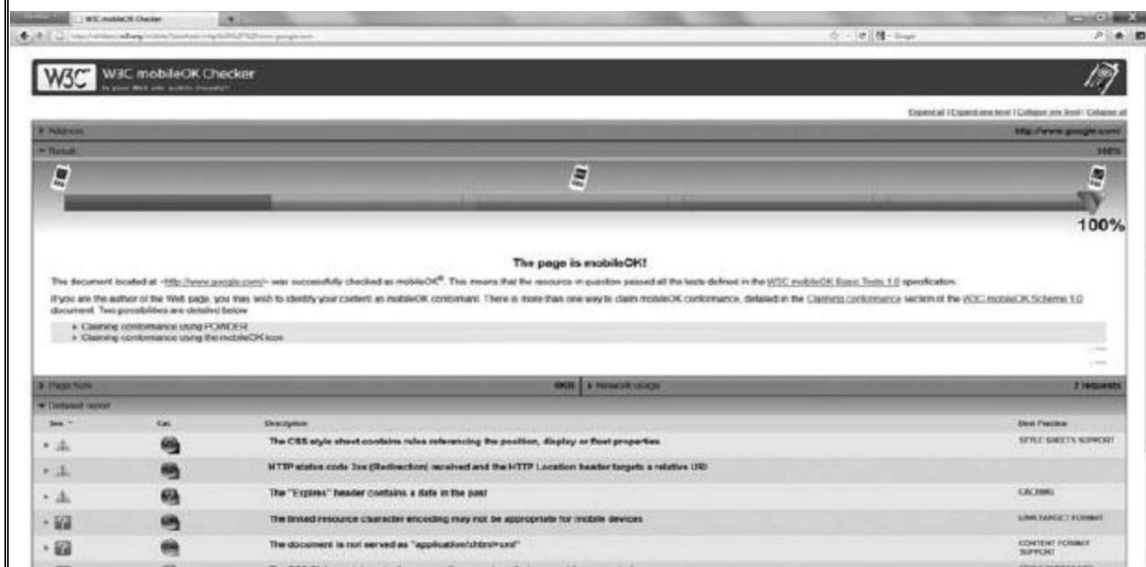


FIGURE 10A page summary created by the W3C Mobile Checker.

Revisiting the Role of HTML5

HTML is the markup language that developers use to build pages for display on the Web and mobile web. When a user browses the Web, either with a computer or mobile device, a web browser downloads and interprets the HTML tags to build the display page the user sees. FIGURE 11, for example, shows a simple HTML page and its resultant display within a web page and mobile device.

HTML5 is the fifth major release of HTML. HTML5 is important because developers can use it to create multimedia pages similar to what they previously created using Flash. Unlike Flash-based pages, which handheld browsers could not display, HTML5 multimedia pages display on all devices. In this way, HTML5 opens a vast new area of development for mobile devices.

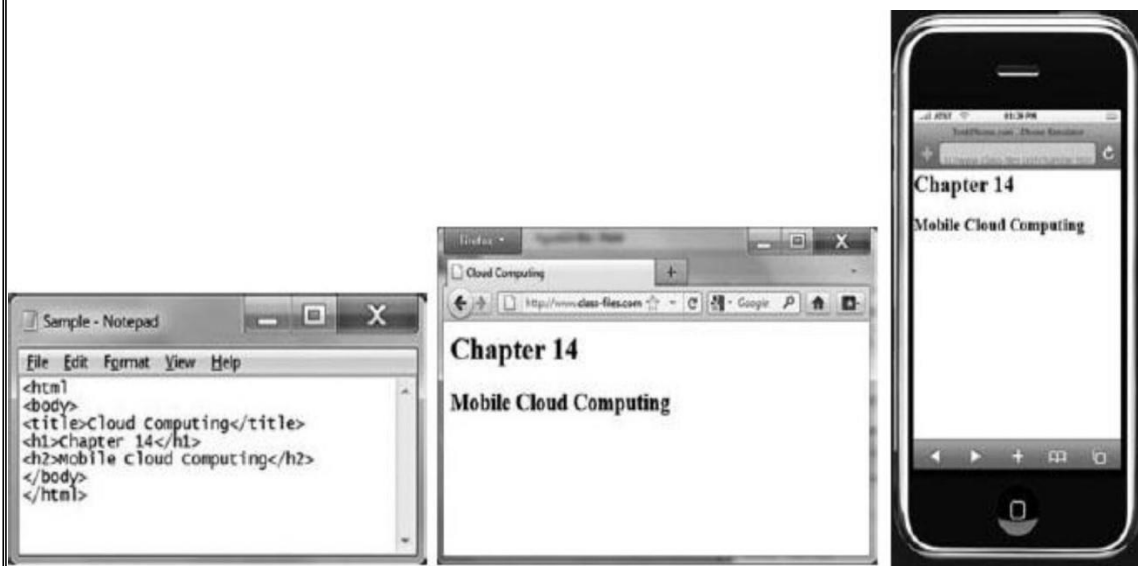


FIGURE 11 An HTML markup file and its display within a web browser and mobile device.

CASE study CISCO IBSG PRESENTATION: “WHEN MOBILE AND CLOUD COLLIDE”

Cisco is one of the key network solution companies driving the infrastructure upon which the cloud resides. Within Cisco, the Internet Business Solutions Group (IBSG) studied mobile solutions and their use of the cloud. The group created five key predictions, which they titled “When Mobile and Cloud Collide.”

Mobile Development Considerations

Coding Cloud-Based Applications, looks at the details of creating a cloud application—it is the chapter for coders and developers. Many people play different roles in the development of a mobile solution (project manager, sales and marketing, technical support, coder, tester, and more). If you work on a mobile solution, remember the following development considerations:

- The mobile web is not the traditional web: You should not expect to use your traditional web pages as mobile solutions. You should optimize your web solutions and then optimize your mobile solutions.
- Fast is good: Mobile data communication is still slower than most computer-based data communication solutions. As such, you should optimize your mobile web layout and design to maximize download performance.
- Remember your goals and requirements: As you design your mobile solutions, keep your original goals and requirements in mind to ensure that your solution matches your business strategy.
- You cannot support everything: Pick your largest market segment (or device) and focus your initial efforts there.
- Do not treat mobile content as an afterthought; create, do not convert, mobile content: Do not simply convert your traditional web content for use on the mobile web. Instead, design your mobile content for optimal performance and market impact.
- Handle different display sizes differently: Mobile applications should query the browser or device to determine the supported display size and then provide matching display content dynamically.

Question Bank**Multiple Choice Questions****(Questions for Application)**

1. Which of the following factors needs to be considered while moving from smartphone execution to a virtual machine running in the cloud?
A. Network bandwidth
B. Memory
C. Processing power
D. storage
2. In which of the following computing, mobile device serves as the presentation platform or the display?
A. Soft
B. Mobile
C. API
D. application
3. Which of the following feature is highly emphasized in open system for mobile application?
A. scalability
B. integrity
C. interoperability
D. readability
4. Which of the following has made the development of mobile Web applications much easier and more powerful ?
A. Web Services
B. Cloud Computing
C. IM
D. analog communication
5. Which of the following is a push mechanism for publishing Web site data that is device-specific?
A. WAP
B. GPRS
C. GAP
D. SAP
6. The process of moving data, applications or other business elements to a cloud computing environment is?
A. Cloud migration
B. Cloud seeding
C. Business continuity Plan
D. Cloud bursting

7. Which is the feature that ensure the authorization of the user to access the server from any where regardless of device.
A. IP whitelisting
B. Automation
C. Scaling
D. Migration
8. Which is a way to reduce the risk of vendor lock-in
A. migration
B. automation
C. open solutions
D. IP whitelisting
9. The application expands into the cloud temporarily, When the on-site application encounters increased user demand, this process is known as _____
A. Cloud migration
B. Cloud Bursting
C. Cloud switching
D. Cloud evolution
10. When the user demand declines, the application leaves the cloud, this is possible because of
A. Cloud migration
B. Cloud Bursting
C. Cloud switching
D. Cloud scaling
11. What does demand period indicate as a performance indicator
A. The cost going high because of excess application use
B. The application have time frame when its usage is high or low
C. The vendor scheme of billing the application as per extensive use
D. The application demand increasing strategy to be observed by user

(Questions for skills)

12. Access control, Auditing and Authentication are the key mechanism for?
A. protecting data
B. business continuity
C. cloud switching
D. capacity plan
13. Which of the following are essential characteristic of smart phone in cloud ?

- A. A smart phone offers advanced calling features such as video calls or conferencing
B. A smart phone has a recognizable operating system
C. A smart phone comes with a touch screen
D. All of the mentioned
14. Mobile Lite is an extension of the _____ cloud service.
A. Salesforce.com
B. Google.com
C. aol.com
D. yahoo.com
15. Which of the following is most popular open-system alternative to system like iOS or BlackBerry OS?
A. Android OS
B. Symbian
C. Solaris
D. windows
16. Which of the following was built to serve as a mobile platform for Internet computing ?
B. Android
A. Windows Phone
C. Symbian
D. Microsoft
17. Difference between an app and widget is
A. Widgets can not be downloaded, app's can be
B. Apps are mobile device independent, widgets are not
C. Apps are always running, widgets are not
D. Widgets are always active and apps are device specific programs
18. What did First-generation phones supported
A. analog communication
B. digital communication.
C. Internet communication
D. Wifi connection
19. Digital communication was introduced in which cell phone generation
A. first
B. second
C. third
D. fourth
20. Which of the following is a not a cloud migration platform
A. RIGHTSCALE
B. 3TERA
C. McAfee

D. CLOUDSWITCH

21. The idea of mobile computing has been started since
A. 2000
B. **1990**
C. 1980
D. 1973
22. Which of the following is not a part of Mobile cloud ecosystem
A. Apps and widget
B. **Compact disks**
C. Transmission speeds for upload and download operations
D. Keyboard interface
23. Abbreviate HIPAA act for health care specifying privacy requirements.
A. Health infrastructure Privacy and Portability
B. Help Insurance Privacy and Portability
C. **Health Insurance Privacy and Portability**
D. Health information Privacy and Portability
24. Abbreviate FERPA act for education which specify privacy requirements
A. Family Enterprise Rights and Privacy
B. Federal Educational Rights and Privacy
C. **Family Educational Rights and Privacy**
D. Forest Educational Rights and Privacy
25. Which changes the document content, such as changing a PNG graphic to a GIF graphic to improve the download performance or device display.
A. **transcoder**
B. Widgets
C. Smart phone
D. W3C Mobile Checker

(Questions for Skills)

1. What do you mean by cloud migration? Explain the benefits of cloud migration.
2. Analyze a case study with reference to cloud migration.
3. What are the characteristics of application which indicate its performance in cloud?
4. What are the IT governance issues to be considered for application migration?
5. Summarize cloud bursting with diagram.
6. What are the necessary measures to be taken to protect data before application migration?
7. Debate on mobile cloud eco system.
8. Debate on evolution of mobile computing.
9. Illustrate the cell phone capabilities in terms of generations.
10. Debate on Mobile players.
11. Illustrate different ways to build mobile solutions?
12. Describe the development consideration for designing Mobile cloud solution.

Reg.No

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SRINIVAS UNIVERSITY
FIFTH SEMESTER BCA DEGREE
EXAMINATION MODEL QUESTION PAPER
Cloud Computing

Time: 2 Hours

Max. Marks: 50

*Instructions: Answer any 10 Questions from PART-A and One Full Questions from Each UNIT
in PART-B*

PART _ A

(10x1 = 10)

1.

a. ____is services on demand software

- A. SaaS
- B. PaaS
- C. IaaS
- D. NaaS

b. A Computer network where every computer shares resource of every other computer is?

- A. Cloud computing
- B. Grid computing
- C. Utility computing
- D. Virtualization

c. ____offers browser-based development environment.

- A. PAAS
- B. SAAS
- C. IAAS
- D. IDAAS

d. Knowledge ,Possession and inheritance is a criteria used in

- A. MIM (Mobile Id Management)
- B. MFA (Multi factor Authentication)
- C. SSO (Single Sign On)
- D. Open Id

e. _____offers online storage space on-demand

- A. Managed cloud storage
- B. Partitioned cloud storage

C. Unmanaged cloud storage

D. Open cloud Storage

f. _____ backups only back up the data that has been changed or updated since the last backup.

A. Incremental Backup

B. Full Backup

C. Differential Backup

D. hybrid back up

g. _____ is a situation in which many under-utilized servers utilize more space or consume more resources than can be justified by their workload

A. Server sprawl

B. Server utilization

C. Server scheduling

D. Server Virtualization

h. The Virtual server running on the physical server are safely separated & don't affect each other is called _____

A. Partitioning

B. Isolation

C. encapsulation

D. Mirroring

i. Which of the following model type is not trusted in terms of security ?

A. Public

B. Private

C. Hybrid

D. community

j. what is the time taken for an issue to be resolved once logged by the service provider.

A. Resilience time

B. Resolution time

C. Response time

D. Reference time

k. The process of moving data, applications or other business elements to a cloud computing environment is?

A. Cloud migration

- B. Cloud seeding
- C. Business continuity Plan
- D. Cloud bursting

1. which is a way to reduce the risk of vendor lock-in

- A. migration
- B. automation
- C. open solutions
- D. IP whitelisting

PART- B

UNIT - I

2. a. Briefly explain all cloud deployment models.

b. What are the benefits of Multitenant nature of saas. (4 + 4)

OR

3. a. Explain private cloud mention its advantage and disadvantages.

b. Explain briefly the cloud service models. (4 + 4)

UNIT-II

4. a. Write about disadvantage of PaaS.

b. What is load balancing? Write its application. (4 + 4)

OR

5.a. What is SSO ? explain the working of SSO.

b. Write a short note on cloud storage system. (4 + 4)

UNIT-III

6 .a. What are the Pros and Cons of network storage.

b. Explain the features of Virtualization. (4 + 4)

OR

- 7 .a. Explain the types of back up.
- b. What is storage virtualization explain its importance. (4 + 4)

UNIT-IV

8. a. What is Business continuity? Explain its Key elements.
- b. Explain Auditing/inspection of a cloud based solution. (4 + 4)

OR

- 9 . a. Explain Disaster Recovery and components of DRaaS.
- b. Write the security advantages of cloud Based solution. (4 + 4)

UNIT V

10. a. Explain cloud bursting with diagram. —
- b. Explain different ways to build mobile solutions. (4 + 4)

OR

- 11 .a. Describe the development consideration for designing Mobile cloud solution.
- b. What is cloud migration? explain the benefits of cloud migration. (4 + 4)